BME302 Unit III – Pipe Flow & Boundary Layer (Expanded Notes)

Lecture Notes · Fluid Mechanics and Fluid Machines

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How to use these notes. Every topic, worked example and question carries one tag. CORE must-know syllabus material at AKTU exam level: master these first. HOTS higher-order thinking: analyse, compare, justify or design (Bloom K4 to K6). ADVANCED beyond the exam level for most students: shaded orange, optional reading.

Unit overview

Unit III is about viscous flow. It covers how friction acts on fluid moving through pipes, and how it acts in the thin boundary layer next to any solid surface. Pipe friction sets the energy a pump must supply. The boundary layer sets the drag on cars, aircraft and turbine blades. The unit maps to **CO3: Evaluate the velocity and pressure variations in various types of simple flows (K3)**, in 8 lectures. These expanded notes give every derivation in full, at least two worked examples per topic, and a formula sheet with practice questions. Notation follows R.K. Bansal and Som, Biswas & Chakraborty, with the Darcy friction factor f used throughout.

AKTU syllabus, Unit III: Equation of motion for laminar flow through pipes, resistance to flow, minor losses, pipes in series and parallel, power transmission through a pipe, siphon, water hammer. Boundary layer thickness, boundary layer over a flat plate, laminar boundary layer, application of the momentum equation, turbulent boundary layer, laminar sublayer, separation and its control, drag and lift.

Lecture	Topic	Key outcome
L1	Laminar flow through circular pipes and between parallel plates	Velocity profile, shear stress and pressure drop
L2	Resistance to flow: Darcy–Weisbach equation, friction factor, turbulent pipe flow	Major head loss; use of the Moody chart
L3	Minor losses; hydraulic gradient and total energy lines	Losses at fittings, entry, exit and area changes
L4	Pipes in series, parallel and branched; power transmission through a pipe	Flow in pipe systems; maximum power condition
L5	Siphon and water hammer	Summit pressure; surge pressure on valve closure
L6	Boundary layer concept, thicknesses, laminar boundary layer, momentum integral equation	Thickness and skin friction on a flat plate
L7	Turbulent boundary layer, laminar sublayer, separation and its control	Where and why flow separates, and how to prevent it
L8	Drag and lift	Forces on immersed bodies

Common notation.

Symbol	Meaning	SI unit
ρ	Mass density	kg/m³
μ	Dynamic viscosity	N·s/m² (Pa·s)
$\nu = \mu/\rho$	Kinematic viscosity	m²/s
D, R	Pipe diameter, pipe radius	m
L	Pipe length	m
V	Mean velocity across a section	m/s
Q	Discharge	m³/s
$\mathrm{Re} = \rho VD/\mu = VD/\nu$	Reynolds number	–
f	Darcy friction factor ($f = 4f_1$, where f_1 is the coefficient of friction used in some texts)	–
h_f	Head lost to friction	m of fluid
K	Minor loss coefficient	–
U	Free-stream velocity	m/s
δ, δ^*, θ	Boundary layer, displacement and momentum thickness	m
τ_0	Wall shear stress	N/m²
$\mathrm{Re}_x = Ux/\nu$	Local Reynolds number at distance x from the leading edge	–

Laminar or turbulent? Reynolds showed with a dye streak in a glass tube that the flow regime depends on one number, $\mathrm{Re} = \rho VD/\mu$, the ratio of inertia to viscous forces. In a pipe the flow is laminar below $\mathrm{Re} \approx 2000$, transitional from 2000 to 4000, and turbulent above 4000. On a flat plate the boundary layer turns turbulent near $\mathrm{Re}_x \approx 5 \times 10^5$.

0. Before you start: basic ideas you need CORE

This section collects the basic ideas that the whole unit uses again and again. Read it first if pipe-flow problems feel hard to start. Every later section assumes you are at ease with units, head, continuity, viscosity and the Reynolds number.

> **IN SIMPLE WORDS**
>
> Think of water flowing from the tank on your roof to a tap downstairs. The water carries energy, and engineers measure that energy as a height, called head. The pipe wall rubs against the water, so some head is lost on the way. This unit is about how much head is lost, and why. To solve almost any problem in it you need only five ideas: units, head, continuity, viscosity and the Reynolds number.

0.1 Quantities and units CORE

Most wrong answers in pipe-flow problems come from units, not from physics. Convert every value to SI units (m, kg, s, N, Pa) before you put it into a formula.

Symbol	Meaning	SI unit	Typical value
D	Pipe diameter	m	0.025 to $1\,\mathrm{m}$
$R = D/2$	Pipe radius	m	Half the diameter
L	Pipe length	m	10 to $5000\,\mathrm{m}$
$A = \pi D^2/4$	Cross-sectional area of the pipe	m^2	$0.0707\,\mathrm{m}^2$ for $D = 0.3\,\mathrm{m}$
V	Mean velocity	$\mathrm{m/s}$	1 to $3\,\mathrm{m/s}$ in water mains
Q	Discharge (volume flow rate)	m^3/s	0.001 to $1\,\mathrm{m}^3/\mathrm{s}$
ρ	Mass density	$\mathrm{kg/m}^3$	Water 1000; oils 800 to 950
μ	Dynamic viscosity	$\mathrm{Pa{\cdot}s}$ (same as $\mathrm{N \cdot s/m}^2$)	Water 1.0×10^{-3} at $20^\circ\mathrm{C}$; oils 0.01 to 1
$\nu = \mu/\rho$	Kinematic viscosity	m^2/s	Water 1.0×10^{-6}
g	Acceleration due to gravity	$\mathrm{m/s}^2$	9.81
p	Pressure	$\mathrm{Pa} = \mathrm{N/m}^2$	Atmosphere about $101.3\,\mathrm{kPa}$
τ	Shear stress	$\mathrm{N/m}^2$	A few $\mathrm{N/m}^2$ in pipes
h_f	Head lost to friction	m of the flowing fluid	A few metres to tens of metres
f	Darcy friction factor	No unit	0.01 to 0.05 in turbulent flow
Re	Reynolds number	No unit	Hundreds (thick oils) to 10^6 (water mains)
P	Power	W	Watts to megawatts

Conversions that students often get wrong.

Given in	Convert to	Rule	Example
mm	m	Divide by 1000	$80\ \text{mm} = 0.08\ \text{m}$
cm^2	m^2	Divide by 10^4	$25\ \text{cm}^2 = 0.0025\ \text{m}^2$
L/s	m^3/s	Divide by 1000	$15\ \text{L/s} = 0.015\ \text{m}^3/\text{s}$
cm^3 (mL)	m^3	Multiply by 10^{-6}	$50\ \text{cm}^3 = 5 \times 10^{-5}\ \text{m}^3$
kPa	m of water	Change to Pa, then $h = p/(\rho g)$	$100\ \text{kPa} \to 100\,000/(1000 \times 9.81) = 10.19\ \text{m}$
poise (P)	Pa·s	$1\ \text{P} = 0.1\ \text{Pa}\cdot\text{s}$	$1.2\ \text{P} = 0.12\ \text{Pa}\cdot\text{s}$
centipoise (cP)	Pa·s	$1\ \text{cP} = 10^{-3}\ \text{Pa}\cdot\text{s}$	$9\ \text{cP} = 0.009\ \text{Pa}\cdot\text{s}$
stokes (St)	m^2/s	$1\ \text{St} = 10^{-4}\ \text{m}^2/\text{s}$	$1\ \text{cSt} = 10^{-6}\ \text{m}^2/\text{s}$ (water)
rpm	rad/s	$\omega = 2\pi N/60$	$1500\ \text{rpm} \to 157.1\ \text{rad/s}$
km/h	m/s	Divide by 3.6	$72\ \text{km/h} = 20\ \text{m/s}$

0.2 Head: energy per unit weight CORE

Engineers measure the energy of flowing water as a height. **Head** is energy per unit weight of fluid. Its unit is J/N=m . There are three kinds:

- **Pressure head**, $p/(\rho g)$: the height to which the pressure could push the liquid up an open tube (a piezometer).
- **Velocity head**, $V^2/(2g)$: the height a jet would reach if it were shot straight up at speed V.
- **Datum head**, z: the height of the point above a chosen reference level.

Derivation from scratch: energy equation with head loss

Aim. Show that the total head at an upstream section 1 equals the total head at a downstream section 2 plus the head lost between them.

Assumptions.

1. The flow is steady and the liquid is incompressible (ρ is constant).
2. Each section has one pressure p, one mean velocity V and one height z above the datum.
3. No pump or turbine lies between sections 1 and 2.

Step 1. Energy carried by a small mass. Take a small mass m of liquid passing section 1. It carries three kinds of energy.

$$\text{Pressure (flow) energy} = p \times \text{volume} = p\,\frac{m}{\rho}$$

$$\text{Kinetic energy} = \frac{1}{2}\,mV^2, \qquad \text{Potential energy} = mgz$$

Step 2. Change to energy per unit weight. The weight of the mass is mg. Divide each energy by mg to get a head, in metres.

$$\frac{p\,m/\rho}{mg} = \frac{p}{\rho g}, \qquad \frac{\frac{1}{2}mV^2}{mg} = \frac{V^2}{2g}, \qquad \frac{mgz}{mg} = z$$

Step 3. Total head at each section. Add the three heads.

$$H_1 = \frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1, \qquad H_2 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

Step 4. Apply conservation of energy. Energy is not created between 1 and 2. Friction and eddies turn some of it into heat, which is lost to the flow. Call this lost energy per unit weight h_L.

$$H_1 = H_2 + h_L$$

Result.

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2 + h_L$$

What it tells us. Head can change form: pressure head can become velocity head or datum head and back. But the total can only fall in the direction of flow, by the loss h_L. Every pipe problem in this unit is this one equation with the right h_L put in.

Here h_L is the head lost to friction and fittings between 1 and 2. It is always positive. With $h_L = 0$ this is Bernoulli's equation.

Example 0.1 CORE A pressure gauge on a water main reads 200 kPa. Express this pressure as a head of water.

1. $p = 200\,\text{kPa} = 200\,000\,\text{Pa}$
2. $h = \dfrac{p}{\rho g} = \dfrac{200\,000}{1000 \times 9.81} = 20.39\,\text{m of water}$

0.3 Discharge and continuity CORE

Discharge is the volume of fluid passing a section each second: $Q = AV$. Water hardly compresses, so in a full pipe the same Q must pass every section. This is **continuity**: $A_1V_1 = A_2V_2$. Since $A \propto D^2$, the velocity changes as $V_2 = V_1\,(\,D_1/D_2\,)^{\,2}$. Halving the diameter makes the water four times faster.

Derivation from scratch: continuity equation

Aim. Show that $A_1V_1 = A_2V_2$ for steady flow of a liquid in a full pipe.

Assumptions.

1. The flow is steady: nothing changes with time at any point.
2. The pipe runs full, and no liquid leaks in or out between sections 1 and 2.
3. The liquid is incompressible, so $\rho_1 = \rho_2 = \rho$.

Step 1. Mass passing a section each second. In one second the liquid at a section moves forward a distance V. The volume that passes is $A \times V$, and its mass is ρAV.

$$\dot{m}_1 = \rho_1 A_1 V_1, \qquad \dot{m}_2 = \rho_2 A_2 V_2$$

Step 2. Conservation of mass. In steady flow, mass cannot pile up between the two sections. So the mass entering at 1 each second equals the mass leaving at 2.

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$$

Step 3. Use incompressibility. Put $\rho_1 = \rho_2 = \rho$ and cancel ρ from both sides.

$$A_1 V_1 = A_2 V_2 = Q$$

Step 4. Write it with diameters. For circular pipes $A = \pi D^2/4$. The factor $\pi/4$ cancels.

$$\frac{\pi}{4} D_1^2 V_1 = \frac{\pi}{4} D_2^2 V_2 \quad \Rightarrow \quad V_2 = V_1 \left(\frac{D_1}{D_2} \right)^2$$

Result.

$$Q = A_1 V_1 = A_2 V_2, \qquad V_2 = V_1 \left(\frac{D_1}{D_2} \right)^2$$

What it tells us. The discharge is the same at every section of a pipe in series. A smaller area means a larger velocity, in the ratio of the areas, or the square of the diameters.

Example 0.2 CORE Water flows at 1 m/s in a 100 mm pipe that narrows to 50 mm. Find the discharge and the velocity in the narrow pipe.

1. $Q = A_1 V_1 = \frac{\pi}{4} \times 0.1^2 \times 1 = 7.85 \times 10^{-3}\ \mathrm{m^3/s}$ (7.85 L/s)
2. $V_2 = V_1 \left(\frac{D_1}{D_2} \right)^2 = 1 \times \left(\frac{0.1}{0.05} \right)^2 = 4\ \mathrm{m/s}$

0.4 Viscosity and the no-slip condition CORE

Viscosity is a fluid's resistance to one layer sliding over another. Honey has a high viscosity, and water a low one. The **no-slip condition** says that fluid touching a solid wall moves with the wall. In a fixed pipe, the fluid at the wall is at rest. The velocity therefore rises from zero at the wall to its largest value at the centre. Neighbouring layers slide past each other, and viscosity resists this sliding with a shear stress. Newton's law of viscosity gives this stress:

$$\tau = \mu \frac{du}{dy}$$

Here τ is the shear stress ($\mathrm{N/m^2}$), μ is the dynamic viscosity ($\mathrm{Pa{\cdot}s}$), and du/dy is the velocity gradient ($\mathrm{s^{-1}}$), with y measured from the wall. The gradient is steepest at the wall, so the shear stress is largest there. The viscosity of a liquid falls as it gets hotter, which is why warm oil flows more easily.

0.5 Laminar and turbulent flow: the Reynolds number CORE

In **laminar** flow the fluid moves in smooth layers, like cards sliding in a pack, and a dye streak stays a thin line. In **turbulent** flow random eddies mix the fluid, and a dye streak spreads at once. The **Reynolds number** decides which one occurs:

$$\mathrm{Re} = \frac{\rho V D}{\mu} = \frac{V D}{\nu} = \frac{\text{inertia force}}{\text{viscous force}}$$

At small Re, viscosity wins and damps out any disturbance. At large Re, inertia wins and eddies grow.

Re **in a pipe**	**Flow regime**
Below 2000	Laminar
2000 to 4000	Transitional
Above 4000	Turbulent

Water in ordinary pipes is almost always turbulent. Thick oils in small pipes are often laminar.

Example 0.3 CORE Water ($\nu = 10^{-6}\ \mathrm{m^2/s}$) flows at 0.5 m/s in a 50 mm pipe. Is the flow laminar or turbulent?

1. $\mathrm{Re} = \dfrac{V D}{\nu} = \dfrac{0.5 \times 0.05}{10^{-6}} = 25\,000$
2. Since $25\,000 > 4000$, the flow is **turbulent**.

Common mistakes and exam tips.

- Putting a diameter in mm straight into a formula. Change it to metres first: $80\ \mathrm{mm} = 0.08\ \mathrm{m}$.
- Using D where R is meant. The area is $A = \pi D^2/4 = \pi R^2$, never πD^2.
- Mixing gauge and absolute pressure. A surface open to the air has $p = 0$ gauge. Use the same kind at both points.
- Mixing μ and ν. Use $\mathrm{Re} = \rho V D/\mu$ or $V D/\nu$, never $\rho V D/\nu$.
- Writing heads of different fluids together. For oil, divide the pressure by $\rho_{\text{oil}}\, g$, not by $1000\, g$.

Check your understanding.

1. CORE Convert 30 L/s to $\mathrm{m^3/s}$. *(It is $0.03\ \mathrm{m^3/s}$.)*
2. CORE A pipe narrows from 200 mm to 100 mm. How does the velocity change? *(It becomes four times as large, since $V \propto 1/D^2$.)*
3. HOTS Water flows in a pipe at $\mathrm{Re} = 1500$. The velocity is doubled. Is the flow still surely laminar? *(No. Re becomes 3000, which is in the transitional range.)*

1. Laminar flow through circular pipes (L1) CORE

In steady, fully developed laminar flow, the velocity across a pipe is a paraboloid. The mean velocity is half the maximum, and the pressure drop is directly proportional to the velocity. This is the Hagen–Poiseuille law. It is the only pipe-flow result that can be derived exactly from first principles.

IN SIMPLE WORDS

Laminar flow is slow, smooth flow in which the fluid moves in neat layers. Picture honey or thick oil creeping through a thin tube. The fluid at the wall stays still, and the fluid at the centre moves fastest. Engineers meet it in oil lines, lubrication systems and blood vessels. For this one case the velocity and the head loss can be worked out exactly, without any experiment.

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

1.1 Entrance length and fully developed flow ADVANCED

Fluid entering a pipe from a reservoir has a nearly flat velocity profile. A boundary layer grows from the wall until it meets at the axis. Beyond this **entrance length** the profile no longer changes along the pipe, and the flow is **fully developed**. The analysis below applies only to the fully developed region.

$$\frac{L_e}{D} \approx 0.058\,\mathrm{Re} \quad (\text{laminar}), \qquad \frac{L_e}{D} \approx 4.4\,\mathrm{Re}^{1/6} \quad (\text{turbulent})$$

At $\mathrm{Re} = 2000$ the laminar entrance length is about 116 diameters. A turbulent entrance is much shorter, typically 20 to 30 diameters.

1.2 Equation of motion CORE

IN SIMPLE WORDS

We find the motion from a force balance on a small cylinder of fluid. The pressure behind it pushes it forward. The shear (friction) on its curved surface holds it back. In steady flow these forces balance, like a car cruising at constant speed. In the result, $\partial p/\partial x$ is the rate at which pressure changes along the pipe, and τ is the shear stress on the cylinder's surface.

Derivation from scratch: equation of motion for laminar pipe flow

Aim. Find the relation between the pressure gradient $\partial p/\partial x$ and the shear stress τ at any radius r in a pipe.

Assumptions.

1. The flow is steady, laminar and fully developed: the velocity profile is the same at every section.
2. The fluid is incompressible and Newtonian.
3. The pipe is straight, of radius R, with its axis along x, inclined at θ to the horizontal.
4. The pressure is the same at all points of a cross-section (it changes only with x).

Step 1. Choose the element. Take a cylinder of fluid of radius r and length Δx, with its axis on the pipe axis. Its end faces each have area πr^2. Its curved surface has area $2\pi r\,\Delta x$. Its volume is $\pi r^2 \Delta x$.

Step 2. Pressure force on the left face. The pressure p acts on the upstream face and pushes the cylinder forward (in the $+x$ direction).

$$F_{\text{left}} = +\, p\,\pi r^2$$

Step 3. Pressure force on the right face. Over the length Δx the pressure changes to $p + \dfrac{\partial p}{\partial x}\Delta x$. It acts on the downstream face and pushes the cylinder backward (in the $-x$ direction).

$$F_{\text{right}} = -\left(p + \frac{\partial p}{\partial x}\,\Delta x\right)\pi r^2$$

Step 4. Shear force on the curved surface. The fluid outside the cylinder is nearer the wall, so it moves more slowly. It drags back on the cylinder with a shear stress τ over the curved area. This force acts in the $-x$ direction.

$$F_{\text{shear}} = -\,\tau\,(\,2\pi r\,\Delta x\,)$$

Step 5. Weight component along the axis. The weight of the cylinder is $\rho g \pi r^2 \Delta x$. For a pipe rising at angle θ, its component along the axis acts backward.

$$F_{\text{weight}} = -\,\rho g\,\pi r^2 \Delta x\,\sin\theta$$

Step 6. Newton's second law. The flow is steady and fully developed, so the cylinder moves at constant velocity. Its acceleration is zero, so the sum of the forces along x is zero.

$$p\,\pi r^2 - \left(p + \frac{\partial p}{\partial x}\,\Delta x\right)\pi r^2 - \tau\,(\,2\pi r\,\Delta x\,) - \rho g\,\pi r^2 \Delta x\,\sin\theta = 0$$

Step 7. Cancel the $p\,\pi r^2$ terms. Open the bracket. The terms $+p\,\pi r^2$ and $-p\,\pi r^2$ cancel.

$$-\frac{\partial p}{\partial x}\,\Delta x\,\pi r^2 - \tau 2\pi r\,\Delta x - \rho g\,\pi r^2 \Delta x\,\sin\theta = 0$$

Step 8. Divide by the volume $\pi r^2 \Delta x$. Note that $\dfrac{2\pi r\,\Delta x}{\pi r^2 \Delta x} = \dfrac{2}{r}$.

$$-\frac{\partial p}{\partial x} - \frac{2\tau}{r} - \rho g \sin\theta = 0$$

Step 9. Rearrange. Move the shear term to the right-hand side.

$$-\frac{\partial p}{\partial x} - \rho g \sin\theta = \frac{2\tau}{r}$$

Step 10. Combine pressure and gravity. Along the pipe the height rises as $dz = \sin\theta\, dx$, so $\sin\theta = \partial z/\partial x$. Then $\rho g \sin\theta = \partial\left(\rho g z\right)/\partial x$, and the left-hand side is the gradient of one quantity, $p^* = p + \rho g z$.

$$-\frac{\partial p}{\partial x} - \frac{\partial\left(\rho g z\right)}{\partial x} = -\frac{\partial}{\partial x}\left(p + \rho g z\right) = -\frac{\partial p^*}{\partial x}$$

Result.

$$-\frac{\partial p}{\partial x} - \rho g \sin\theta = \frac{2\tau}{r} \quad \Rightarrow \quad -\frac{\partial p^*}{\partial x} = \frac{2\tau}{r}, \qquad p^* = p + \rho g z$$

What it tells us. The fall in piezometric pressure along the pipe is what pushes the fluid against friction. For flow in the $+x$ direction the pressure falls along x, so $\partial p/\partial x$ is negative and $-\partial p/\partial x$ is positive. A steeper pressure fall means a larger shear stress.

The flow is driven by the gradient of the **piezometric pressure** $p^* = p + \rho g z$. In a horizontal pipe p^* is simply p, and the results below are written for that case. For an inclined pipe, replace p by $p + \rho g z$ throughout.

1.3 Shear stress distribution CORE

IN SIMPLE WORDS

Shear stress is the friction force per unit area between fluid layers. In a pipe it is zero at the centre and grows in a straight line to its largest value at the wall. Think of a crowd walking down a corridor: people in the middle move freely, and those brushing the walls feel the most rubbing. The wall value, τ_0, is the stress that actually drags on the pipe.

Derivation from scratch: shear stress distribution

Aim. Find how the shear stress τ varies with the radius r, and its value τ_0 at the wall.

Assumptions.

1. All the assumptions of the equation of motion (Section 1.2) hold.
2. The pipe is horizontal, so $\theta = 0$ and $p^* = p$.

Step 1. Start from the force balance. With $\sin\theta = 0$, the result of Section 1.2 becomes:

$$-\frac{\partial p}{\partial x} = \frac{2\tau}{r}$$

Step 2. Make τ the subject. Multiply both sides by $r/2$.

$$\tau = -\frac{\partial p}{\partial x}\frac{r}{2}$$

Step 3. Note that $\partial p/\partial x$ does not depend on r. The pressure is the same over a cross-section, so $\partial p/\partial x$ is one constant for the whole section. Hence τ is a constant times r: a straight line through the origin.

Step 4. Value at the axis. Put $r = 0$.

$$\tau\big|_{r=0} = -\frac{\partial p}{\partial x}\times\frac{0}{2} = 0$$

Step 5. Value at the wall. Put $r = R$.

$$\tau_0 = -\frac{\partial p}{\partial x}\frac{R}{2}$$

Result.

$$\tau = -\frac{\partial p}{\partial x}\frac{r}{2}, \qquad \tau_0 = -\frac{\partial p}{\partial x}\frac{R}{2} \quad (\text{at the wall})$$

What it tells us. Dividing the two gives $\tau/\tau_0 = r/R$. The shear stress grows in a straight line from zero at the centre to τ_0 at the wall. Over a length L, $-\partial p/\partial x = \left(p_1 - p_2\right)/L$, so $\tau_0 = \left(p_1 - p_2\right)D/\left(4L\right)$.

The shear stress is **zero at the axis and largest at the wall**, varying linearly with r. This result uses only equilibrium, so it holds for turbulent flow too.

1.4 Velocity distribution CORE

IN SIMPLE WORDS

The velocity profile shows how fast each layer moves. In laminar pipe flow it has the shape of a bullet nose (a paraboloid). The velocity is zero at the wall, because of no-slip, and largest at the axis, where it is $u_{\max}$. In the formula, r is the distance from the axis and R is the pipe radius. So $u = u_{\max}$ at $r = 0$, and $u = 0$ at $r = R$.

Derivation from scratch: velocity distribution in laminar pipe flow

Aim. Find the velocity u at any radius r, and the maximum velocity $u_{\max}$.

Assumptions.

1. The shear stress result of Section 1.3 holds: $\tau = -\frac{\partial p}{\partial x}\frac{r}{2}$.
2. The flow is laminar and the fluid is Newtonian, so $\tau = \mu\, du/dy$.
3. No slip at the wall: $u = 0$ at $r = R$.
4. $\partial p/\partial x$ is a constant across the section.

Step 1. Write Newton's law in terms of r. Newton's law uses y, the distance from the wall. Here $y = R - r$, so $dy = -dr$.

$$\tau = \mu \frac{du}{dy} = \mu \frac{du}{-dr} = -\mu \frac{du}{dr}$$

The minus sign is needed because u falls as r grows, so du/dr is negative while τ is positive.

Step 2. Equate the two expressions for τ. One comes from the force balance, the other from viscosity.

$$-\mu \frac{du}{dr} = -\frac{\partial p}{\partial x} \frac{r}{2}$$

Step 3. Make du/dr the subject. Divide both sides by $-\mu$.

$$\frac{du}{dr} = \frac{1}{2\mu} \frac{\partial p}{\partial x} r$$

Step 4. Integrate with respect to r. The factor $\frac{1}{2\mu}\frac{\partial p}{\partial x}$ is a constant. Since $\int r\, dr = r^2/2$, we add a constant of integration C.

$$u = \frac{1}{2\mu} \frac{\partial p}{\partial x} \frac{r^2}{2} + C = \frac{1}{4\mu} \frac{\partial p}{\partial x} r^2 + C$$

Step 5. Apply the boundary condition. At the wall, $r = R$ and $u = 0$.

$$0 = \frac{1}{4\mu} \frac{\partial p}{\partial x} R^2 + C \quad \Rightarrow \quad C = -\frac{1}{4\mu} \frac{\partial p}{\partial x} R^2$$

Step 6. Substitute C back.

$$u = \frac{1}{4\mu} \frac{\partial p}{\partial x} r^2 - \frac{1}{4\mu} \frac{\partial p}{\partial x} R^2$$

Step 7. Take out the common factor. Take out $-\frac{1}{4\mu}\frac{\partial p}{\partial x}$, which reverses the order inside the bracket.

$$u = -\frac{1}{4\mu} \frac{\partial p}{\partial x} \left(R^2 - r^2 \right)$$

Step 8. Find where u is largest. From Step 3, $du/dr = 0$ only at $r = 0$. The second derivative is $\frac{d^2u}{dr^2} = \frac{1}{2\mu}\frac{\partial p}{\partial x}$, which is negative since $\partial p/\partial x < 0$. So $r = 0$ (the axis) is a maximum.

$$u_{\max} = u\big|_{r=0} = -\frac{1}{4\mu} \frac{\partial p}{\partial x} R^2$$

Step 9. Write u in terms of $u_{\max}$. Take R^2 out of the bracket in Step 7.

$$u = -\frac{1}{4\mu} \frac{\partial p}{\partial x} R^2 \left(1 - \frac{r^2}{R^2} \right) = u_{\max} \left[1 - \left(\frac{r}{R} \right)^2 \right]$$

Result.

$$u = -\frac{1}{4\mu}\frac{\partial p}{\partial x}\left(R^2 - r^2\right) = u_{\max}\left[1 - \left(\frac{r}{R}\right)^2\right], \qquad u_{\max} = -\frac{1}{4\mu}\frac{\partial p}{\partial x}R^2$$

What it tells us. The velocity varies as r^2, so the profile is a parabola in any diametral plane. The maximum velocity grows with the pressure fall and with R^2, and falls as the viscosity rises.

The velocity profile is a **paraboloid of revolution**, with the maximum at the axis.

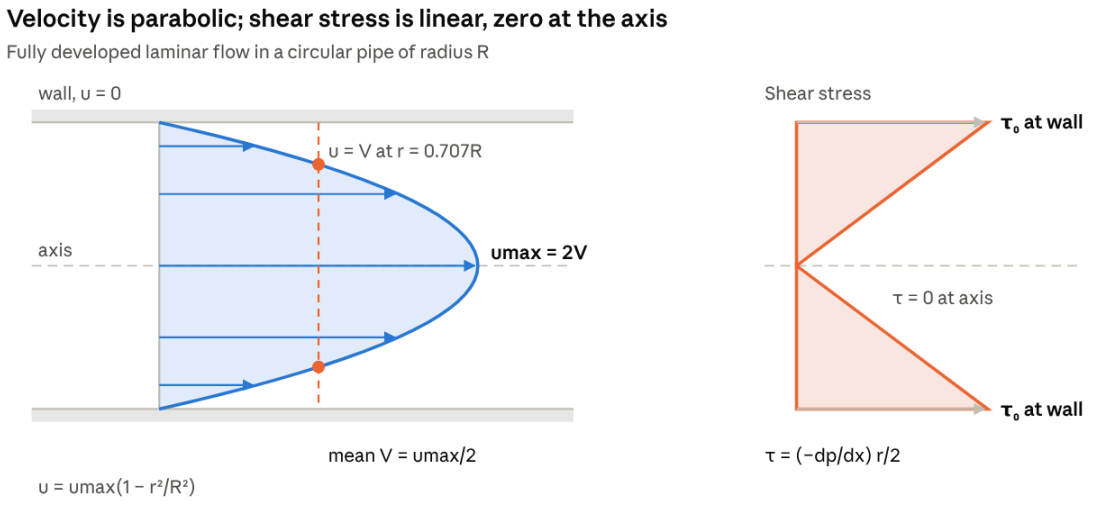


Figure 1. Laminar pipe flow: velocity profile and shear stress distribution

The velocity is greatest at the axis and zero at the wall, while the shear stress does the opposite.

1.5 Discharge and mean velocity CORE

IN SIMPLE WORDS

The discharge Q is found by adding up the flow through thin rings, like the rings of an onion cut across. The mean velocity V is the single uniform speed that would carry the same Q. In laminar pipe flow, V is exactly half of $u_{\max}$. If the centre moves at 2 m/s, the mean velocity is 1 m/s.

Derivation from scratch: discharge and mean velocity

Aim. Find the discharge Q and the mean velocity V, and show that $V = u_{\max}/2$.

Assumptions.

1. The velocity profile of Section 1.4 holds: $u = -\frac{1}{4\mu}\frac{\partial p}{\partial x}\left(R^2 - r^2\right)$.
2. The velocity is the same all round a thin ring at radius r.

Step 1. Choose a thin ring. Take a ring of radius r and small width dr. If it is cut open and laid flat, it is a strip of length $2\pi r$ and width dr.

$$dA = 2\pi r\, dr$$

Step 2. Flow through the ring. The velocity over the ring is u, so the discharge through it is velocity times area.

$$dQ = u\, dA = u\, 2\pi r\, dr$$

Step 3. Add up all the rings. Integrate from the axis ($r = 0$) to the wall ($r = R$).

$$Q = \int_0^R u\, 2\pi r\, dr$$

Step 4. Put in the profile. Take the constants $-\frac{1}{4\mu}\frac{\partial p}{\partial x}$ and 2π outside the integral.

$$Q = -\frac{1}{4\mu}\frac{\partial p}{\partial x} 2\pi \int_0^R \left(R^2 - r^2 \right) r\, dr = -\frac{\pi}{2\mu}\frac{\partial p}{\partial x} \int_0^R \left(R^2 r - r^3 \right) dr$$

Step 5. Evaluate the integral term by term.

$$\int_0^R \left(R^2 r - r^3 \right) dr = \left[\frac{R^2 r^2}{2} - \frac{r^4}{4} \right]_0^R = \frac{R^4}{2} - \frac{R^4}{4} - 0 = \frac{R^4}{4}$$

Step 6. Substitute back.

$$Q = -\frac{\pi}{2\mu}\frac{\partial p}{\partial x} \times \frac{R^4}{4} = -\frac{\pi}{8\mu}\frac{\partial p}{\partial x} R^4$$

Step 7. Mean velocity. By definition, V is the discharge divided by the full area πR^2. One π and R^2 cancel.

$$V = \frac{Q}{\pi R^2} = -\frac{\pi R^4}{8\mu\, \pi R^2}\frac{\partial p}{\partial x} = -\frac{1}{8\mu}\frac{\partial p}{\partial x} R^2$$

Step 8. Compare with $u_{\max}$. From Section 1.4, $u_{\max} = -\frac{1}{4\mu}\frac{\partial p}{\partial x} R^2$. Divide.

$$\frac{V}{u_{\max}} = \frac{1/8}{1/4} = \frac{1}{2}$$

Step 9. Where does the local velocity equal V? Put $u = V = u_{\max}/2$ in $u = u_{\max}\left[1 - \left(r/R\right)^2\right]$ and cancel $u_{\max}$.

$$1 - \left(\frac{r}{R}\right)^2 = \frac{1}{2} \quad \Rightarrow \quad \left(\frac{r}{R}\right)^2 = \frac{1}{2} \quad \Rightarrow \quad r = \frac{R}{\sqrt{2}} = 0.707R$$

Result.

$$Q = \int_0^R u\, 2\pi r\, dr = -\frac{\pi}{8\mu}\frac{\partial p}{\partial x} R^4, \qquad V = \frac{Q}{\pi R^2} = -\frac{1}{8\mu}\frac{\partial p}{\partial x} R^2 = \frac{u_{\max}}{2}$$

What it tells us. The discharge grows as R^4: doubling the radius passes sixteen times the flow at the same pressure gradient. The mean velocity is exactly half the centre-line velocity in laminar flow.

The local velocity equals the mean velocity at $r = R/\sqrt{2} = 0.707R$. A Pitot tube placed there reads the mean velocity directly.

1.6 Hagen–Poiseuille equation CORE

IN SIMPLE WORDS

The Hagen–Poiseuille equation gives the pressure drop needed to push a laminar flow through a pipe. The drop grows with the viscosity μ, the velocity V and the length L. It falls quickly as the diameter D grows: as D^2 for a given velocity, and as D^4 for a given discharge. That is why a syringe with a thin needle needs a hard push. In laminar flow the friction factor is simply $f = 64/\mathrm{Re}$.

Derivation from scratch: Hagen–Poiseuille equation and $f = 64/\mathrm{Re}$

Aim. Find the pressure drop $p_1 - p_2$ and head loss h_f over a length L of pipe in laminar flow, and show that $f = 64/\mathrm{Re}$.

Assumptions.

1. The mean velocity result of Section 1.5 holds: $V = -\dfrac{1}{8\mu}\dfrac{\partial p}{\partial x}R^2$.
2. The pipe is horizontal, of diameter D, and the flow is fully developed, so the pressure falls at a steady rate along it.
3. Section 1 is upstream and section 2 is a length L downstream.

Step 1. Write the pressure gradient over the length L. The pressure falls uniformly from p_1 to p_2 over L.

$$\frac{\partial p}{\partial x} = \frac{p_2 - p_1}{L} \quad \Rightarrow \quad -\frac{\partial p}{\partial x} = \frac{p_1 - p_2}{L}$$

Since $p_1 > p_2$, the quantity $-\partial p/\partial x$ is positive, as it must be for flow in the $+x$ direction.

Step 2. Substitute into the mean velocity.

$$V = \frac{1}{8\mu}\frac{p_1 - p_2}{L}R^2$$

Step 3. Change from radius to diameter. Put $R = D/2$, so $R^2 = D^2/4$.

$$V = \frac{1}{8\mu}\frac{p_1 - p_2}{L}\frac{D^2}{4} = \frac{\left(p_1 - p_2\right)D^2}{32\,\mu L}$$

Step 4. Make the pressure drop the subject. Multiply both sides by $32\mu L/D^2$.

$$p_1 - p_2 = \frac{32\,\mu V L}{D^2}$$

Step 5. Write it in terms of discharge. By continuity $V = \dfrac{Q}{\pi D^2/4} = \dfrac{4Q}{\pi D^2}$.

$$p_1 - p_2 = \frac{32\,\mu L}{D^2} \times \frac{4Q}{\pi D^2} = \frac{128\,\mu Q L}{\pi D^4}$$

Step 6. Change pressure drop to head loss. In a horizontal pipe of uniform diameter, $z_1 = z_2$ and $V_1 = V_2$. The energy equation of Section 0.2 then gives $h_f = \left(p_1 - p_2\right) / \left(\rho g\right)$.

$$h_f = \frac{p_1 - p_2}{\rho g} = \frac{32\,\mu V L}{\rho g D^2}$$

Step 7. Compare with the Darcy–Weisbach form. Section 3 writes every friction loss as $h_f = fLV^2 / \left(2gD\right)$. Set the two expressions equal.

$$\frac{fLV^2}{2gD} = \frac{32\,\mu V L}{\rho g D^2}$$

Step 8. Solve for f. Multiply both sides by $\dfrac{2gD}{LV^2}$. The factors g, L, one V and one D cancel.

$$f = \frac{32\,\mu V L}{\rho g D^2} \times \frac{2gD}{LV^2} = \frac{64\,\mu}{\rho V D}$$

Step 9. Recognise the Reynolds number. Since $\mathrm{Re} = \rho V D/\mu$, the fraction $\mu / \left(\rho V D\right)$ is $1/\mathrm{Re}$. With $f = 4f_1$, the coefficient of friction is $f_1 = f/4$.

$$f = \frac{64}{\mathrm{Re}}, \qquad f_1 = \frac{f}{4} = \frac{16}{\mathrm{Re}}$$

Result.

$$p_1 - p_2 = \frac{32\,\mu V L}{D^2} = \frac{128\,\mu Q L}{\pi D^4}, \qquad h_f = \frac{p_1 - p_2}{\rho g} = \frac{32\,\mu V L}{\rho g D^2}$$

$$f = \frac{64}{\mathrm{Re}} \qquad \left(f_1 = \frac{16}{\mathrm{Re}}\right)$$

What it tells us. In laminar flow the head loss is directly proportional to V, μ and L, and inversely proportional to D^2. At a fixed discharge it varies as $1/D^4$, so a slightly wider pipe needs far less pressure. The friction factor depends on the Reynolds number alone, not on the wall roughness.

The head loss is proportional to V and to μ, and inversely proportional to D^2. In laminar flow, the friction factor depends on Reynolds number alone.

The power needed to keep the flow going is $P = \left(p_1 - p_2\right) Q = \rho g Q h_f$.

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

1.7 Kinetic energy and momentum correction factors ADVANCED

The energy and momentum equations are written with the mean velocity V. Because the true profile is not flat, they need correction factors:

$$\alpha = \frac{1}{AV^3}\int_A u^3\,dA, \qquad \beta = \frac{1}{AV^2}\int_A u^2\,dA$$

For the laminar paraboloid, with $u = 2V\left(1 - r^2/R^2\right)$, the integrals give $\alpha = 2$ and $\beta = 4/3$. Turbulent profiles are much flatter, so $\alpha \approx 1.03$ to 1.1 and $\beta \approx 1.01$ to 1.04. Taking both as 1 is acceptable for turbulent flow, but not for laminar flow.

1.8 Application: capillary-tube viscometer CORE

IN SIMPLE WORDS

A capillary viscometer measures viscosity by timing a slow flow through a very thin tube. We know D, L, the pressure difference and the volume collected in a measured time. The Hagen–Poiseuille equation is then turned round to give μ. It is like judging how thick a syrup is by how long it takes to drain through a straw.

The Hagen–Poiseuille equation is the basis of the capillary-tube viscometer. Liquid flows through a fine tube of known D and L under a measured pressure difference, and the discharge is timed. Then:

$$\mu = \frac{\pi D^4\left(p_1 - p_2\right)}{128\,Q\,L}$$

The tube must be long enough for the entrance region to be negligible, and Re must stay well below 2000.

Worked examples

Step-by-step method.

1. Convert all data to SI units: D in m, Q in $\mathrm{m^3/s}$, μ in $\mathrm{Pa{\cdot}s}$.
2. Find the mean velocity from $V = Q/A$, with $A = \pi D^2/4$.
3. Check the regime: $\mathrm{Re} = \rho V D/\mu$ must be below 2000 for these formulas.
4. Find the pressure drop from $p_1 - p_2 = 32\mu VL/D^2$.
5. Change it to head loss, $h_f = \left(p_1 - p_2\right)/(\,\rho g\,)$, in metres of the flowing fluid.
6. If asked, find $u_{\max} = 2V$, $\tau_0 = \left(p_1 - p_2\right)D/(\,4L\,)$ and the power $\left(p_1 - p_2\right)Q$.
7. For an inclined pipe, use the drop in $p + \rho g z$, not in p alone.

Example 1.0 (warm-up) CORE Oil ($\rho = 900\ \text{kg/m}^3$, $\mu = 0.09\ \text{N·s/m}^2$) flows at a mean velocity of 1 m/s through a 50 mm pipe, 10 m long. Check that the flow is laminar, and find the pressure drop and the head loss.

1. $\text{Re} = \dfrac{\rho V D}{\mu} = \dfrac{900 \times 1 \times 0.05}{0.09} = 500$, which is below 2000, so the flow is laminar
2. $p_1 - p_2 = \dfrac{32 \mu V L}{D^2} = \dfrac{32 \times 0.09 \times 1 \times 10}{0.05^2} = 11.52\ \text{kPa}$
3. $h_f = \dfrac{p_1 - p_2}{\rho g} = \dfrac{11\,520}{900 \times 9.81} = 1.30\ \text{m of oil}$

Example 1.1 CORE Oil ($\rho = 900\ \text{kg/m}^3$, $\mu = 0.12\ \text{N·s/m}^2$) flows through an 80 mm pipe at 4 L/s. For a 50 m length, find the Reynolds number, the pressure drop, the head loss, the maximum velocity, the wall shear stress and the pumping power. Also check the entrance length.

1. $A = \dfrac{\pi}{4} \times 0.08^2 = 5.027 \times 10^{-3}\ \text{m}^2$; $V = \dfrac{0.004}{5.027 \times 10^{-3}} = 0.796\ \text{m/s}$
2. $\text{Re} = \dfrac{900 \times 0.796 \times 0.08}{0.12} = 477$, so the flow is laminar
3. $p_1 - p_2 = \dfrac{32 \times 0.12 \times 0.796 \times 50}{0.08^2} = 23.87\ \text{kPa}$
4. $h_f = \dfrac{23\,873}{900 \times 9.81} = 2.70\ \text{m of oil}$. Check with $f = 64/477 = 0.134$: $h_f = 0.134 \times \dfrac{50}{0.08} \times \dfrac{0.796^2}{19.62} = 2.70\ \text{m}$
5. $u_{\max} = 2V = 1.59\ \text{m/s}$
6. $\tau_0 = \dfrac{(p_1 - p_2)\, D}{4L} = \dfrac{23\,873 \times 0.08}{200} = 9.55\ \text{N/m}^2$
7. $\text{Power} = (p_1 - p_2)\, Q = 23\,873 \times 0.004 = 95.5\ \text{W}$
8. $L_e = 0.058 \times 477 \times 0.08 = 2.2\ \text{m}$, so the flow is fully developed over most of the 50 m length

Example 1.2 CORE In a capillary viscometer the tube is 1.5 mm in diameter and 1.0 m long. A liquid of density 850 kg/m³ flows through it under a pressure difference of 30 kPa, and 50 cm³ is collected in 120 s. Find the viscosity.

1. $Q = \dfrac{50 \times 10^{-6}}{120} = 4.167 \times 10^{-7}\ \text{m}^3/\text{s}$
2. $\mu = \dfrac{\pi \times (1.5 \times 10^{-3})^4 \times 30\,000}{128 \times 4.167 \times 10^{-7} \times 1.0} = 8.95 \times 10^{-3}\ \text{N·s/m}^2$ (about 9 centipoise)
3. Check: $V = Q/A = 0.236\ \text{m/s}$, so $\text{Re} = \dfrac{850 \times 0.236 \times 0.0015}{0.00895} = 34$. The flow is laminar, and the formula applies.

Example 1.3 HOTS Heavy oil ($\rho = 880\ \text{kg/m}^3$, $\mu = 0.25\ \text{N·s/m}^2$) flows up a 60 mm pipe inclined at 30°. Two gauges 20 m apart along the pipe read 400 kPa (lower) and 250 kPa (upper). Find the discharge.

1. Rise between the gauges: $z_2 - z_1 = 20 \sin 30^\circ = 10\ \text{m}$

2. Drop in piezometric pressure:
$\left(p_1 - p_2\right) - \rho g\left(z_2 - z_1\right) = 150\,000 - 880 \times 9.81 \times 10 = 63\,672\ \text{Pa}$
3. $V = \dfrac{63\,672 \times 0.06^2}{32 \times 0.25 \times 20} = 1.433\ \text{m/s}$, so $Q = 1.433 \times \dfrac{\pi}{4} \times 0.06^2 = 4.05\ \text{L/s}$, flowing upward
4. Check: $\text{Re} = \dfrac{880 \times 1.433 \times 0.06}{0.25} = 303$, laminar

In Example 1.3, 86 kPa of the 150 kPa difference only lifts the oil through 10 m. Using $p_1 - p_2$ alone would overestimate the discharge 2.4 times.

Common mistakes and exam tips.

- Using R where the formula has D. The form $32\mu VL/D^2$ uses the diameter; the $u_{\max}$ formula in 1.4 uses the radius.
- Forgetting to check $\text{Re} < 2000$. The Hagen–Poiseuille equation holds only for laminar flow.
- Mixing the mean and the maximum velocity. Use V in Re and in Hagen–Poiseuille; $u_{\max} = 2V$.
- Dividing by the density of water when the fluid is oil. h_f is in metres of the flowing fluid.
- In an inclined pipe, taking the gauge difference as the driving pressure. First subtract $\rho g\left(z_2 - z_1\right)$, as in Example 1.3.

Check your understanding.

1. CORE In laminar flow, what happens to h_f if the velocity doubles? *(It doubles, since $h_f \propto V$.)*
2. CORE Where is the shear stress zero in laminar pipe flow, and where is it largest? *(Zero at the axis, largest at the wall.)*
3. HOTS In laminar flow the diameter is doubled at the same discharge. How does the pressure drop change? *(It falls 16 times, since $p_1 - p_2 \propto Q/D^4$.)*

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

2. Laminar flow between parallel plates (L1) ADVANCED

The narrow gaps in bearings, piston clearances, valve seats and hydraulic seals behave like two parallel plates. With both plates fixed, the velocity profile is a parabola. The mean velocity is two-thirds of the maximum, and the pressure drop is $12\mu VL/t^2$. When one plate moves, a linear shear profile is added to it (Couette flow).

2.1 Both plates fixed ADVANCED

Let the plates be a distance t apart, with y measured from the lower plate. Take a fluid element of length Δx, height Δy and unit width. A force balance in the flow direction gives $\partial\tau/\partial y = \partial p/\partial x$. With $\tau = \mu\, du/dy$, integrating twice gives

$u = (1/2\mu)(\partial p/\partial x)\, y^2 + C_1 y + C_2$. The no-slip conditions $u = 0$ at $y = 0$ and at $y = t$ fix the constants:

$$u = \frac{1}{2\mu}\left(-\frac{\partial p}{\partial x}\right)(t\,y - y^2), \qquad \tau = \left(-\frac{\partial p}{\partial x}\right)\left(\frac{t}{2} - y\right)$$

The velocity is greatest at mid-gap ($y = t/2$), and the shear stress is zero there and largest at the plates:

$$u_{\max} = \frac{1}{8\mu}\left(-\frac{\partial p}{\partial x}\right)t^2, \qquad V = \frac{1}{t}\int_0^t u\,dy = \frac{1}{12\mu}\left(-\frac{\partial p}{\partial x}\right)t^2 = \frac{2}{3}u_{\max}$$

Over a length L, $-\partial p/\partial x = (p_1 - p_2)/L$, so:

$$p_1 - p_2 = \frac{12\,\mu V L}{t^2}, \qquad h_f = \frac{12\,\mu V L}{\rho g t^2}, \qquad q = Vt = \frac{(p_1 - p_2)\,t^3}{12\,\mu L} \quad (\text{per unit width})$$

The leakage q through a gap rises with the **cube** of the clearance. Doubling the clearance of a hydraulic piston raises the leakage eight times, which is why such clearances are held to a few micrometres.

2.2 One plate moving: Couette flow ADVANCED

If the upper plate moves at velocity U and the lower plate is fixed, the boundary conditions become $u = 0$ at $y = 0$ and $u = U$ at $y = t$:

$$u = \frac{U\,y}{t} + \frac{1}{2\mu}\left(-\frac{\partial p}{\partial x}\right)(t\,y - y^2), \qquad q = \frac{U\,t}{2} + \frac{t^3}{12\mu}\left(-\frac{\partial p}{\partial x}\right)$$

Case	Velocity profile	Typical example
No pressure gradient	Linear, $u = Uy/t$; $\tau = \mu U/t$ everywhere	Lightly loaded journal bearing; the viscometer of Newton's law
Favourable gradient ($\partial p/\partial x < 0$)	Linear plus parabola, bulging forward	Flow helped along by pressure
Adverse gradient ($\partial p/\partial x > 0$)	Backflow near the fixed plate if the gradient is large	Loaded bearing wedge; seal under back-pressure

The net flow is zero when $\partial p/\partial x = 6\mu U/t^2$. Beyond that adverse gradient, fluid near the fixed plate flows backward.

Worked examples

Example 2.1 ADVANCED Oil ($\rho = 900\ \mathrm{kg/m^3}$, $\mu = 0.2\ \mathrm{N{\cdot}s/m^2}$) flows between two fixed parallel plates 10 mm apart. The pressure falls by 2 kPa per metre of length. Find the maximum and mean velocities, the discharge per metre width and the wall shear stress.

1. $u_{\max} = \dfrac{2000 \times 0.01^2}{8 \times 0.2} = 0.125\ \mathrm{m/s}$
2. $V = \dfrac{2}{3} \times 0.125 = 0.0833\ \mathrm{m/s}$
3. $q = Vt = 0.0833 \times 0.01 = 8.33 \times 10^{-4}\ \mathrm{m^3/s}$ per metre width (0.83 L/s)
4. $\tau_0 = \left(-\dfrac{\partial p}{\partial x}\right)\dfrac{t}{2} = 2000 \times 0.005 = 10\ \mathrm{N/m^2}$
5. Check: $\mathrm{Re} = \rho V t/\mu = \dfrac{900 \times 0.0833 \times 0.01}{0.2} = 3.75$, laminar

Example 2.2 ADVANCED A plate of area 0.5 m² slides at 1.5 m/s over a fixed surface, separated by a 2 mm film of oil ($\mu = 0.3\ \mathrm{N{\cdot}s/m^2}$). (a) With no pressure gradient, find the drag force and the power lost. (b) Find the pressure gradient at which the net flow in the gap becomes zero.

1. (a) $\tau = \mu U/t = \dfrac{0.3 \times 1.5}{0.002} = 225\ \mathrm{N/m^2}$; $F = 225 \times 0.5 = 112.5\ \mathrm{N}$; power $= FU = 169\ \mathrm{W}$
2. (b) Net flow zero: $\dfrac{Ut}{2} = \dfrac{t^3}{12\mu}\dfrac{\partial p}{\partial x}$, so $\dfrac{\partial p}{\partial x} = \dfrac{6\mu U}{t^2} = \dfrac{6 \times 0.3 \times 1.5}{0.002^2} = 675\ \mathrm{kPa/m}$ (pressure rising in the direction of the plate's motion)

   Example 2.2(b) is the principle of the hydrodynamic bearing. The moving surface drags oil into a narrowing gap, and the pressure that builds up there carries the load.

Common mistakes and exam tips (Section 2).

- Using a pipe radius or diameter in the plate formulas. Plate formulas use the gap t.
- Taking the mean velocity as half the maximum. Between fixed plates $V = \frac{2}{3}u_{\max}$; the half is for a pipe.
- Forgetting that q is per unit width ($\mathrm{m^2/s}$). Multiply by the width to get $\mathrm{m^3/s}$.
- Using the pipe Re when a plate gap is meant. For plates use $\rho V t/\mu$, as in Example 2.1.
- In Couette flow, putting $u = 0$ at both plates. The moving plate has $u = U$.

Check your understanding.

1. CORE Between two fixed plates, what is the ratio of mean to maximum velocity? *(Two-thirds, compared with one-half in a pipe.)*

2. HOTS The clearance of a seal is doubled at the same pressure difference. How does the leakage change? *(It rises 8 times, since $q \propto t^3$.)*
3. CORE With no pressure gradient and one plate moving, what shape is the velocity profile? *(A straight line, $u = Uy/t$.)*

3. Resistance to flow: Darcy–Weisbach equation and turbulent pipe flow (L2) CORE

In turbulent flow, the head lost to friction in a straight pipe grows with length and roughly with the square of the velocity. The Darcy–Weisbach equation expresses this loss, for laminar and turbulent flow alike, through one coefficient: the friction factor f. In turbulent flow, f depends on the Reynolds number and on the relative roughness k/D, and is read from the Moody chart.

> **IN SIMPLE WORDS**
>
> In most real pipes, water flows turbulently. Friction at the wall eats up head all along the pipe, which is why a long garden hose gives a weak jet. The Darcy–Weisbach equation $h_f = fLV^2/(2gD)$ tells us how much head is lost. Here L/D is the length of the pipe counted in diameters, $V^2/(2g)$ is the velocity head, and f is a pure number, often about 0.02. A longer, thinner pipe or a faster flow means more loss.

3.1 Laws of fluid friction CORE

> **IN SIMPLE WORDS**
>
> The laws of fluid friction compare friction in laminar and in turbulent flow. In laminar flow, friction depends on viscosity and grows in step with the velocity. In turbulent flow, friction grows roughly as the square of the velocity and depends strongly on wall roughness. Like the push you feel on your hand out of a moving bus window, turbulent resistance rises sharply with speed.

Property	Laminar flow	Turbulent flow
Dependence on velocity	Friction $\propto V$	Friction $\propto V^n$, $n \approx 1.75$ to 2
Surface roughness	No effect	Strong effect at high Re
Temperature	Strong effect, through viscosity	Slight effect
Fluid density	No effect	Friction $\propto \rho$
Pressure	No effect	No effect
Wetted area	Friction $\propto$ area	Friction $\propto$ area

3.2 Derivation of the Darcy–Weisbach equation CORE

IN SIMPLE WORDS

We derive Darcy–Weisbach by balancing two forces on the water in a straight pipe. The pressure difference pushes the water forward. Wall friction over the wetted area holds it back. The velocity does not change, so the two forces balance. Watch the coefficient: the Darcy f is about 0.02, while some books use $f_1 = f/4$, about 0.005.

Derivation from scratch: Darcy–Weisbach equation

Aim. Find the head lost to friction, h_f, in a straight pipe of length L and diameter D, first in terms of the velocity V and then of the discharge Q.

Assumptions.

1. The flow is steady and the fluid is incompressible.
2. The pipe is horizontal and of uniform diameter D, so $z_1 = z_2$ and $V_1 = V_2 = V$.
3. Following Froude's experiments, the frictional resistance per unit wetted area is $f'V^2$, where f' is the frictional resistance per unit area at unit velocity.

Step 1. Choose the element. Take the whole column of fluid between section 1 and section 2, a length L apart. Its end faces have area $A = \pi D^2/4$. Its wetted surface (the area in contact with the pipe wall) is the circumference times the length, πDL.

Step 2. Pressure force on the upstream face. The pressure p_1 pushes the column forward.

$$F_1 = + p_1 \frac{\pi D^2}{4}$$

Step 3. Pressure force on the downstream face. The pressure p_2 pushes the column backward.

$$F_2 = - p_2 \frac{\pi D^2}{4}$$

Step 4. Friction force at the wall. Resistance per unit area times the wetted area. It opposes the flow.

$$F_{\text{friction}} = - f'V^2 \ (\pi DL)$$

The pipe is horizontal, so the weight has no component along the axis.

Step 5. Force balance. The velocity is the same at both sections, so the momentum of the column does not change. The net force on it is zero.

$$p_1 \frac{\pi D^2}{4} - p_2 \frac{\pi D^2}{4} - f'V^2 \ (\pi DL) = 0 \quad \Rightarrow \quad \left(p_1 - p_2 \right) \frac{\pi D^2}{4} = f'V^2 \ (\pi DL)$$

Step 6. Find the pressure drop. Divide both sides by $\pi D^2/4$. Note that $\dfrac{\pi DL}{\pi D^2/4} = \dfrac{4L}{D}$.

$$p_1 - p_2 = \frac{4 f' L V^2}{D}$$

Step 7. Change to head loss. With $z_1 = z_2$ and $V_1 = V_2$, the energy equation of Section 0.2 gives $h_f = \left(p_1 - p_2 \right) / \left(\rho g \right)$.

$$h_f = \frac{p_1 - p_2}{\rho g} = \frac{4 f' L V^2}{\rho g D}$$

Step 8. Bring in the dimensionless coefficient. Put $f'/\rho = f_1/2$, where f_1 is the dimensionless coefficient of friction (this gives Bansal's form).

$$h_f = 4 \times \frac{f'}{\rho} \times \frac{L V^2}{g D} = 4 \times \frac{f_1}{2} \times \frac{L V^2}{g D} = \frac{4 f_1 L V^2}{2 g D}$$

Step 9. Change to the Darcy friction factor. Write $f = 4 f_1$.

$$h_f = \frac{4 f_1 L V^2}{2 g D} = \frac{f L V^2}{2 g D}$$

Step 10. Write V in terms of Q. By continuity $V = \dfrac{4Q}{\pi D^2}$, so $V^2 = \dfrac{16\,Q^2}{\pi^2 D^4}$.

$$h_f = \frac{f L}{2 g D} \times \frac{16\,Q^2}{\pi^2 D^4} = \frac{16 f L Q^2}{2 \pi^2 g D^5} = \frac{8 f L Q^2}{\pi^2 g D^5}$$

Step 11. Put in the numbers for $g = 9.81\ \mathrm{m/s^2}$. The constant is $\pi^2 g/8 = \pi^2 \times 9.81/8 = 12.1$.

$$h_f = \frac{f L Q^2}{\left(\pi^2 g/8 \right)\, D^5} \approx \frac{f L Q^2}{12.1\, D^5}$$

Result.

$$h_f = \frac{4 f_1 L V^2}{2 g D} = \frac{f L V^2}{2 g D}, \qquad f = 4 f_1$$

$$h_f = \frac{8 f L Q^2}{\pi^2 g D^5} \approx \frac{f L Q^2}{12.1\, D^5}$$

What it tells us. The friction loss is proportional to the length and to the velocity head $V^2/\left(2g \right)$. At a fixed velocity it falls as $1/D$, but at a fixed discharge it falls as $1/D^5$. This equation holds for laminar and turbulent flow; only the value of f changes.

Check which coefficient a problem uses: $f \approx 0.02$ is the Darcy f, and $f \approx 0.005$ is f_1.

The D^5 term means that doubling the diameter cuts the friction loss about 32 times at the same discharge.

Derivation from scratch: wall shear stress $\tau_0 = f\rho V^2/8$ and shear velocity

Aim. Express the wall shear stress τ_0 and the shear velocity u_* in terms of f and the mean velocity V.

Assumptions.

1. The same column of fluid as above: horizontal pipe, diameter D, length L, steady flow, $V_1 = V_2$.
2. The friction at the wall is now written as a wall shear stress τ_0, the same all over the wetted surface.
3. The Darcy–Weisbach equation $h_f = fLV^2/\left(\,2gD\,\right)$ holds.

Step 1. Force balance with τ_0. The pressure force on the column balances the shear force $\tau_0 \times \pi DL$ at the wall.

$$\left(\,p_1 - p_2\,\right)\frac{\pi D^2}{4} = \tau_0\ \left(\,\pi DL\,\right)$$

Step 2. Make τ_0 the subject. Divide both sides by πDL. One π and one D cancel.

$$\tau_0 = \frac{\left(\,p_1 - p_2\,\right)\,D}{4L}$$

This agrees with Section 1.3, where $\tau_0 = -\dfrac{\partial p}{\partial x}\dfrac{R}{2} = \dfrac{p_1 - p_2}{L}\dfrac{D}{4}$.

Step 3. Put in the Darcy–Weisbach pressure drop. From $h_f = \left(\,p_1 - p_2\,\right)/\left(\,\rho g\,\right)$:

$$p_1 - p_2 = \rho g\, h_f = \rho g \frac{fLV^2}{2gD} = \frac{f\rho LV^2}{2D}$$

Step 4. Substitute and cancel. The factors L and D cancel.

$$\tau_0 = \frac{f\rho LV^2}{2D} \times \frac{D}{4L} = \frac{f}{8}\rho V^2$$

Step 5. Define the shear velocity. The quantity $\sqrt{\tau_0/\rho}$ has the units of velocity. Put in τ_0 from Step 4 and cancel ρ.

$$u_* = \sqrt{\frac{\tau_0}{\rho}} = \sqrt{\frac{f\rho V^2}{8\rho}} = \sqrt{\frac{fV^2}{8}} = V\sqrt{\frac{f}{8}}$$

Result.

$$\tau_0 = \frac{f}{8}\rho V^2, \qquad u_* = \sqrt{\frac{\tau_0}{\rho}} = V\sqrt{\frac{f}{8}}$$

What it tells us. For a given f, the wall shear stress grows as the square of the velocity and in proportion to the density. Measuring the pressure drop along a pipe therefore gives the wall shear stress directly.

The quantity u_* has units of velocity and is called the **shear (friction) velocity**. It is the velocity scale of turbulent flow near a wall.

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

3.3 Shear stress in turbulent flow ADVANCED

In turbulent flow the velocity at a point fluctuates: $u = \bar{u} + u'$, $v = v'$. Fluid lumps moving across the layers carry momentum with them. This adds an apparent stress, the **Reynolds stress**, to the viscous stress:

$$\tau = \mu \frac{d\bar{u}}{dy} - \rho\, \bar{u'v'} = (\mu + \eta)\, \frac{d\bar{u}}{dy}$$

Here η is the **eddy viscosity** (Boussinesq). Unlike μ, it is not a fluid property: it depends on the flow and on the distance from the wall. Away from the wall the turbulent part is hundreds of times the viscous part.

Prandtl's mixing length theory. Prandtl assumed that a lump of fluid travels a distance l (the mixing length) across the flow before it mixes with its new surroundings. Both u' and v' are then of order $l\,(d\bar{u}/dy)$, so:

$$\tau = \rho\, l^2 \left(\frac{d\bar{u}}{dy}\right)^2, \qquad l = \kappa y, \quad \kappa = 0.4$$

Near the wall, $\tau \approx \tau_0$. Integrating $d\bar{u}/dy = u_*/(\kappa y)$ gives a **logarithmic velocity profile**. This is the basis of all the turbulent pipe-flow formulas below.

3.4 Velocity distribution in turbulent pipe flow ADVANCED

With y measured from the wall, the integrated profile is $\bar{u} = (u_*/\kappa)\, \ln y + C$. Using $u = u_{\max}$ at $y = R$ gives the **velocity defect law**, which holds for smooth and rough pipes alike:

$$\frac{u_{\max} - \bar{u}}{u_*} = 5.75\, \log_{10} \frac{R}{y}, \qquad u_{\max} = V + 3.75\, u_*$$

The constant C depends on the wall:

$$\text{Smooth: } \frac{\bar{u}}{u_*} = 5.75\, \log_{10} \frac{u_* y}{\nu} + 5.5, \qquad \text{Rough: } \frac{\bar{u}}{u_*} = 5.75\, \log_{10} \frac{y}{k} + 8.5$$

A simpler empirical fit, good for Re up to about 10^5, is the **one-seventh power law**, $u/u_{\max} = (y/R)^{1/7}$, which gives $V = 0.817\, u_{\max}$. The turbulent profile is much flatter than the laminar paraboloid, because the strong mixing evens out the velocity across the core.

3.5 Laminar sublayer: hydraulically smooth and rough pipes CORE

IN SIMPLE WORDS

Even in turbulent flow, a very thin layer next to the wall stays laminar. This is the laminar sublayer, of thickness δ'. If the bumps on the wall, of height k, stay hidden inside it, the pipe behaves as smooth. If they poke through it, the pipe behaves as rough. Think of road bumps under a thick carpet: you cannot feel them until the carpet gets thin. A faster flow makes the sublayer thinner.

Right at the wall, the turbulent fluctuations die out and viscous shear dominates. This thin layer is the **laminar (viscous) sublayer**. Its thickness is:

$$\delta' = \frac{11.6\,\nu}{u_*}$$

Whether a pipe behaves as smooth or rough depends on how its roughness height k compares with δ', not on k alone:

k/δ'	**Behaviour**	**Friction factor depends on**
< 0.25	Hydraulically smooth: roughness buried in the sublayer	Re only
0.25 to 6	Transition	Re and k/D
> 6	Hydraulically rough: roughness pokes through the sublayer	k/D only

The same pipe can be smooth at low velocity and rough at high velocity, because δ' shrinks as u_* grows.

3.6 Friction factor formulas and the Moody chart CORE

IN SIMPLE WORDS

The friction factor f is the key number in Darcy–Weisbach. In laminar flow, $f = 64/\mathrm{Re}$. In turbulent flow, f depends on Re and on the relative roughness k/D. The Moody chart is a ready-made graph: find Re on the bottom axis, follow the curve for your k/D, and read f on the side. In most exam problems f is simply given, often as 0.02.

Regime	Friction factor f (Darcy)	Source
Laminar, $\mathrm{Re} < 2000$	$f = 64/\mathrm{Re}$	Hagen–Poiseuille (exact)
Smooth pipe, $4000 < \mathrm{Re} < 10^5$	$f = 0.316/\mathrm{Re}^{0.25}$	Blasius
Smooth pipe, any turbulent Re	$1/\sqrt{f} = 2.0 \log_{10}\left(\mathrm{Re}\sqrt{f}\right) - 0.8$	Prandtl–von Kármán
Fully rough pipe	$1/\sqrt{f} = 2.0 \log_{10}\left(D/k\right) + 1.14$	von Kármán–Nikuradse
All turbulent flow, commercial pipes	$1/\sqrt{f} = -2.0 \log_{10}\left[\left(k/D\right)/3.7 + 2.51/\left(\mathrm{Re}\sqrt{f}\right)\right]$	Colebrook–White
Explicit approximation to Colebrook	$1/\sqrt{f} = -1.8 \log_{10}\left[\left(\left(k/D\right)/3.7\right)^{1.11} + 6.9/\mathrm{Re}\right]$	Haaland (within about 2%)

The **Moody chart** plots the Colebrook equation: f against Re on log axes, with one curve for each value of k/D. It shows four zones: the laminar line, the critical zone (2000 to 4000), the transition zone where the curves bend, and the fully rough zone where they flatten out.

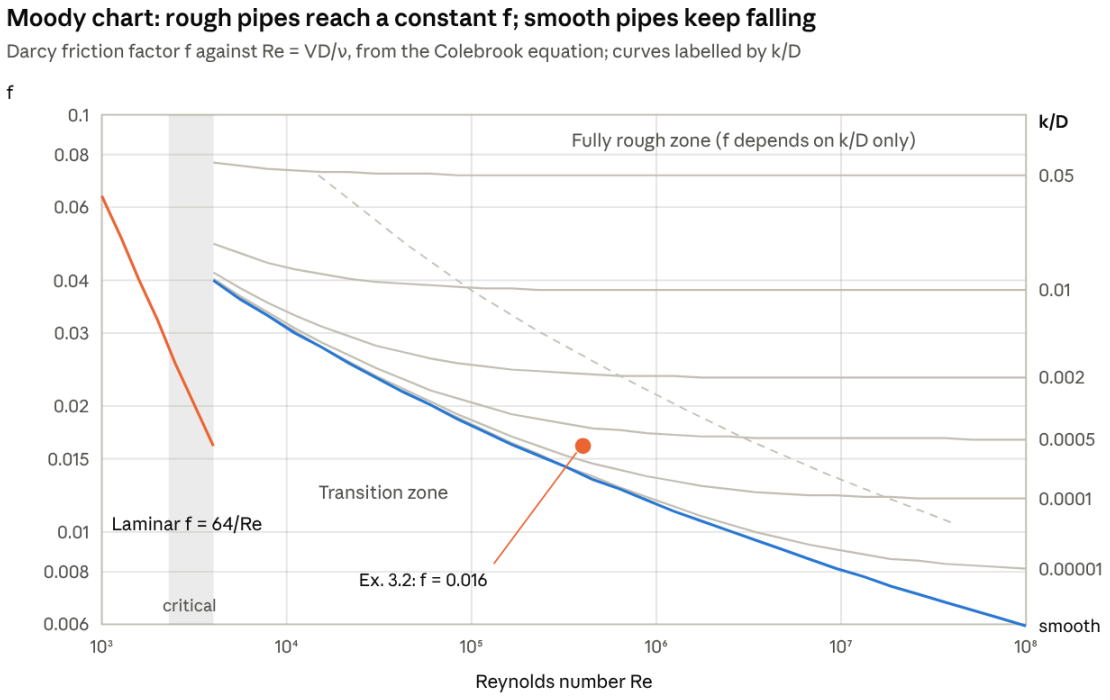


Figure 2. Moody chart drawn from the Colebrook equation

Right of the dashed line, each roughness curve is flat: f depends on k/D alone. The smooth-pipe curve keeps falling with Re.

Pipe material	Equivalent roughness k (mm)
Drawn tubing, glass, PVC	0.0015
Commercial steel, wrought iron	0.045
Galvanised iron	0.15
Cast iron	0.26
Concrete	0.3 to 3.0
Riveted steel	0.9 to 9.0

Worked examples

Step-by-step method.

1. Convert all data to SI units.
2. Find the velocity from $V = Q/A$ (or Q from $Q = AV$).
3. Find $\mathrm{Re} = VD/\nu$ to see whether the flow is laminar or turbulent.
4. Get f: use the given value (check whether it is f or f_1), or $64/\mathrm{Re}$ if laminar, or the Moody chart with k/D if turbulent.
5. Find the loss from $h_f = fLV^2/(2gD)$, or $h_f = 8fLQ^2/\left(\pi^2 g D^5\right)$ when Q is given.
6. If asked, find the power lost, $P = \rho g Q h_f$.
7. To find a diameter, rearrange step 5 for D^5, then choose the next standard size.

Example 3.0 (warm-up) CORE Water flows at 2 m/s through a 100 mm pipe, 100 m long. Taking $f = 0.02$, find the head lost to friction and the power lost.

1. $h_f = \dfrac{fLV^2}{2gD} = \dfrac{0.02 \times 100 \times 2^2}{2 \times 9.81 \times 0.1} = 4.08\ \mathrm{m}$
2. $Q = AV = \dfrac{\pi}{4} \times 0.1^2 \times 2 = 0.0157\ \mathrm{m^3/s}$
3. $P = \rho g Q h_f = 1000 \times 9.81 \times 0.0157 \times 4.08 = 628\ \mathrm{W}$

Example 3.1 CORE Water flows at 0.15 m³/s through a 300 mm pipe 1500 m long. Taking $f = 0.02$, find the head lost to friction and the power lost.

1. $V = \dfrac{0.15}{(\pi/4) \times 0.3^2} = 2.12\ \mathrm{m/s}$
2. $h_f = \dfrac{0.02 \times 1500 \times 2.12^2}{2 \times 9.81 \times 0.3} = 22.95\ \mathrm{m}$
3. Power lost $= \rho g Q h_f = 1000 \times 9.81 \times 0.15 \times 22.95 = 33.8\ \mathrm{kW}$

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

Example 3.2 ADVANCED Water ($\nu = 1.0 \times 10^{-6}\ \mathrm{m^2/s}$) flows at 2 m/s through a commercial steel pipe ($k = 0.045\ \mathrm{mm}$) of 200 mm diameter and 500 m length. Find the friction factor and the head loss.

1. $\mathrm{Re} = \dfrac{2 \times 0.2}{10^{-6}} = 4.0 \times 10^5$, turbulent; $k/D = 0.045/200 = 2.25 \times 10^{-4}$
2. Haaland: $\dfrac{1}{\sqrt{f}} = -1.8 \log_{10}\left[\left(\dfrac{2.25 \times 10^{-4}}{3.7}\right)^{1.11} + \dfrac{6.9}{4 \times 10^5}\right]$, giving $f = 0.0158$
3. Colebrook, iterated from that start: $f = 0.0160$ (the Moody chart gives about 0.016)
4. $h_f = 0.0160 \times \dfrac{500}{0.2} \times \dfrac{2^2}{19.62} = 8.16\ \mathrm{m}$
5. The Blasius smooth-pipe value would be $0.316/\left(4 \times 10^5\right)^{0.25} = 0.0126$, which underestimates the loss by about 21%.

Example 3.3 HOTS For the flow of Example 3.2, find the wall shear stress, the shear velocity, the laminar sublayer thickness and the centre-line velocity. Is the pipe hydraulically smooth or rough?

1. $\tau_0 = \dfrac{f}{8}\rho V^2 = \dfrac{0.016}{8} \times 1000 \times 2^2 = 8.00\ \mathrm{N/m^2}$
2. $u_* = \sqrt{\tau_0/\rho} = 0.0894\ \mathrm{m/s}$
3. $\delta' = \dfrac{11.6 \times 10^{-6}}{0.0894} = 0.130\ \mathrm{mm}$
4. $k/\delta' = 0.045/0.130 = 0.35$, which lies between 0.25 and 6: the pipe is in the **transition** zone, neither fully smooth nor fully rough
5. $u_{\max} = V + 3.75u_* = 2 + 3.75 \times 0.0894 = 2.34\ \mathrm{m/s}$, only 17% above the mean (compared with 100% in laminar flow)

Example 3.4 CORE Find the diameter of a pipe to carry 0.1 m³/s of water over 1000 m with a friction loss of 10 m. Take $f = 0.02$.

1. $D^5 = \dfrac{8fLQ^2}{\pi^2 g h_f} = \dfrac{8 \times 0.02 \times 1000 \times 0.1^2}{\pi^2 \times 9.81 \times 10} = 1.653 \times 10^{-3}$
2. $D = 0.278\ \mathrm{m}$, so choose the next standard size, 300 mm
3. Check: $V = 1.65\ \mathrm{m/s}$ and $\mathrm{Re} = 4.6 \times 10^5$ (taking $\nu = 10^{-6}\ \mathrm{m^2/s}$), so the flow is turbulent and $f = 0.02$ is reasonable

Common mistakes and exam tips.

- Mixing the Darcy f with f_1. If the given value is near 0.005, it is f_1: use $f = 4f_1$.
- Dropping the 2 in $2g$. The denominator is $2gD$, with $2g = 19.62\ \mathrm{m/s^2}$.
- Getting powers wrong in the Q form. $h_f \propto 1/D^5$, so a 1% error in D gives about a 5% error in h_f.
- Using D in mm in Re. The ratio k/D only needs both in the same unit, but Re needs metres.
- Calling a pipe smooth or rough from k alone. It depends on k/δ', which changes with velocity.

Check your understanding.

1. CORE In turbulent flow with a fixed f, what happens to h_f if the velocity doubles? *(It becomes four times as large, since $h_f \propto V^2$.)*
2. CORE A problem gives a coefficient of friction of 0.005. What is the Darcy f? *(It is $f = 4f_1 = 0.02$.)*
3. HOTS At the same discharge and f, the diameter is doubled. How does the friction loss change? *(It falls 32 times, since $h_f \propto 1/D^5$.)*

4. Minor losses; hydraulic gradient and total energy lines (L3)

CORE

Every change in the size, shape or direction of a pipe disturbs the flow and dissipates energy in eddies. These are the **minor losses**, written as $KV^2/(2g)$. In long pipelines ($L/D > 1000$) they are small beside friction. In short pipes, pump suction lines and building services they can exceed it.

IN SIMPLE WORDS

Minor losses are the extra head lost at fittings: bends, valves, entries, exits and changes of size. Each fitting stirs the water into eddies, and the eddies turn useful energy into heat. Think of a bus that loses speed at every speed breaker and sharp turn. Each loss is written as $KV^2/(2g)$: a coefficient K times the velocity head. The HGL and TEL are graphs that show how the head falls along the pipe.

4.1 Loss due to sudden enlargement (derivation) CORE

IN SIMPLE WORDS

When a pipe suddenly becomes wider, the fast stream from the small pipe cannot turn the corner. It shoots into the large pipe, and swirling eddies form in the corners. The loss is $(V_1 - V_2)^2/(2g)$, so it depends on the drop in velocity. A pipe emptying into a tank is the extreme case: the whole velocity head is lost.

Derivation from scratch: head loss at a sudden enlargement

Aim. Show that the head lost where a pipe of area A_1 enlarges suddenly to area A_2 is $h_e = (V_1 - V_2)^2/(2g)$.

Assumptions.

1. The flow is steady and the liquid is incompressible; the pipe is horizontal, so $z_1 = z_2$.
2. Velocities are uniform over each section: V_1 in the small pipe, V_2 in the large pipe.
3. The jet from the small pipe separates and eddies form in the corners. Experiments show that the pressure on the annular face of the step is p_1.
4. The length between the sections is short, so wall shear between them is neglected.

Step 1. Choose the control volume. Section 1 is at the step, across the full large-pipe area A_2 (the jet area A_1 plus the annular face $A_2 - A_1$). Section 2 is downstream, where the flow has filled the larger pipe again. Take x in the direction of flow.

Step 2. Forces on the control volume along x. On section 1 the pressure p_1 acts on the jet area A_1 and on the annular face $A_2 - A_1$, pushing forward. On section 2 the pressure p_2 acts on A_2, pushing backward.

$$\sum F_x = p_1 A_1 + p_1 (A_2 - A_1) - p_2 A_2 = p_1 A_2 - p_2 A_2$$

Step 3. Rate of change of momentum. Mass ρQ enters each second with velocity V_1 and leaves with velocity V_2.

$$\text{Momentum out} - \text{momentum in} = \rho Q V_2 - \rho Q V_1 = \rho Q (V_2 - V_1)$$

Step 4. Impulse–momentum equation. Net force equals the rate of change of momentum.

$$p_1 A_2 - p_2 A_2 = \rho Q (V_2 - V_1)$$

Step 5. Put $Q = A_2 V_2$ and cancel A_2.

$$(p_1 - p_2) A_2 = \rho A_2 V_2 (V_2 - V_1) \quad \Rightarrow \quad p_1 - p_2 = \rho V_2 (V_2 - V_1)$$

Step 6. Write the pressure rise as a head. Multiply by -1 and divide by ρg.

$$\frac{p_2 - p_1}{\rho g} = \frac{V_2 (V_1 - V_2)}{g}$$

Step 7. Energy equation with a loss h_e. Apply the energy equation of Section 0.2 between sections 1 and 2, with $z_1 = z_2$.

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + h_e$$

Step 8. Make h_e the subject.

$$h_e = \frac{V_1^2 - V_2^2}{2g} - \frac{p_2 - p_1}{\rho g}$$

Step 9. Eliminate the pressures. Put in the result of Step 6.

$$h_e = \frac{V_1^2 - V_2^2}{2g} - \frac{V_2\,(\,V_1 - V_2\,)}{g}$$

Step 10. Use a common denominator $2g$**.** Multiply the second term above and below by 2, then expand.

$$h_e = \frac{V_1^2 - V_2^2 - 2V_1V_2 + 2V_2^2}{2g} = \frac{V_1^2 - 2V_1V_2 + V_2^2}{2g} = \frac{(\,V_1 - V_2\,)^{\,2}}{2g}$$

Step 11. Write it with areas. By continuity $V_2 = V_1A_1/A_2$, so $V_1 - V_2 = V_1\,(\,1 - A_1/A_2\,)$.

$$h_e = \frac{V_1^2\,(\,1 - A_1/A_2\,)^{\,2}}{2g} = \left(1 - \frac{A_1}{A_2}\right)^2 \frac{V_1^2}{2g}$$

Result.

$$h_e = \frac{V_1^2 - V_2^2}{2g} - \frac{V_2\,(\,V_1 - V_2\,)}{g} = \frac{(\,V_1 - V_2\,)^{\,2}}{2g} = \left(1 - \frac{A_1}{A_2}\right)^2 \frac{V_1^2}{2g}$$

What it tells us. The loss depends on the velocity that is "destroyed", $V_1 - V_2$, and grows as its square. Even so, the pressure rises across the enlargement (Step 6), because some velocity head is recovered as pressure head.

This is the **Borda–Carnot** result. When a pipe discharges into a large tank, $A_2 \to \infty$ and $V_2 \to 0$, so the exit loss is the whole velocity head, $V_1^2/\,(\,2g\,)$.

4.2 Loss due to sudden contraction (derivation) CORE

IN SIMPLE WORDS

At a sudden contraction the water squeezes into a narrow neck, the vena contracta, just inside the smaller pipe. It then spreads out again to fill the pipe, and this spreading causes most of the loss. It is like a crowd squeezing through a narrow gate and then spreading out. The loss is about $0.5\,V_2^2/\,(\,2g\,)$, where V_2 is the velocity in the smaller pipe.

Derivation from scratch: head loss at a sudden contraction

Aim. Show that the head lost where a pipe contracts suddenly to area A_2 is $h_c = (\,1/C_c - 1\,)^{\,2}V_2^2/\,(\,2g\,)$.

Assumptions.

1. At a sudden contraction, the flow does not fill the smaller pipe at once. It converges to a **vena contracta** of area $A_c = C_cA_2$, then expands again to fill the pipe.
2. The loss in the converging part (accelerating flow) is small and is neglected.
3. Almost all the loss occurs in the expansion from the vena contracta to the full pipe, which is a sudden enlargement.

Step 1. Name the sections. Section 1 is in the large pipe, section c is at the vena contracta (area A_c, velocity V_c), and section 2 is downstream in the small pipe, where it runs full (area A_2, velocity V_2).

Step 2. Apply the sudden-enlargement result from c to 2. In Section 4.1 replace V_1 by V_c and keep V_2.

$$h_c = \frac{(V_c - V_2)^2}{2g}$$

Step 3. Find V_c from continuity. The same discharge passes the vena contracta and section 2.

$$A_c V_c = A_2 V_2 \quad \Rightarrow \quad C_c A_2 V_c = A_2 V_2 \quad \Rightarrow \quad V_c = \frac{V_2}{C_c}$$

Step 4. Substitute and take out V_2.

$$h_c = \frac{(V_2/C_c - V_2)^2}{2g} = \frac{V_2^2 (1/C_c - 1)^2}{2g} = \left(\frac{1}{C_c} - 1\right)^2 \frac{V_2^2}{2g}$$

Step 5. Write it as a loss coefficient. Call the bracket k.

$$k = \left(\frac{1}{C_c} - 1\right)^2$$

Result.

$$h_c = \frac{(V_c - V_2)^2}{2g} = \left(\frac{1}{C_c} - 1\right)^2 \frac{V_2^2}{2g} = k\frac{V_2^2}{2g}$$

What it tells us. The loss comes from the re-expansion after the vena contracta, not from the squeezing itself. A smaller C_c (a sharper corner) gives a narrower neck and a larger loss.

With the typical $C_c = 0.62$, $k = 0.376$. Where C_c is not given, $k = 0.5$ is used. The loss is based on the velocity in the **smaller** pipe.

4.3 Other minor losses CORE

IN SIMPLE WORDS

Every fitting has its own loss coefficient K, found by experiment. A large K means a large loss: a fully open globe valve ($K \approx 10$) wastes far more head than a fully open gate valve ($K \approx 0.2$). Look up K in the table and multiply it by the velocity head $V^2/(2g)$. The equivalent length $L_e = KD/f$ turns a fitting into an extra length of straight pipe.

Cause	Head loss	Notes
Sudden enlargement	$(V_1 - V_2)^2/(2g)$	Derived above
Sudden contraction	$0.5\,V_2^2/(2g)$	V_2 = velocity in the smaller pipe
Entrance, sharp-edged	$0.5\,V^2/(2g)$	Well-rounded entry: 0.04 to $0.1\,V^2/(2g)$
Exit into a tank	$V^2/(2g)$	All kinetic energy is lost
Obstruction in a pipe (area a in a pipe of area A)	$\left[A/(C_c(A-a)) - 1\right]^2 V^2/(2g)$	Orifice plates, partly open valves
Gradual enlargement (diffuser)	$k(V_1 - V_2)^2/(2g)$	k is smallest, about 0.13, at a cone angle near 7°
Bend or elbow	$kV^2/(2g)$	90° standard elbow $k \approx 0.75$ to 0.9; long-radius bend 0.3 to 0.5
Gate valve, fully open	0.15 to $0.2\,V^2/(2g)$	Half open: about 2 to 5
Globe valve, fully open	about $10\,V^2/(2g)$	
Tee, flow through the branch	about $1.8\,V^2/(2g)$	

Derivation from scratch: exit and entrance losses as special cases

Aim. Obtain the exit loss $V^2/(2g)$ and the sharp-edged entrance loss $0.5\,V^2/(2g)$ from the enlargement and contraction results, and the equivalent length $L_e = KD/f$.

Assumptions.

1. The results of Sections 4.1 and 4.2 hold.
2. A reservoir is so large that the velocity in it is practically zero.
3. V is the mean velocity in the pipe.

Step 1. Exit loss: a pipe discharging into a tank. This is a sudden enlargement from the pipe area A to the tank area $A_2 \to \infty$. By continuity $V_2 = VA/A_2 \to 0$.

$$h_{\text{exit}} = \frac{(V - V_2)^2}{2g} \;\to\; \frac{(V - 0)^2}{2g} = \frac{V^2}{2g}$$

Step 2. Entrance loss: a tank feeding a pipe through a sharp edge. This is a sudden contraction from a very large area into the pipe. A vena contracta forms just inside the pipe, so the result of Section 4.2 applies with $V_2 = V$.

$$h_{\text{entry}} = \left(\frac{1}{C_c} - 1\right)^2 \frac{V^2}{2g}$$

Step 3. Put in a value of C_c. With $C_c \approx 0.62$ the bracket is about 0.376. Tests on sharp-edged entrances, which also include the small loss in the converging part, give a coefficient close to 0.5. This is the value used in practice.

$$h_{\text{entry}} = 0.5\,\frac{V^2}{2g}$$

Step 4. Equivalent length of a fitting. Set the minor loss equal to the friction loss in a length L_e of the same pipe, and cancel $V^2/(2g)$.

$$K\frac{V^2}{2g} = \frac{fL_e}{D}\frac{V^2}{2g} \quad \Rightarrow \quad K = \frac{fL_e}{D} \quad \Rightarrow \quad L_e = \frac{KD}{f}$$

Result.

$$h_{\text{exit}} = \frac{V^2}{2g}, \qquad h_{\text{entry}} = 0.5\,\frac{V^2}{2g}, \qquad L_e = \frac{KD}{f}$$

What it tells us. At an exit all the kinetic energy of the stream is lost in the tank. At a sharp entrance about half a velocity head is lost; rounding the entrance removes most of this loss.

Equivalent length. A minor loss $KV^2/(2g)$ can be expressed as the length of straight pipe that gives the same friction loss: $L_e = KD/f$. A globe valve with $K = 10$ in a pipe with $f = 0.02$ is worth 500 diameters of straight pipe.

4.4 Hydraulic gradient line and total energy line CORE

IN SIMPLE WORDS

The TEL shows the total head at each point of a pipe. The HGL shows the level that water would reach in small open tubes fixed along the pipe. The gap between them is the velocity head, $V^2/(2g)$. Think of the TEL as a slope that only goes downhill in the direction of flow, unless a pump lifts it. Where the HGL falls below the pipe, the pressure there is below atmospheric.

The **total energy line (TEL)** plots the total head, $p/(\rho g) + V^2/(2g) + z$, along the pipe. The **hydraulic gradient line (HGL)** plots the piezometric head, $p/(\rho g) + z$: the height to which water would rise in a piezometer tube. The HGL lies one velocity head, $V^2/(2g)$, below the TEL.

- The TEL always falls in the direction of flow, unless a pump adds energy. Its slope is h_f/L along a uniform pipe, and it drops suddenly at every minor loss.
- The HGL is parallel to the TEL in a uniform pipe. It steps up where the pipe enlarges (pressure is recovered) and down where it contracts.
- At a reservoir surface both lines start at the water level. At an exit into a reservoir the HGL ends at the water surface, and the TEL ends one velocity head above it before dropping by the exit loss.
- Where the HGL drops below the pipe centre line, the pressure in the pipe is below atmospheric. This is the condition in a siphon (Section 7).

Worked examples

Step-by-step method.

1. Convert all data to SI units and find each velocity from $V = Q/A$.
2. List every loss along the path: entry, friction in each pipe, enlargements and contractions, valves and bends, and exit.
3. Write each loss as $KV^2/(2g)$ with the correct velocity: the smaller pipe for a contraction, and $V_1 - V_2$ for an enlargement.
4. Write the energy equation between two convenient points, often two free surfaces where $p = 0$ (gauge) and $V \approx 0$.
5. Add up the losses and solve for the unknown (H, V or Q).
6. Check that the answer is sensible: in a long pipe, minor losses are small beside friction.

Example 4.0 (warm-up) CORE A pipe enlarges suddenly from 100 mm to 200 mm diameter. The velocity in the small pipe is 4 m/s. Find the head lost at the enlargement.

1. $V_2 = V_1 \left(\dfrac{D_1}{D_2}\right)^2 = 4 \times \left(\dfrac{0.1}{0.2}\right)^2 = 1 \text{ m/s}$
2. $h_e = \dfrac{(V_1 - V_2)^2}{2g} = \dfrac{(4-1)^2}{2 \times 9.81} = 0.459 \text{ m}$

Example 4.1 CORE A horizontal pipe enlarges suddenly from 150 mm to 300 mm diameter. It carries 0.05 m³/s of water. Find the head loss and the rise in pressure across the enlargement.

1. $V_1 = \dfrac{0.05}{(\pi/4) \times 0.15^2} = 2.83 \text{ m/s}$; $V_2 = 0.71 \text{ m/s}$
2. $h_e = \dfrac{(2.83 - 0.71)^2}{19.62} = 0.230 \text{ m}$
3. Pressure rise: $\dfrac{p_2 - p_1}{\rho g} = \dfrac{V_1^2 - V_2^2}{2g} - h_e = 0.383 - 0.230 = 0.153 \text{ m}$, so $p_2 - p_1 = 1.50 \text{ kPa}$
4. Without the loss, the pressure would rise by 0.383 m (3.76 kPa). The eddies waste 60% of the possible recovery.

Example 4.2 CORE Two reservoirs are joined by a pipe that is 200 mm in diameter for its first 300 m and then enlarges suddenly to 300 mm for the remaining 200 m. The entrance is sharp-edged, and $f = 0.02$ throughout. Find the difference in water levels needed for a flow of 0.06 m³/s.

$V_1 = 0.06/\left[(\pi/4) \times 0.2^2\right] = 1.91 \text{ m/s}$; $V_2 = 0.85 \text{ m/s}$. The difference in levels equals the sum of all losses:

Loss	Expression	Head (m)
Entrance	$0.5\,V_1^2/\,(\,2g\,)$	0.093
Friction, 200 mm pipe	$f\,(\,L_1/D_1\,)\,V_1^2/\,(\,2g\,)$	5.577
Sudden enlargement	$(\,V_1 - V_2\,)^{\,2}/\,(\,2g\,)$	0.057
Friction, 300 mm pipe	$f\,(\,L_2/D_2\,)\,V_2^2/\,(\,2g\,)$	0.490
Exit	$V_2^2/\,(\,2g\,)$	0.037
Total $= H$		6.25

The minor losses total 0.19 m, only 3% of the whole. In a pipeline 500 m long, friction dominates.

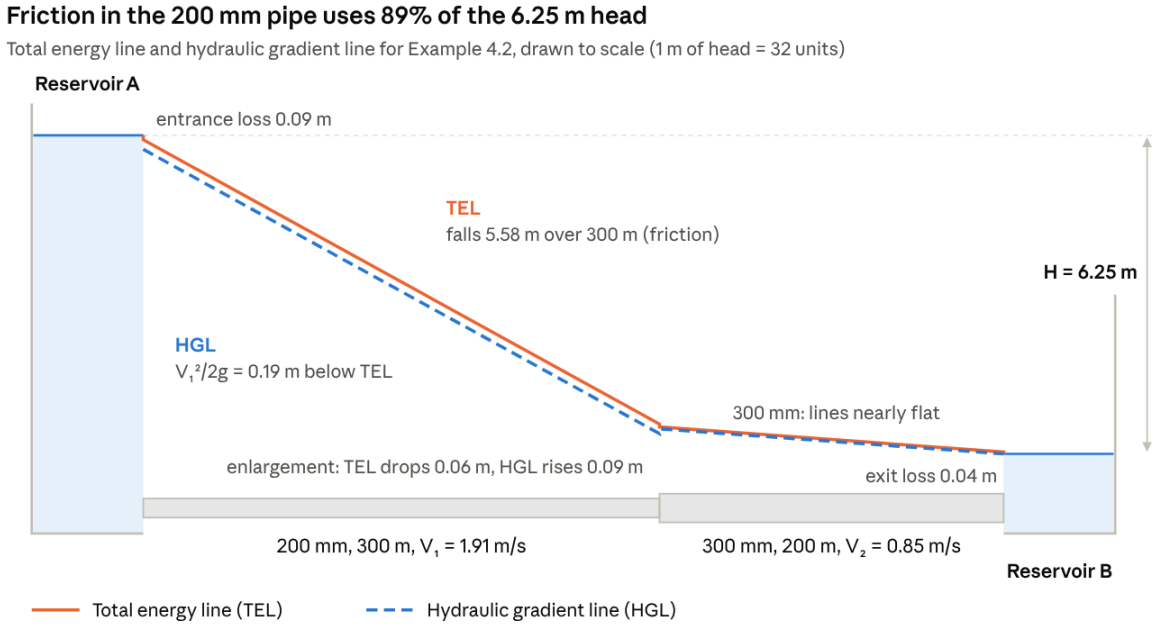


Figure 3. Total energy line and hydraulic gradient line for Example 4.2

The two lines run one velocity head apart. At the enlargement the TEL drops by the loss, while the HGL steps up as velocity head turns back into pressure.

Example 4.3 HOTS A 100 mm pipe, 100 m long ($f = 0.025$), draws water from a tank and discharges to the atmosphere 10 m below the tank's water level. It has a sharp entrance and a partly closed valve with $K = 5$. Find the discharge.

1. Energy from the tank surface to the jet: $10 = \,(\,0.5 + fL/D + 5 + 1\,)\,\dfrac{V^2}{2g}$, where the 1 is the velocity head of the free jet
2. $fL/D = 0.025 \times 100/0.1 = 25$, so the sum of coefficients is 31.5
3. $V = \sqrt{\dfrac{2 \times 9.81 \times 10}{31.5}} = 2.50\ \text{m/s}$; $Q = 2.50 \times \dfrac{\pi}{4} \times 0.1^2 = \ 19.6\ \text{L/s}$
4. The valve's equivalent length is $KD/f = 5 \times 0.1/0.025 = 20\ \text{m}$ of pipe.

Common mistakes and exam tips.

- Using the larger pipe's velocity for a contraction loss. Use the velocity in the smaller pipe.
- Writing the enlargement loss as $\left(V_1^2 - V_2^2\right)/(2g)$. It is $(V_1 - V_2)^2/(2g)$.
- Forgetting the exit loss into a tank, or counting the jet's velocity head at a free outlet twice. Count it once.
- Giving a loss a negative sign. Losses are always positive and go on the downstream side of the energy equation.
- Drawing the HGL above the TEL. The HGL is always $V^2/(2g)$ below the TEL.

Check your understanding.

1. CORE Water leaves a pipe into a tank at 3 m/s. What is the exit loss? *(It is $V^2/(2g) = 9/19.62 = 0.459$ m.)*
2. CORE Which velocity is used in the loss at a sudden contraction? *(The velocity in the smaller pipe.)*
3. HOTS At a sudden enlargement the TEL drops, but the HGL rises. Why? *(Velocity head turns back into pressure head, and this gain is larger than the loss.)*

5. Pipes in series, parallel and branched systems (L4) CORE

Pipe systems are solved with two rules, the same ones used for electrical circuits. Continuity: the flow into any junction equals the flow out. Energy: the head loss between two points is the same along every path that joins them. In series the discharge is common and the losses add. In parallel the loss is common and the discharges add.

IN SIMPLE WORDS

Real water systems join many pipes together. In series, pipes are joined end to end, like garden hoses coupled one after another. The same water passes through all of them, and their losses add up. In parallel, the flow splits into branches and joins again, like traffic sharing two roads between the same two towns. Every branch has the same head loss, and the branch flows add up.

5.1 Pipes in series (compound pipe) CORE

IN SIMPLE WORDS

In pipes in series, the discharge Q is the same in every pipe. The total head H is shared out as losses in each pipe. The narrowest pipe takes the biggest share, because the loss grows as $1/D^5$. An equivalent pipe is a single pipe of one diameter that gives the same total loss at the same discharge.

Derivation from scratch: total head for pipes in series

Aim. Show that the difference H between two reservoir levels joined by pipes in series equals the sum of all the losses.

Assumptions.

1. Pipes of different diameters joined end to end carry the same discharge Q (continuity).
2. The flow is steady; both reservoirs are large and open to the air.
3. Pipe i has length L_i, diameter D_i, velocity V_i and Darcy friction factor (f_A for pipe 1, f_B for pipe 2, and so on; letters are used so they are not confused with $f_1 = f/4$). Here the diameters increase at each joint (sudden enlargements); a joint where the pipe narrows would use the contraction loss $0.5\,V^2/\left(2g\right)$ of the smaller pipe instead.

Step 1. Energy equation between the two free surfaces. At each surface the gauge pressure is zero and the velocity is practically zero. Let the upper level be z_A and the lower level z_B.

$$0 + 0 + z_A = 0 + 0 + z_B + h_L \quad \Rightarrow \quad H = z_A - z_B = h_L$$

Step 2. List the losses in the order the water meets them. The terms below are, from left to right: entrance to pipe 1, friction in pipe 1, enlargement from 1 to 2, friction in pipe 2, and so on, and finally the exit from pipe n.

$$h_L = \frac{0.5V_1^2}{2g} + \frac{f_A L_1 V_1^2}{2gD_1} + \frac{\left(V_1 - V_2\right)^2}{2g} + \frac{f_B L_2 V_2^2}{2gD_2} + \ldots + \frac{V_n^2}{2g}$$

Step 3. Long pipes: drop the minor losses. When L/D is large, the friction terms are much larger than the others. Write each friction term with the discharge, as in Section 3.2, using the same Q in every pipe.

$$H = \frac{8fL_1Q^2}{\pi^2 g D_1^5} + \frac{8fL_2Q^2}{\pi^2 g D_2^5} + \ldots = \sum \frac{8fL_iQ^2}{\pi^2 g D_i^5}$$

Result.

$$H = \frac{0.5V_1^2}{2g} + \frac{f_A L_1 V_1^2}{2gD_1} + \frac{\left(V_1 - V_2\right)^2}{2g} + \frac{f_B L_2 V_2^2}{2gD_2} + \ldots + \frac{V_n^2}{2g}$$

What it tells us. In series the discharge is common and the losses add, like resistors in series. Because each friction term varies as $1/D^5$, the narrowest pipe takes the largest share of the head.

In long pipes the minor losses are usually neglected, so $H = \sum 8fL_iQ^2/\left(\pi^2 g D_i^5\right)$.

Equivalent pipe (Dupuit's equation). A compound pipe can be replaced by a single pipe of uniform diameter D and the same total length $L = L_1 + L_2 + \ldots$ that gives the same loss at the same discharge. With the same f in every pipe, the derivation is as follows.

Derivation from scratch: Dupuit's equation for an equivalent pipe

Aim. Find the diameter D of a single pipe of length $L = L_1 + L_2 + L_3 + ...$ that gives the same head loss as the compound pipe at the same discharge.

Assumptions.

1. Minor losses are neglected (long pipes).
2. The friction factor f is the same in every pipe and in the equivalent pipe.
3. The equivalent pipe carries the same discharge Q as the compound pipe.

Step 1. Loss in the compound pipe. Add the friction loss of each pipe, written with Q.

$$H = \frac{8fL_1Q^2}{\pi^2 g D_1^5} + \frac{8fL_2Q^2}{\pi^2 g D_2^5} + \frac{8fL_3Q^2}{\pi^2 g D_3^5} + ...$$

Step 2. Loss in the equivalent pipe.

$$H = \frac{8fLQ^2}{\pi^2 g D^5}$$

Step 3. Set the two losses equal.

$$\frac{8fLQ^2}{\pi^2 g D^5} = \frac{8fL_1Q^2}{\pi^2 g D_1^5} + \frac{8fL_2Q^2}{\pi^2 g D_2^5} + \frac{8fL_3Q^2}{\pi^2 g D_3^5} + ...$$

Step 4. Cancel the common factor. Every term contains $\frac{8fQ^2}{\pi^2 g}$. Divide both sides by it.

$$\frac{L}{D^5} = \frac{L_1}{D_1^5} + \frac{L_2}{D_2^5} + \frac{L_3}{D_3^5} + ...$$

Result.

$$\frac{L}{D^5} = \frac{L_1}{D_1^5} + \frac{L_2}{D_2^5} + \frac{L_3}{D_3^5} + ...$$

What it tells us. Each pipe adds a "resistance" L_i/D_i^5. Once L is fixed as the total length, the equation gives the single diameter D that has the same resistance.

5.2 Pipes in parallel CORE

IN SIMPLE WORDS

Parallel pipes start at one point and end at another. Both ends have fixed heads, so every branch loses the same head h_f. The wider, shorter branch carries more flow, just as more cars choose the wider, shorter road. The branch flows add up to the flow in the main.

Derivation from scratch: flow split in parallel pipes

Aim. Show that parallel branches have equal head losses and that, for two branches with the same f, $Q_1/Q_2 = \sqrt{(L_2/L_1)\,(D_1/D_2)^5}$.

Assumptions.

1. When a main pipe divides into two or more branches that join again downstream, the branches start at the same point A and end at the same point B.
2. Minor losses in the branches are neglected.
3. Branch i has length L_i, diameter D_i, Darcy friction factor (f_A for pipe 1, f_B for pipe 2, and so on; letters are used so they are not confused with $f_1 = f/4$) and discharge Q_i.

Step 1. Continuity at the junction A. The flow arriving in the main must leave through the branches.

$$Q = Q_1 + Q_2 + \ldots$$

Step 2. One head at each end. Point A has one total head H_A and point B has one total head H_B. Every branch joins A to B, so the energy equation for each branch gives the same loss.

$$h_{f1} = h_{f2} = \ldots = H_A - H_B = h_f$$

Step 3. Write each loss with its discharge. Use the result of Section 3.2 for each branch.

$$h_f = \frac{8 f_A L_1 Q_1^2}{\pi^2 g D_1^5} = \frac{8 f_B L_2 Q_2^2}{\pi^2 g D_2^5} = \ldots$$

Step 4. Two branches with the same f. Put $f_A = f_B = f$ and cancel $\frac{8f}{\pi^2 g}$.

$$\frac{L_1 Q_1^2}{D_1^5} = \frac{L_2 Q_2^2}{D_2^5}$$

Step 5. Form the ratio of the flows. Divide both sides by $L_1 Q_2^2 / D_1^5$.

$$\frac{Q_1^2}{Q_2^2} = \frac{L_2}{L_1} \times \frac{D_1^5}{D_2^5} = \frac{L_2}{L_1}\left(\frac{D_1}{D_2}\right)^5$$

Step 6. Take the square root.

$$\frac{Q_1}{Q_2} = \sqrt{\frac{L_2}{L_1}\left(\frac{D_1}{D_2}\right)^5}$$

Step 7. Find each flow. Write $Q_1 = \left(Q_1/Q_2\right) Q_2$ and put it in $Q = Q_1 + Q_2$.

$$Q = Q_2\left(1 + \frac{Q_1}{Q_2}\right) \quad \Rightarrow \quad Q_2 = \frac{Q}{1 + Q_1/Q_2}, \qquad Q_1 = Q - Q_2$$

Result.

$$Q = Q_1 + Q_2 + \ldots, \qquad h_f = \frac{8 f_A L_1 Q_1^2}{\pi^2 g D_1^5} = \frac{8 f_B L_2 Q_2^2}{\pi^2 g D_2^5} = \ldots$$

$$\frac{Q_1}{Q_2} = \sqrt{\frac{L_2}{L_1}\left(\frac{D_1}{D_2}\right)^5} \qquad (\text{same } f)$$

What it tells us. In parallel the head loss is common and the discharges add, like resistors in parallel. The wider and shorter branch carries more flow, and the diameter matters far more than the length because it enters as $D^{5/2}$.

A second pipe laid alongside an old one in this way is a common means of raising the capacity of a water main.

5.3 Branched pipes: the three-reservoir problem HOTS

IN SIMPLE WORDS

Three tanks at different heights are joined to one junction. Water always flows out of the highest tank and into the lowest. The middle tank may supply water or receive it, depending on the head at the junction. We guess the junction head, check whether the flow in equals the flow out, and correct the guess.

Three reservoirs at levels $z_A > z_B > z_C$ are joined by three pipes that meet at a junction J. The flow from A always goes to J, and the flow from J always goes to C. Whether B supplies the junction or is fed by it depends on the piezometric head h_J at the junction:

- if $h_J > z_B$, the flow is from J into B, and $Q_1 = Q_2 + Q_3$;
- if $h_J < z_B$, B supplies the junction, and $Q_1 + Q_2 = Q_3$;
- if $h_J = z_B$, there is no flow in pipe 2.

Method. First assume $h_J = z_B$, so that $Q_2 = 0$. Find Q_1 from the loss $z_A - z_B$ and Q_3 from the loss $z_B - z_C$. If $Q_1 > Q_3$, the junction head must be higher than z_B, so water flows into B. Then adjust h_J by trial until continuity is satisfied at J.

Worked examples

Step-by-step method.

1. Convert all data to SI units. For each pipe write $h_f = rQ^2$, with $r = 8fL/\left(\pi^2 g D^5\right)$.
2. Decide whether the pipes are in series, in parallel or branched.
3. Series: the same Q flows in each pipe. Add the losses, set the sum equal to H and solve for Q.
4. Parallel: every branch has the same h_f. Find Q_1/Q_2 from equal losses, then use $Q_1 + Q_2 = Q$.
5. Three reservoirs: guess the junction head, find each Q, check continuity at the junction and adjust.
6. Check: the losses add up to H (series), or come out equal (parallel).

Example 5.0 (warm-up) CORE Two pipes in series carry $0.02\,\mathrm{m^3/s}$ of water: pipe 1 is 100 m of 200 mm pipe, and pipe 2 is 100 m of 100 mm pipe. Taking $f = 0.02$ and neglecting minor losses, find the total head loss.

1. $V_1 = \dfrac{0.02}{(\pi/4)\times 0.2^2} = 0.637\,\mathrm{m/s}$; by continuity $V_2 = 4V_1 = 2.546\,\mathrm{m/s}$
2. $h_{f1} = \dfrac{0.02\times 100\times 0.637^2}{2\times 9.81\times 0.2} = 0.207\,\mathrm{m}$
3. $h_{f2} = \dfrac{0.02\times 100\times 2.546^2}{2\times 9.81\times 0.1} = 6.61\,\mathrm{m}$
4. $H = h_{f1} + h_{f2} = 0.207 + 6.61 = 6.82\,\mathrm{m}$; the thin pipe takes 97% of the loss

Example 5.1 HOTS Three pipes are laid in series between two reservoirs whose water levels differ by 15 m: 300 m of 300 mm pipe, 170 m of 200 mm pipe and 210 m of 150 mm pipe. Taking $f = 0.02$ and neglecting minor losses, find the discharge and the diameter of an equivalent single pipe.

1. $h_f = 8fLQ^2/\left(\pi^2 g D^5\right)$ for each pipe, so $15 = Q^2 \times \sum \dfrac{8\times 0.02\times L_i}{\pi^2\times 9.81\times D_i^5}$
2. The three coefficients are 204.0, 877.9 and 4570.0 (s²/m⁵), which sum to 5651.9
3. $Q = \sqrt{15/5651.9} = 0.0515\,\mathrm{m^3/s}$ (51.5 L/s)
4. The losses are 0.54 m, 2.33 m and 12.13 m: the smallest pipe takes 81% of the head
5. Equivalent pipe: $\dfrac{L}{D^5} = \dfrac{300}{0.3^5} + \dfrac{170}{0.2^5} + \dfrac{210}{0.15^5} = 3.420\times 10^6$, with $L = 680\,\mathrm{m}$, so $D = 0.182\,\mathrm{m}$

Example 5.2 CORE A main carrying 0.12 m³/s divides into two branches that rejoin downstream: branch 1 is 500 m of 200 mm pipe, and branch 2 is 400 m of 150 mm pipe. Taking $f = 0.02$ in both, find the flow in each branch and the head loss.

1. $\dfrac{Q_1}{Q_2} = \sqrt{\dfrac{400}{500}\left(\dfrac{0.2}{0.15}\right)^5} = \sqrt{0.8\times 4.214} = 1.836$
2. $Q_2 = 0.12/2.836 = 0.0423\,\mathrm{m^3/s}$; $Q_1 = 0.0777\,\mathrm{m^3/s}$

3. $h_f = \dfrac{8 \times 0.02 \times 500 \times 0.0777^2}{\pi^2 \times 9.81 \times 0.2^5} = 15.6\,\text{m}$; branch 2 gives the same value, which checks the split
4. The velocities are 2.47 m/s and 2.39 m/s, nearly equal

Example 5.3 HOTS Reservoirs A, B and C have water levels of 100 m, 80 m and 50 m. They are joined to a common junction by pipe 1 (1000 m, 300 mm), pipe 2 (800 m, 250 mm) and pipe 3 (1200 m, 200 mm). Taking $f = 0.02$ and neglecting minor losses, find the flow in each pipe.

1. The coefficients $8fL/\left(\pi^2 g D^5\right)$ are 680.0 for pipe 1, 1353.8 for pipe 2 and 6197.4 for pipe 3
2. Trial $h_J = 80\,\text{m}$: $Q_1 = \sqrt{20/680} = 0.1715$ and $Q_3 = \sqrt{30/6197.4} = 0.0696\,\text{m}^3/\text{s}$. Since $Q_1 > Q_3$, the junction head is above 80 m and water flows into B
3. Adjust h_J until $Q_1 = Q_2 + Q_3$: $h_J = 86.06\,\text{m}$
4. $Q_1 = 0.1432\,\text{m}^3/\text{s}$ out of A, $Q_2 = 0.0669\,\text{m}^3/\text{s}$ into B and $Q_3 = 0.0763\,\text{m}^3/\text{s}$ into C. Check: $0.0669 + 0.0763 = 0.1432$

Common mistakes and exam tips.

- Mixing up the rules. In series Q is common and losses add; in parallel h_f is common and flows add.
- Adding the losses of parallel branches. The loss between the two junctions is the loss of any one branch.
- Slips with fifth powers. For example, $0.2^5 = 3.2 \times 10^{-4}$; check them on the calculator.
- In the three-reservoir problem, assuming the direction of flow in pipe 2. First try $h_J = z_B$ to find it.
- In Dupuit's equation, forgetting that L is the total length $L_1 + L_2 + \ldots$, with the same f in all pipes.

Check your understanding.

1. CORE In pipes in series, which quantity is the same in every pipe? *(The discharge Q.)*
2. CORE In pipes in parallel, which quantity is the same in every branch? *(The head loss h_f.)*
3. HOTS A second identical pipe is laid beside an old one, at the same total flow. How does the head loss change? *(Each pipe carries $Q/2$, so the loss falls to one-quarter, since $h_f \propto Q^2$.)*

6. Power transmission through a pipe (L4) CORE

A pipe that brings water from a high reservoir to a turbine or a nozzle transmits power. Part of the head is used up by friction on the way. The power delivered is greatest when friction takes one-third of the head, and the efficiency is then 66.7%. This is why penstocks are

always designed to run with much less than one-third loss.

IN SIMPLE WORDS

A pipe can carry power, not just water, much as a cable carries electric power from a dam to a town. The head left at the end of the pipe is the gross head minus the friction loss. The power delivered is $\rho g Q\ (\ H - h_f\ )$: the weight of water arriving each second times the head it still has. Push too much water through and friction eats most of the head, so there is a best flow.

6.1 Power delivered and efficiency CORE

IN SIMPLE WORDS

In $P = \rho g Q\ (\ H - h_f\ )$, the term $\rho g Q$ is the weight of water delivered per second. The term $H - h_f$ is the head still available at the outlet. The efficiency $\eta = \ (\ H - h_f\ )\ / H$ is the fraction of the head that reaches the end. A short, wide pipe gives a high efficiency.

Derivation from scratch: power transmitted through a pipe and efficiency

Aim. Find the power P delivered at the outlet of a pipe and the efficiency η of transmission.

Assumptions.

1. Let H be the total head at the inlet of a pipe of length L and diameter D, and h_f the friction loss.
2. Minor losses are neglected.
3. The flow is steady with discharge Q and mean velocity V.

Step 1. Head left at the outlet. Friction uses up h_f of the head on the way.

$$\text{Head at outlet} = H - h_f$$

Step 2. Weight of water delivered each second. Mass per second is ρQ, so weight per second is $\rho g Q$ (in N/s).

Step 3. Power = weight per second times head. Head is energy per unit weight (Section 0.2), so this product is energy per second, in J/s = W.

$$P = \rho g Q\ \left(\ H - h_f\ \right)$$

Step 4. Write P in terms of V. Put $Q = \dfrac{\pi D^2}{4} V$ and $h_f = \dfrac{f L V^2}{2 g D}$.

$$P = \rho g \frac{\pi D^2}{4} V \left(H - \frac{f L V^2}{2 g D} \right)$$

Step 5. Efficiency. Efficiency is the power delivered divided by the power supplied at the inlet, $\rho g Q H$. The factor $\rho g Q$ cancels.

$$\eta = \frac{\rho g Q\ \left(\ H - h_f\ \right)}{\rho g Q H} = \frac{H - h_f}{H}$$

Result.

$$P = \rho g Q\ (\ H - h_f\ )\ = \rho g \frac{\pi D^2}{4} V\left(H - \frac{fLV^2}{2gD}\right), \qquad \eta = \frac{H - h_f}{H}$$

What it tells us. The power rises with the flow, but the friction loss rises as V^2 and eats into the head. The efficiency is simply the fraction of the head that survives to the outlet.

6.2 Condition for maximum power (derivation) CORE

IN SIMPLE WORDS

If the flow is small, little power is carried. If the flow is large, friction takes most of the head. Somewhere between lies a maximum. Calculus shows that it occurs when friction takes one-third of the head, $h_f = H/3$. The efficiency is then only 66.7%.

Derivation from scratch: condition for maximum power transmission

Aim. Find the velocity (and so the friction loss) at which a given pipe delivers the greatest power, and the efficiency at that point.

Assumptions.

1. The pipe (L, D, f) and the inlet head H are fixed; only the flow can change.
2. f is taken as constant.
3. The power is given by Section 6.1: $P = \rho g \frac{\pi D^2}{4} V\left(H - \frac{fLV^2}{2gD}\right)$.

Step 1. Separate the constant part. For a given pipe and head, P depends on V alone. Write $k = \rho g\, \pi D^2/4$, which is a positive constant, and multiply out the bracket.

$$P = k\left(HV - \frac{fLV^3}{2gD}\right)$$

Step 2. Why there is a maximum. Raising V raises the flow, which increases the first term. But it also raises the friction loss as V^2, so the second term grows as V^3 and wins at high V. So P rises, reaches a peak and then falls.

Step 3. First derivative. Differentiate term by term: $\frac{d}{dV}(\ HV) = H$ and $\frac{d}{dV}\left(\ V^3\ \right) = 3V^2$.

$$\frac{dP}{dV} = k\frac{d}{dV}\left(HV - \frac{fLV^3}{2gD}\right) = k\left(H - 3\frac{fLV^2}{2gD}\right)$$

Step 4. Set the first derivative to zero. Since $k \neq 0$, the bracket must be zero.

$$H - 3\frac{fLV^2}{2gD} = 0$$

Step 5. Recognise h_f. The term $\frac{fLV^2}{2gD}$ is exactly the friction loss h_f.

$$H - 3h_f = 0 \quad \Rightarrow \quad h_f = \frac{H}{3}$$

Step 6. Second derivative check. Differentiate $\frac{dP}{dV}$ again: $\frac{d}{dV}\left(3V^2\right) = 6V$.

$$\frac{d^2P}{dV^2} = k\left(-\frac{6fLV}{2gD}\right) = -k\frac{6fLV}{2gD}$$

Every quantity here is positive, so $\frac{d^2P}{dV^2} < 0$. The stationary point is therefore a **maximum**.

Step 7. Efficiency at maximum power. Put $h_f = H/3$ in $\eta = (H - h_f)/H$.

$$\eta_{\text{max power}} = \frac{H - H/3}{H} = \frac{2H/3}{H} = \frac{2}{3} = 66.7\%$$

Result.

$$\frac{d}{dV}\left(HV - \frac{fLV^3}{2gD}\right) = H - 3\frac{fLV^2}{2gD} = 0 \quad \Rightarrow \quad h_f = \frac{H}{3}$$

$$\eta_{\text{max power}} = \frac{H - H/3}{H} = \frac{2}{3} = 66.7\%$$

What it tells us. A pipe delivers its greatest power when friction takes one-third of the inlet head, and the power is then $P_{\max} = \rho g Q\,(2H/3)$. The efficiency at that point is $2/3$. Smaller flows give a higher efficiency but less power.

The condition gives the most power from a given pipe, not the best use of the water. Real hydro schemes trade this off against the cost of a larger pipe and usually accept 5 to 10% loss.

6.3 Nozzle diameter for maximum power CORE

IN SIMPLE WORDS

When the pipe ends in a nozzle, the water leaves as a fast jet that drives a Pelton wheel. The nozzle size sets the flow, much as a thumb pressed over a hose end changes the jet. A tiny nozzle passes too little water; a large one lets friction take too much head. The best nozzle diameter d makes the pipe friction equal to $H/3$.

Derivation from scratch: nozzle diameter for maximum power

Aim. Find the nozzle diameter d at the end of a pipe that makes the jet carry the greatest power.

Assumptions.

1. When the pipe ends in a nozzle (as in a Pelton wheel supply), the power is carried by the jet's kinetic energy.
2. The pipe has length L, diameter D and friction factor f; the inlet head is H.
3. Let d be the nozzle diameter and v the jet velocity. The jet leaves at atmospheric pressure.
4. The loss in the nozzle and other minor losses are neglected.

Step 1. Energy equation from inlet to jet. The head at the inlet is used up by pipe friction, and what remains appears as the velocity head of the jet.

$$H = h_f + \frac{v^2}{2g}$$

Step 2. Power condition. The jet power is $\rho g Q \left(v^2/2g \right) = \rho g Q \ (\ H - h_f \ )$, the same form as in Section 6.1. Changing d only changes the flow, so the maximum again occurs at $h_f = H/3$ (Section 6.2).

Step 3. Split the head. Put $h_f = H/3$ in Step 1.

$$\frac{v^2}{2g} = H - \frac{H}{3} = \frac{2H}{3}, \qquad h_f = \frac{H}{3} = \frac{1}{2} \times \frac{2H}{3} = \frac{1}{2}\frac{v^2}{2g}$$

Step 4. Continuity between pipe and nozzle. The same discharge passes both.

$$\frac{\pi}{4} D^2 V = \frac{\pi}{4} d^2 v \quad \Rightarrow \quad V = v \left(\frac{d}{D} \right)^2$$

Step 5. Write h_f with the jet velocity. Put $V^2 = v^2 \ (\ d/D \ )^{\ 4}$ in the Darcy–Weisbach equation.

$$h_f = \frac{f L V^2}{2gD} = \frac{fL}{D} \left(\frac{d}{D} \right)^4 \frac{v^2}{2g}$$

Step 6. Apply the condition of Step 3.

$$\frac{fL}{D} \left(\frac{d}{D} \right)^4 \frac{v^2}{2g} = \frac{1}{2}\frac{v^2}{2g}$$

Step 7. Cancel $v^2/\ (\ 2g \ )$ and solve for d^4.

$$\frac{fL}{D}\frac{d^4}{D^4} = \frac{1}{2} \quad \Rightarrow \quad d^4 = \frac{D^5}{2fL}$$

Step 8. Take the fourth root, and convert to f_1. With $f = 4f_1$, $2fL = 8f_1 L$.

$$d = \left(\frac{D^5}{2fL} \right)^{1/4} = \left(\frac{D^5}{8 f_1 L} \right)^{1/4}$$

Result.

$$\frac{fL}{D}\left(\frac{d}{D}\right)^4\frac{v^2}{2g}=\frac{1}{2}\frac{v^2}{2g}\quad\Rightarrow\quad d=\left(\frac{D^5}{2fL}\right)^{1/4}=\left(\frac{D^5}{8f_1L}\right)^{1/4}$$

What it tells us. The best nozzle grows with the pipe diameter (as $D^{5/4}$) and shrinks slowly for a longer or rougher pipe. With this nozzle, two-thirds of the head reaches the jet, so $v=\sqrt{2g\times 2H/3}$.

The second form uses the coefficient of friction $f_1=f/4$, as in Bansal.

Worked examples

Step-by-step method.

1. Convert all data to SI units and find $V=Q/A$.
2. Find the friction loss $h_f=fLV^2/\left(2gD\right)$.
3. Find the head available at the outlet, $H-h_f$.
4. Find the power $P=\rho gQ\left(H-h_f\right)$ and the efficiency $\eta=\left(H-h_f\right)/H$.
5. For maximum power, put $h_f=H/3$, solve for V and Q, and use $P_{\max}=\rho gQ\left(2H/3\right)$.
6. For a nozzle, use $d=\left(D^5/\left(2fL\right)\right)^{1/4}$ and the jet velocity $v=\sqrt{2g\times 2H/3}$.

Example 6.0 (warm-up) CORE A pipe carries $0.1\ \mathrm{m^3/s}$ of water from a reservoir with a gross head of 100 m. The friction loss in the pipe is 20 m. Find the power delivered and the efficiency of transmission.

1. Head available at the outlet: $H-h_f=100-20=80\ \mathrm{m}$
2. $P=\rho gQ\left(H-h_f\right)=1000\times 9.81\times 0.1\times 80=78.48\ \mathrm{kW}$
3. $\eta=\dfrac{H-h_f}{H}=\dfrac{80}{100}=80\%$

Example 6.1 CORE A pipe 3 km long and 300 mm in diameter ($f=0.02$) carries water from a reservoir to a turbine, with a gross head of 250 m. Find the power delivered and the efficiency of transmission for discharges of 0.12 m³/s and 0.25 m³/s.

Q (m³/s)	V (m/s)	h_f (m)	Power delivered (kW)	Efficiency (%)
0.12	1.70	29.4	259.7	88.2
0.25	3.54	127.5	300.4	49.0

Here $h_f=fLV^2/\left(2gD\right)=0.02\times 10\,000\times V^2/19.62$, and $P=9.81\,Q\left(250-h_f\right)$ kW. Doubling the flow raises the power by only 16%, while friction wastes more than half the head.

Example 6.2 HOTS For the pipe of Example 6.1, find the maximum power that can be delivered and the discharge at which it occurs.

1. $h_f = H/3 = 83.3\,\text{m}$, so $V = \sqrt{\dfrac{2g \times 83.3 \times D}{fL}} = \sqrt{\dfrac{19.62 \times 83.3 \times 0.3}{60}} = 2.86\,\text{m/s}$
2. $Q = 2.86 \times \dfrac{\pi}{4} \times 0.3^2 = 0.202\,\text{m}^3/\text{s}$
3. $P_{\max} = \rho g Q \left(\dfrac{2H}{3}\right) = 9810 \times 0.202 \times 166.7 = 330.4\,\text{kW}$, at 66.7% efficiency

Example 6.3 HOTS The pipe of Example 6.1 ends in a nozzle that discharges to a Pelton wheel. Find the nozzle diameter for maximum power, and the jet velocity and power.

1. $d = \left(\dfrac{D^5}{2fL}\right)^{1/4} = \left(\dfrac{0.3^5}{120}\right)^{1/4} = 67.1\,\text{mm}$
2. Jet velocity: $v = \sqrt{2g \times 2H/3} = \sqrt{19.62 \times 166.7} = 57.2\,\text{m/s}$
3. $Q = 57.2 \times \dfrac{\pi}{4} \times 0.0671^2 = 0.202\,\text{m}^3/\text{s}$, the same as Example 6.2, as it should be
4. Jet power $= \frac{1}{2}\rho Q v^2 = 0.5 \times 1000 \times 0.202 \times 57.2^2 = 330.4\,\text{kW}$

Common mistakes and exam tips.

- Using the gross head H for the delivered power. Use the head left at the outlet, $H - h_f$.
- Writing the maximum-power condition as $h_f = H/2$. It is $h_f = H/3$.
- Calling 66.7% the maximum efficiency. It is the efficiency at maximum power; smaller flows give higher efficiency.
- Mixing coefficients in the nozzle formula. With the Darcy f use $2fL$; with f_1 use $8f_1L$.
- Forgetting to divide watts by 1000 when the answer is asked in kW.

Check your understanding.

1. CORE What is the condition for maximum power transmission through a pipe? *(The friction loss equals one-third of the gross head, $h_f = H/3$.)*
2. CORE What is the efficiency at maximum power? *(It is $2/3$, or 66.7%.)*
3. HOTS Why are real penstocks not designed for maximum power? *(One-third of the head would be wasted, so engineers use larger pipes and accept only 5 to 10% loss.)*

7. Siphon (L5) CORE

A siphon is a long bent pipe that carries liquid from a higher reservoir to a lower one over a ridge higher than the supply level. The flow is driven only by the difference in water levels. Its limit is the pressure at the summit: this must not fall to the point where dissolved air comes out of solution, or the flow stops. For water this caps the summit at about 7 to 8 m above the supply level.

IN SIMPLE WORDS

Think of sucking juice through a bent straw over the rim of a glass, then letting it run into a lower cup. Once the pipe is full, the water keeps flowing by itself because the outlet is lower than the inlet. Engineers like it because it moves water over a ridge with no pump and no power. The weak point is the top of the bend, where the pressure is below atmospheric. If it drops too far, air bubbles form and the flow stops. In the formulas, H is the drop in water level that drives the flow and h_s is the height of the summit above the supply level.

7.1 Working and uses CORE

IN SIMPLE WORDS

Priming means filling the pipe completely with water first, like sucking on the straw. After that, the longer and heavier column on the outlet side pulls the water over the top. No pump is needed once the flow starts. If air gets in and the pipe empties, the siphon must be primed again.

The siphon is first filled with liquid (primed). The weight of the longer column on the outlet side then keeps the liquid moving over the summit. Its uses include:

- carrying water over a hill or ridge between two reservoirs;
- emptying a tank or channel that has no outlet at the bottom;
- draining irrigation water from a canal over its bank;
- siphon spillways on dams, which start automatically as the water level rises.

7.2 Summit pressure and the limit on summit height CORE

IN SIMPLE WORDS

On the way up to the summit, the water spends pressure head to gain height, to gain velocity and to pay for losses. So the pressure at the top is below atmospheric, a partial vacuum. In the equation, $p_a/(\rho g)$ is atmospheric pressure as a head (about 10.3 m of water) and $p_s/(\rho g)$ is the summit pressure head. h_s is the summit height, and the $V^2/(2g)$ terms are the velocity head and the losses. The summit pressure must stay above about 2.5 m of water absolute. For the discharge, the whole level difference H is used up in entry, friction and exit losses.

Derivation from scratch: discharge through a siphon and pressure at its summit

Aim. To find the velocity (and so the discharge) in a siphon, and the pressure at its summit S.

Assumptions.

1. The flow is steady and the pipe runs full. Its diameter D is the same everywhere, so the velocity V is the same everywhere.
2. Both reservoirs are large. Their free surfaces are almost at rest ($V \approx 0$) and at atmospheric pressure p_a.
3. The friction factor f is constant. The entrance loss is $0.5V^2/(2g)$ and the exit loss is $V^2/(2g)$.
4. The total pipe length is L. The summit S is a height h_s above the supply level A, at a pipe length L_1 from the inlet.
5. All pressures are absolute, and all heads are in metres of the flowing liquid.

Step 1. Energy equation with losses. Between any two points 1 and 2 on the flow path, the energy per unit weight at 1 equals the energy per unit weight at 2 plus the head lost between them.

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2 + h_L$$

Step 2. Discharge: from the upper free surface A to the lower free surface B. Take the datum at the level of B, so $z_1 = H$ and $z_2 = 0$. Both surfaces are at p_a and nearly at rest.

$$\frac{p_a}{\rho g} + 0 + H = \frac{p_a}{\rho g} + 0 + 0 + h_L$$

Step 3. Cancel the atmospheric head. The term $p_a/(\rho g)$ appears on both sides, so it cancels. The whole difference in levels is used up in losses.

$$H = h_L$$

Step 4. Write the loss term by term. The water loses head at the entrance, by friction over the whole length L, and at the exit.

$$h_L = 0.5\,\frac{V^2}{2g} + \frac{fL}{D}\,\frac{V^2}{2g} + 1 \times \frac{V^2}{2g}$$

Step 5. Take out the common factor $V^2/(2g)$.

$$H = \left(0.5 + \frac{fL}{D} + 1\right)\frac{V^2}{2g}$$

Step 6. Solve for the velocity and the discharge. Multiply both sides by $2g$, divide by the bracket, and take the square root.

$$V = \sqrt{\frac{2gH}{1.5 + fL/D}}, \qquad Q = \frac{\pi}{4}D^2\,V$$

Step 7. Summit: from the free surface A to the summit S. Now take the datum at the level of A, so $z_1 = 0$ and $z_2 = h_s$. At S the pressure is p_s and the velocity is V. Between A and S the losses are the entrance loss and friction over the length L_1 only.

$$\frac{p_a}{\rho g} + 0 + 0 = \frac{p_s}{\rho g} + \frac{V^2}{2g} + h_s + 0.5\,\frac{V^2}{2g} + \frac{fL_1}{D}\,\frac{V^2}{2g}$$

Step 8. Solve for the summit pressure head. Move every term except $p_s/(\rho g)$ to the left side. Collect the three $V^2/(2g)$ terms: $1 + 0.5 + fL_1/D = 1.5 + fL_1/D$.

$$\frac{p_s}{\rho g} = \frac{p_a}{\rho g} - h_s - \left(1.5 + \frac{fL_1}{D}\right)\frac{V^2}{2g}$$

Result.

$$\frac{p_a}{\rho g} = \frac{p_s}{\rho g} + h_s + \frac{V^2}{2g} + 0.5\,\frac{V^2}{2g} + \frac{fL_1}{D}\,\frac{V^2}{2g}$$

$$H = \left(0.5 + \frac{fL}{D} + 1\right)\frac{V^2}{2g}$$

What it tells us. The velocity rises as $\sqrt{H}$ and falls when the pipe is longer or rougher. The summit pressure falls by 1 m of water for every 1 m the summit is raised. It also falls as the velocity or the length L_1 up to the summit increases. Setting $p_s/(\rho g)$ equal to the lowest safe absolute head (about 2.5 m of water) gives the greatest allowed summit height.

The summit pressure p_s is therefore below atmospheric by the summit height plus the velocity head plus the losses up to the summit. In theory the summit could reach 10.3 m above A (one atmosphere of water). In practice water releases dissolved air when the absolute pressure drops to about 2.5 m of water. The air collects at the summit and breaks the flow. Practical siphons keep the summit within about 7 to 8 m of the supply level, and include an air valve at the summit for priming.

Worked examples

Step-by-step method.

1. Convert the diameter and all lengths to metres.
2. Add the loss coefficients: 0.5 (entry) plus fL/D (friction) plus 1 (exit).
3. Find V from $H = (\text{sum of coefficients}) \times V^2/(2g)$, then $Q = (\pi/4)\,D^2 V$.
4. For the summit, apply the energy equation from the upper water surface to the summit, using only the length L_1 up to the summit.
5. Write the summit pressure as an absolute head: $10.3 - h_s - (1 + 0.5 + fL_1/D)\,V^2/(2g)$.
6. Compare it with the limit (about 2.5 m of water absolute) to judge whether the siphon will run.

Example 7.0 (warm-up) CORE A siphon 100 mm in diameter and 40 m long joins two tanks whose water levels differ by 2 m. Take $f = 0.02$, an entrance loss of $0.5V^2/(2g)$ and an exit loss of $V^2/(2g)$. Find the discharge.

1. $\dfrac{fL}{D} = \dfrac{0.02 \times 40}{0.1} = 8$; sum of coefficients $= 0.5 + 8 + 1 = 9.5$
2. $V = \sqrt{\dfrac{2 \times 9.81 \times 2}{9.5}} = 2.032\ \text{m/s}$
3. $Q = \dfrac{\pi}{4} \times 0.1^2 \times 2.032 = 0.0160\ \text{m}^3/\text{s} = 16.0\ \text{L/s}$

Example 7.1 CORE A siphon of 200 mm diameter and 500 m total length joins two reservoirs whose water levels differ by 4 m. The summit is 3 m above the upper water level, and 100 m of pipe lies upstream of it. Take $f = 0.02$, an entrance loss of $0.5V^2/(2g)$ and an atmospheric pressure of 10.3 m of water. Find the discharge and the absolute pressure at the summit.

1. Sum of coefficients: $0.5 + \dfrac{0.02 \times 500}{0.2} + 1 = 51.5$
2. $V = \sqrt{\dfrac{2 \times 9.81 \times 4}{51.5}} = 1.234\ \text{m/s}$; $Q = 1.234 \times \dfrac{\pi}{4} \times 0.2^2 = 38.8\ \text{L/s}$
3. Velocity head: $\dfrac{V^2}{2g} = 0.0777\ \text{m}$
4. Summit: $\dfrac{p_s}{\rho g} = 10.3 - 3 - 0.0777\left(1 + 0.5 + \dfrac{0.02 \times 100}{0.2}\right)$
   $= 10.3 - 3 - 0.893 = 6.41\ \text{m of water (absolute)}$, that is, 3.89 m below atmospheric

Example 7.2 HOTS For the siphon of Example 7.1, find the greatest height of the summit above the upper water level if the absolute pressure there must not fall below 2.5 m of water.

1. $2.5 = 10.3 - h_s - 0.893$
2. $h_{s,\,\max} = 6.91\ \text{m}$
3. Moving the summit nearer the inlet (smaller L_1) reduces the loss up to the summit and allows a slightly higher ridge

Common mistakes and exam tips.

- Work in absolute pressure at the summit: start from 10.3 m of water. Do not mix gauge and absolute heads.
- Up to the summit, use only the length L_1 in the friction term, not the full length L.
- The exit loss $V^2/(2g)$ belongs to the discharge equation; at the summit the velocity head $V^2/(2g)$ appears instead.
- Convert the diameter from mm to m before finding fL/D.
- In theory questions, state the practical limit: summit within about 7 to 8 m of the supply level.

Check your understanding.

1. CORE What drives the flow in a siphon? *(The difference H between the two water levels.)*
2. CORE Why is the summit pressure below atmospheric? *(Pressure head is spent to lift the water, give it velocity and overcome losses up to the summit.)*
3. HOTS What happens to the summit pressure if the summit is raised by 1 m, with the same flow? *(It falls by 1 m of water, since $p_s/(\rho g)$ contains $-h_s$.)*

8. Water hammer (L5) CORE

When a valve at the end of a long pipeline is closed quickly, the moving column of water is brought to rest. Its momentum is converted into a pressure rise that travels up the pipe as a wave at about the speed of sound in water. The rise is ρVC for a sudden closure, about 1.4 MPa for every 1 m/s of flow stopped. It can burst pipes, crack valve bodies and lift pipe supports, and is heard as a hammering knock.

IN SIMPLE WORDS

Imagine a fast-moving column of water suddenly meeting a closed door. The water stops, and its momentum turns into a sharp rise in pressure. You can hear it as a bang in house pipes when a tap is shut quickly. Engineers care because the rise can reach hundreds of metres of water and burst pipes. In $p = \rho VC$, ρ is the water density, V is the velocity that is stopped and C is the speed of the pressure wave.

8.1 Gradual closure (rigid water column) CORE

IN SIMPLE WORDS

If the valve is closed slowly, the whole water column slows down together, like a car braking gently. Newton's second law gives the force: mass times deceleration. Here L is the pipe length, V the initial velocity and T the closing time. A longer closing time T gives a smaller pressure rise.

Derivation from scratch: pressure rise for gradual closure of a valve

Aim. To find the pressure rise p at the valve when the flow is stopped slowly in a time T .

Assumptions.

1. The closure is gradual ($T > 2L/C$), so the whole water column slows down together, like a rigid body.
2. The pipe is rigid, of area A and length L; the water is treated as incompressible.
3. The valve is closed uniformly, so the deceleration is constant.
4. Friction is neglected, and the pressure at the reservoir end does not change.

Step 1. Choose the body. Take the whole column of water in the pipe, from the reservoir to the valve. Its mass is density times volume.

$$m = \rho \times AL = \rho AL$$

Step 2. Find the deceleration. The velocity falls uniformly from V to 0 in time T.

$$a = \frac{V - 0}{T} = \frac{V}{T}$$

Step 3. Find the force that stops the column. At the valve the pressure rises by p. This extra pressure acts on the end face of area A and pushes the column backwards (against the flow) with force pA. No other force acts along the pipe.

$$F = p\,A$$

Step 4. Apply Newton's second law, $F = m\,a$.

$$p\,A = \rho A L \cdot \frac{V}{T}$$

Step 5. Cancel the area A on both sides.

$$p = \frac{\rho L V}{T}$$

Step 6. Convert to a head. Divide by ρg; the ρ cancels.

$$h = \frac{p}{\rho g} = \frac{\rho L V}{\rho g T} = \frac{L V}{g T}$$

Result.

$$p = \frac{\rho L V}{T}, \qquad h = \frac{p}{\rho g} = \frac{L V}{g T}$$

What it tells us. The pressure rise is proportional to the pipe length L and to the velocity V stopped. It is inversely proportional to the closing time T: closing twice as slowly halves the rise. The formula holds only when $T > 2L/C$ (Section 8.2).

8.2 Speed of the pressure wave and critical time CORE

IN SIMPLE WORDS

Water is not perfectly rigid. It squeezes a little, so a closure travels along the pipe as a wave, like a pulse along a slinky spring. K is the bulk modulus (how hard the water is to compress) and C is the wave speed, about 1450 m/s. The wave goes up the pipe and comes back in time $2L/C$. If the valve closes faster than this round trip, the closure counts as sudden.

Water is slightly compressible, so a closure is felt along the pipe as a pressure wave that travels at speed C. In a rigid pipe $C = \sqrt{K/\rho}$, about 1450 m/s for water ($K = 2.1$ GPa). The wave runs up to the reservoir, is reflected as a pressure drop, and returns to the valve after time $2L/C$. This round trip sets the **critical time**:

- $T > 2L/C$: **gradual closure**. The reflected wave arrives before closure is complete and relieves part of the pressure. Use $p = \rho L V/T$.
- $T < 2L/C$: **sudden closure**. The full pressure rise builds before any relief arrives. Use the formulas below.

Derivation from scratch: speed of the pressure wave and critical time

Aim. To show that the pressure wave in a rigid pipe travels at $C = \sqrt{K/\rho}$, and that the critical time is $2L/C$.

Assumptions.

1. The pipe is rigid, of area A and length L. The valve is closed instantly.
2. Water is slightly compressible, with bulk modulus K (pressure rise divided by volumetric strain).
3. Friction is neglected, and V is very small compared with C.

Step 1. Picture the wave. When the valve shuts, the water next to it stops and its pressure rises by p. The boundary between stopped water and moving water (the wave front) runs upstream at speed C. In a short time dt the front moves a distance $C\,dt$. So a slice of water of length $C\,dt$ is brought from V to rest.

Step 2. Impulse–momentum on the slice. The mass of the slice is $\rho A C\,dt$. Its momentum falls from $\rho A C\,dt \times V$ to zero. The force that stops it is the extra pressure p on area A, acting for time dt.

$$p\,A\,dt = (\,\rho A C\,dt\,)\,V$$

Cancel $A\,dt$ on both sides:

$$p = \rho C V$$

Step 3. Compression of the slice. During dt, water behind the front still arrives at speed V. So an extra length $V dt$ of water is squeezed into the slice of length $C\,dt$. The volumetric strain is:

$$\frac{\text{change in volume}}{\text{volume}} = \frac{A V dt}{A C\,dt} = \frac{V}{C}$$

By the definition of the bulk modulus, pressure rise $= K \times$ volumetric strain:

$$p = K\frac{V}{C}$$

Step 4. Equate the two expressions for p.

$$\rho C V = \frac{K V}{C}$$

Cancel V and multiply both sides by C/ρ:

$$C^2 = \frac{K}{\rho} \quad \Rightarrow \quad C = \sqrt{\frac{K}{\rho}}$$

For water, $C = \sqrt{2.1 \times 10^9 / 1000} = 1449\ \text{m/s}$.

Step 5. Critical time. The wave takes L/C to reach the reservoir. There it is reflected as a pressure drop, which takes another L/C to come back to the valve.

$$T_c = \frac{L}{C} + \frac{L}{C} = \frac{2L}{C}$$

Result.

$$C = \sqrt{\frac{K}{\rho}}, \qquad \text{critical time} = \frac{2L}{C}$$

What it tells us. A stiffer fluid (larger K) carries the wave faster. A longer pipe gives a longer critical time, so a long pipeline is harder to close "gradually". Step 2 already shows that a sudden closure gives $p = \rho V C$; Section 8.3 obtains the same result by an energy method.

8.3 Sudden closure in a rigid pipe (derivation) CORE

IN SIMPLE WORDS

In a sudden closure there is no time for relief. All the kinetic energy of the moving water is stored as squeeze energy in the water, like compressing a spring. Equating the two gives $p = \rho V C$. Every 1 m/s of velocity stopped gives about 1.4 MPa of pressure rise.

Derivation from scratch: pressure rise for sudden closure in a rigid pipe (energy method)

Aim. To find the pressure rise p when a valve is closed suddenly at the end of a rigid pipe.

Assumptions.

1. The closure is sudden ($T < 2L/C$), so no relief wave arrives before the flow stops.
2. The pipe is rigid; it does not stretch. Its area is A and its length is L.
3. Water is compressible, with bulk modulus K, and obeys $p = K \times$ volumetric strain.
4. Friction is neglected. All the kinetic energy of the water is stored as strain energy of compression in the water.

Step 1. Kinetic energy of the moving column. The mass of water is $\rho A L$ and it moves at V.

$$\text{KE} = \frac{1}{2} m V^2 = \frac{1}{2} \rho A L \, V^2$$

Step 2. Volumetric strain of the water. When the pressure rises by p, the water is squeezed. From the definition of the bulk modulus:

$$\text{volumetric strain} = \frac{p}{K}, \qquad \text{change in volume} = \frac{p}{K} A L$$

Step 3. Strain energy stored in the water. The pressure grows from 0 to p while the volume shrinks, so the average pressure is $p/2$. Work done = average pressure $\times$ change in volume.

$$\text{SE} = \frac{p}{2} \times \frac{p}{K} A L = \frac{p^2}{2K} A L$$

Step 4. Equate kinetic energy lost and strain energy stored.

$$\frac{1}{2}\rho AL\,V^2 = \frac{p^2}{2K}\,AL$$

Step 5. Cancel the common factor $\frac{1}{2}AL$ on both sides.

$$\rho V^2 = \frac{p^2}{K}$$

Step 6. Solve for p. Multiply by K and take the square root.

$$p^2 = \rho K V^2 \quad \Rightarrow \quad p = V\sqrt{\rho K}$$

Step 7. Write it with the wave speed. Since $C = \sqrt{K/\rho}$, we have $\sqrt{\rho K} = \rho\sqrt{K/\rho} = \rho C$.

$$p = \rho V C, \qquad h = \frac{p}{\rho g} = \frac{VC}{g}$$

Result.

$$\frac{1}{2}\rho AL\,V^2 = \frac{p^2}{2K}\,AL \quad \Rightarrow \quad p = V\sqrt{\rho K} = \rho VC$$

What it tells us. The pressure rise is proportional to the velocity stopped. It does not depend on the pipe length or on the closing time, as long as the closure is sudden. For water, stopping 1 m/s gives $1000 \times 1 \times 1449 \approx 1.45\,\text{MPa}$, about 148 m of water.

8.4 Sudden closure in an elastic pipe HOTS

IN SIMPLE WORDS

A real pipe wall stretches a little under pressure, like a balloon. The wall then stores part of the energy, so less is left to raise the pressure. E is the Young's modulus of the pipe material, D the diameter and t the wall thickness. A thin, flexible pipe gives a smaller pressure rise than a rigid one.

In a real pipe the walls stretch under the pressure rise and store part of the energy. For a pipe of diameter D, wall thickness t and Young's modulus E, the hoop strain energy of the wall is added to the strain energy of the water.

Derivation from scratch: pressure rise for sudden closure in an elastic pipe

Aim. To find the pressure rise p for a sudden closure when the pipe wall also stretches and stores energy.

Assumptions.

1. The closure is sudden. Friction is neglected.
2. The pipe is a thin cylinder (t much smaller than D) of length L, area $A = \pi D^2/4$, wall thickness t and Young's modulus E.
3. Only the hoop (circumferential) stress in the wall is counted; the longitudinal stress and Poisson's ratio are neglected.
4. All the kinetic energy of the water is stored as strain energy, partly in the water and partly in the pipe wall.

Step 1. Kinetic energy of the water (as in Section 8.3).

$$\mathrm{KE} = \frac{1}{2}\,\rho A L\,V^2$$

Step 2. Strain energy in the water (as in Section 8.3).

$$\mathrm{SE}_{\mathrm{water}} = \frac{p^2}{2K}\,AL$$

Step 3. Hoop stress in the wall. Cut the pipe along a diameter and look at one half, of length L. The pressure p pushes the half apart with a force equal to p times the projected area DL. Two strips of wall, each of area tL, resist it with stress σ.

$$p \times DL = \sigma \times 2tL \quad \Rightarrow \quad \sigma = \frac{pD}{2t}$$

Step 4. Strain energy per unit volume of the wall. The wall stretches elastically, with strain σ/E. The energy stored per unit volume is the area under the straight stress–strain line.

$$\frac{1}{2} \times \sigma \times \frac{\sigma}{E} = \frac{\sigma^2}{2E}$$

Step 5. Volume of the wall. The wall is a thin sheet: circumference $\times$ thickness $\times$ length.

$$\text{volume of wall} = \pi D \times t \times L = \pi DtL$$

Step 6. Strain energy in the wall. Multiply Step 4 by Step 5, and put in $\sigma = pD/\,(\,2t\,)$.

$$\mathrm{SE}_{\mathrm{wall}} = \frac{1}{2E}\left(\frac{pD}{2t}\right)^2 \pi DtL = \frac{p^2D^2}{8Et^2}\,\pi DtL = \frac{\pi p^2 D^3 L}{8Et}$$

Step 7. Write it in terms of AL. Split $\pi D^3/8$ as $\left(\,\pi D^2/4\,\right) \times (\,D/2\,)$, and use $A = \pi D^2/4$.

$$\mathrm{SE}_{\mathrm{wall}} = \frac{\pi D^2}{4}\,L \times \frac{p^2 D}{2Et} = \frac{p^2 D}{2Et}\,AL$$

Step 8. Energy balance. Kinetic energy lost = strain energy in water + strain energy in wall.

$$\frac{1}{2}\rho AL\,V^2 = \frac{p^2}{2K}\,AL + \frac{p^2 D}{2Et}\,AL$$

Step 9. Cancel $\frac{1}{2}AL$ and take p^2 out as a common factor.

$$\rho V^2 = p^2\left(\frac{1}{K} + \frac{D}{Et}\right)$$

Step 10. Solve for p. Divide by the bracket and take the square root.

$$p = V\sqrt{\frac{\rho}{\dfrac{1}{K} + \dfrac{D}{Et}}}$$

Result.

$$\frac{1}{2}\rho AL\,V^2 = \frac{p^2}{2K}\,AL + \frac{p^2 D}{2Et}\,AL \quad\Rightarrow\quad p = V\sqrt{\frac{\rho}{\dfrac{1}{K} + \dfrac{D}{Et}}}$$

What it tells us. The wall term $D/(Et)$ adds to $1/K$, so the denominator grows and the pressure rise falls. A wide, thin or flexible pipe (large D, small t or small E) gives more relief. For a rigid pipe ($E \to \infty$) the wall term vanishes and the result returns to $p = V\sqrt{\rho K} = \rho VC$.

The effective wave speed is lower than in a rigid pipe, so the pressure rise is smaller. The hoop stress produced in the wall is $\sigma = pD/(2t)$, from the thin-cylinder formula.

8.5 Effects and protection CORE

IN SIMPLE WORDS

Protection works in two ways: stop the water more slowly, or give the surge somewhere to go. A surge tank is an open standpipe where the water level can rise instead of the pressure. An air vessel works like a cushion. A relief valve lets some water out when the pressure gets too high.

Measure	How it helps
Slow valve closure ($T > 2L/C$)	Turns a sudden closure into a gradual one; the rise falls as $1/T$
Surge tank near the turbine or valve	An open standpipe that absorbs the column's momentum; the water level oscillates instead of the pressure rising
Air vessel	A trapped air cushion compresses and absorbs the shock; used on pump delivery lines
Pressure relief valve	Opens at a set pressure and releases water
Stronger pipe walls	Accept the rise; checked with $\sigma = pD/(2t)$

Worked examples

Step-by-step method.

1. Convert D and t to metres, and K and E to $\mathrm{N/m^2}$ (Pa).
2. Find the wave speed $C = \sqrt{K/\rho}$.
3. Find the critical time $2L/C$ and compare it with the closing time T.
4. If $T > 2L/C$ (gradual), use $p = \rho LV/T$.
5. If $T < 2L/C$ (sudden), use $p = \rho VC$ for a rigid pipe, or the elastic-pipe formula.
6. Convert to head with $h = p/(\rho g)$, and find the hoop stress $\sigma = pD/(2t)$ if asked.

Example 8.0 (warm-up) CORE Water flows at 1.5 m/s in a pipe 1000 m long. The valve at its end is closed uniformly in 20 s. Take $C = 1449\ \mathrm{m/s}$. Find the pressure rise.

1. $\dfrac{2L}{C} = \dfrac{2 \times 1000}{1449} = 1.38\ \mathrm{s}$; $T = 20\ \mathrm{s} > 1.38\ \mathrm{s}$, so the closure is **gradual**
2. $p = \dfrac{\rho LV}{T} = \dfrac{1000 \times 1000 \times 1.5}{20} = 75\ \mathrm{kPa}$
3. $h = \dfrac{p}{\rho g} = \dfrac{75\,000}{1000 \times 9.81} = 7.65\ \mathrm{m\ of\ water}$

Example 8.1 HOTS Water flows at 2 m/s through a pipe 2000 m long and 500 mm in diameter. The valve at its end is closed in 30 s. Take $K = 2.1\ \mathrm{GPa}$. Is the closure gradual or sudden? Find the pressure rise.

1. $C = \sqrt{\dfrac{2.1 \times 10^9}{1000}} = 1449\ \mathrm{m/s}$; critical time $\dfrac{2L}{C} = \dfrac{4000}{1449} = 2.76\ \mathrm{s}$
2. $T = 30\ \mathrm{s} > 2.76\ \mathrm{s}$, so the closure is **gradual**
3. $p = \dfrac{\rho LV}{T} = \dfrac{1000 \times 2000 \times 2}{30} = 133.3\ \mathrm{kPa}$ (13.6 m of water)

Example 8.2 HOTS The valve of Example 8.1 is closed suddenly. Find the pressure rise (a) if the pipe is rigid, and (b) if the pipe is steel ($E = 200\ \mathrm{GPa}$) with a 10 mm wall. Find also the hoop stress in case (b).

1. (a) $p = \rho VC = 1000 \times 2 \times 1449 = 2.90\ \mathrm{MPa}$ (295 m of water)
2. (b) $\dfrac{1}{K} + \dfrac{D}{Et} = \dfrac{1}{2.1 \times 10^9} + \dfrac{0.5}{200 \times 10^9 \times 0.01} = 7.262 \times 10^{-10}\ \mathrm{m^2/N}$
3. $p = 2 \times \sqrt{\dfrac{1000}{7.262 \times 10^{-10}}} = 2.35\ \mathrm{MPa}$ (239 m of water); the effective wave speed is 1173 m/s
4. Hoop stress $\sigma = \dfrac{pD}{2t} = \dfrac{2.35 \times 0.5}{0.02} = 58.7\ \mathrm{MPa}$

   Stopping a 2 m/s flow suddenly in this steel pipe raises the pressure about 18 times more than closing the valve over 30 s. Slow closure is the cheapest protection.

Common mistakes and exam tips.

- Always decide first whether the closure is gradual or sudden, by comparing T with $2L/C$.
- Use K in pascals: $2.1\ \text{GPa} = 2.1 \times 10^9\ \text{Pa}$.
- In the elastic-pipe formula, $D/(Et)$ uses the diameter, not the radius, with t in metres.
- The hoop stress is $\sigma = pD/(2t)$; do not drop the 2.
- State whether the answer is in Pa or in m of water; $h = p/(\rho g)$ converts between them.

Check your understanding.

1. CORE What is the critical time of closure? *(The round-trip time $2L/C$ of the pressure wave.)*
2. HOTS If the velocity stopped is doubled, how does the sudden-closure pressure rise change? *(It doubles, since $p = \rho VC$.)*
3. HOTS Why does an elastic pipe give a smaller pressure rise than a rigid one? *(The stretching wall stores part of the energy and lowers the wave speed.)*

9. Boundary layer: concept, thicknesses and the laminar boundary layer (L6) CORE

When a fluid of small viscosity, such as air or water, flows past a body, viscosity matters only in a thin layer next to the surface. In this layer the velocity rises from zero at the wall (no slip) to nearly the free-stream value U. Prandtl (1904) called it the **boundary layer**. Outside it the flow can be treated as inviscid. On a flat plate a laminar layer grows as $\sqrt{x}$, as $\delta \approx 5x/\sqrt{\text{Re}_x}$.

IN SIMPLE WORDS

Put your hand out of a moving car window. The air touching your skin moves with your hand, but a little way off it rushes past. The thin zone where the speed changes from zero to the full stream speed is the boundary layer. Engineers care because the friction drag on ships, aircraft and plates comes from this layer. U is the free-stream speed, x is the distance from the leading edge and δ is the layer thickness. $\text{Re}_x = Ux/\nu$ tells whether the layer is laminar or turbulent at that point.

9.1 Development over a flat plate CORE

IN SIMPLE WORDS

At the leading edge the layer starts with zero thickness and grows downstream, like a carpet of slow fluid getting thicker. At first it is smooth (laminar). Once Re_x reaches about 5×10^5, it becomes turbulent and grows faster. Even then, a very thin smooth film stays next to the wall.

Consider a uniform stream U meeting a thin flat plate aligned with the flow:

- **Laminar boundary layer.** From the leading edge, the layer is laminar and grows as $\sqrt{x}$. Fluid moves in smooth layers.
- **Transition.** At a distance where $\mathrm{Re}_x = Ux/\nu$ reaches about 5×10^5 (the critical Reynolds number), the layer becomes unstable and turns turbulent. A rough surface, a sharp leading edge or a disturbed free stream brings transition earlier.
- **Turbulent boundary layer.** The layer thickens much faster, as $x^{0.8}$, and the velocity profile becomes fuller.
- **Laminar sublayer.** Even under a turbulent layer, a very thin layer next to the wall remains laminar (Section 10).

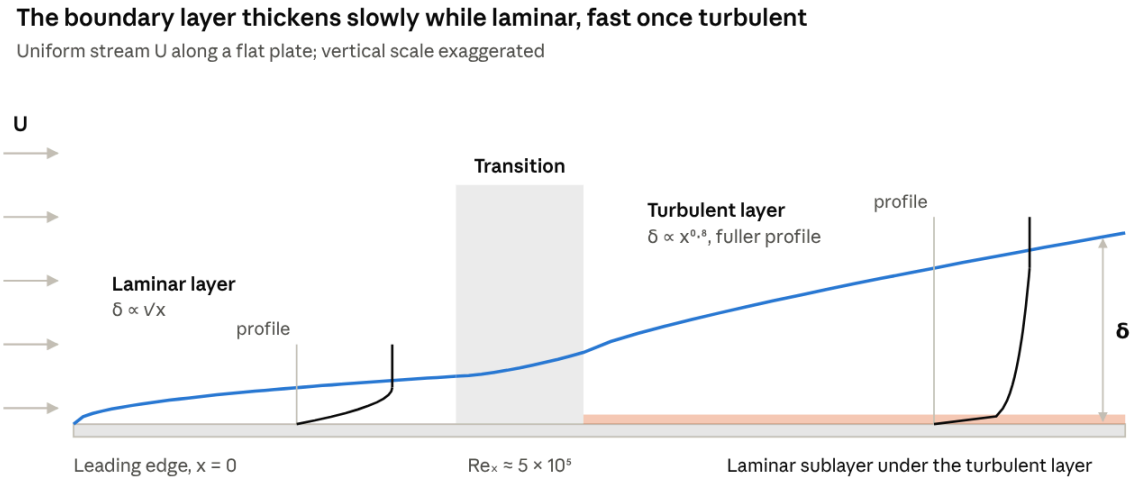


Figure 4. Boundary layer development along a flat plate

Beyond transition the layer thickens faster and its profile fills out, while a thin laminar sublayer stays next to the wall.

9.2 Boundary layer thicknesses CORE

IN SIMPLE WORDS

The boundary layer has no sharp edge, so we define thicknesses that are easy to calculate. δ is where the speed reaches 99 per cent of U. δ^* tells how far the slow layer pushes the outer flow away from the wall. θ measures the momentum lost to friction. The ratio $H = \delta^*/\theta$ hints whether the layer is healthy or close to separating.

Thickness	Definition	Physical meaning
Boundary layer thickness δ	Distance from the wall where $u = 0.99U$	Edge of the region slowed by viscosity
Displacement thickness δ^*	$\int_0^\delta \left(1 - u/U\right) dy$	Distance the wall would have to be moved outward for an inviscid flow to carry the same discharge
Momentum thickness θ	$\int_0^\delta \left(u/U\right)\left(1 - u/U\right) dy$	Thickness of a layer of free-stream fluid whose momentum equals the momentum lost in the boundary layer
Energy thickness δ^{**}	$\int_0^\delta \left(u/U\right)\left(1 - u^2/U^2\right) dy$	Same idea for kinetic energy

Always $\delta > \delta^* > \delta^{**} > \theta$. The ratio $H = \delta^*/\theta$ is the **shape factor**. It is about 2.6 for a laminar layer and 1.3 to 1.4 for a turbulent one. A rising H warns that the layer is approaching separation.

Derivation from scratch: displacement, momentum and energy thicknesses

Aim. To obtain the formulas for δ^*, θ and δ^{**} from the deficits of mass, momentum and kinetic energy in the boundary layer.

Assumptions.

1. Steady, two-dimensional, incompressible flow (ρ constant) over a plate; we work per unit width.
2. At a section x, the velocity rises from $u = 0$ at the wall to $u = U$ at $y = \delta$.
3. Without viscosity the velocity would be U right down to the wall.

Step 1. Mass deficit. Take a thin strip of height dy at height y. The mass flow through it is $\rho u\, dy$. With no viscosity it would be $\rho U\, dy$. Add the losses over the whole layer:

$$\text{mass deficit} = \int_0^\delta \rho U\, dy - \int_0^\delta \rho u\, dy = \int_0^\delta \rho\, \left(U - u\right)\, dy$$

Step 2. Define δ^*. δ^* is the thickness of a layer of free-stream fluid (velocity U) that would carry this same mass. That layer carries $\rho U \delta^*$.

$$\rho U \delta^* = \int_0^\delta \rho\, \left(U - u\right)\, dy$$

Divide both sides by ρU (both are constants):

$$\delta^* = \int_0^\delta \left(1 - \frac{u}{U}\right) dy$$

Step 3. Momentum deficit. Follow the fluid that actually passes through the strip; its mass flow is $\rho u\,dy$. If it moved at U it would carry momentum $(\rho u\,dy)\,U$. It really carries $(\rho u\,dy)\,u$. The loss over the whole layer is:

$$\text{momentum deficit} = \int_0^\delta \rho u\,(U-u)\,dy$$

Step 4. Define θ. θ is the thickness of a layer of free-stream fluid whose momentum flow equals this deficit. That layer has mass flow $\rho U\theta$ and velocity U, so momentum flow $\rho U^2\theta$.

$$\rho U^2\theta = \int_0^\delta \rho u\,(U-u)\,dy$$

Divide both sides by ρU^2:

$$\theta = \int_0^\delta \frac{u}{U}\left(1-\frac{u}{U}\right)dy$$

Step 5. Kinetic energy deficit. The same fluid, mass flow $\rho u\,dy$, would carry kinetic energy $\frac{1}{2}(\rho u\,dy)\,U^2$ at speed U. It really carries $\frac{1}{2}(\rho u\,dy)\,u^2$.

$$\text{energy deficit} = \int_0^\delta \frac{1}{2}\rho u\left(U^2-u^2\right)dy$$

Step 6. Define δ^{**}. A free-stream layer of thickness δ^{**} carries kinetic energy $\frac{1}{2}\left(\rho U\delta^{**}\right)U^2 = \frac{1}{2}\rho U^3\delta^{**}$. Equate and divide by $\frac{1}{2}\rho U^3$:

$$\frac{1}{2}\rho U^3\delta^{**} = \int_0^\delta \frac{1}{2}\rho u\left(U^2-u^2\right)dy \quad\Rightarrow\quad \delta^{**} = \int_0^\delta \frac{u}{U}\left(1-\frac{u^2}{U^2}\right)dy$$

Result.

$$\delta^* = \int_0^\delta \left(1-\frac{u}{U}\right)dy, \qquad \theta = \int_0^\delta \frac{u}{U}\left(1-\frac{u}{U}\right)dy, \qquad \delta^{**} = \int_0^\delta \frac{u}{U}\left(1-\frac{u^2}{U^2}\right)dy$$

What it tells us. δ^* is how far the slow layer pushes the outer flow away from the wall. θ measures the momentum taken away by wall friction, and it is the thickness used in the momentum integral equation (Section 9.4). All three are smaller than δ, because the fluid near the edge of the layer is already moving almost at U.

9.3 Drag on a flat plate CORE

IN SIMPLE WORDS

The fluid rubs on the plate and drags it along; this rubbing force is skin friction drag. τ_0 is the shear stress at the wall, b is the plate width and L is the plate length. Adding up τ_0 over the whole plate gives the drag force F_D. The coefficients C_{fx} and C_D express the stress or force as a fraction of the dynamic pressure $\frac{1}{2}\rho U^2$.

Derivation from scratch: drag force on a flat plate and the drag coefficients

Aim. To find the total friction drag on one side of a flat plate from the wall shear stress, and to define the local and average coefficients.

Assumptions.

1. A flat plate of length L (along the flow) and width b lies parallel to a uniform stream U, so $dp/dx = 0$ and there is no pressure drag.
2. Only one side of the plate is considered.
3. The wall shear stress τ_0 varies with x but not across the width.

Step 1. Shear stress at the wall. By Newton's law of viscosity, applied at the wall:

$$\tau_0 = \mu \left(\frac{\partial u}{\partial y} \right)_{y=0}$$

The fluid above moves forward, so it drags the plate forward (in the flow direction) with this stress.

Step 2. Force on a small strip. Take a strip of the plate between x and $x + dx$. Its area is $b\,dx$. The shear force on it is stress $\times$ area:

$$dF_D = \tau_0 \times b\,dx$$

Step 3. Add the strips from the leading edge to the trailing edge.

$$F_D = \int_0^L \tau_0\, b\, dx$$

Step 4. Local skin-friction coefficient. Compare the local wall stress with the dynamic pressure $\frac{1}{2}\rho U^2$ of the free stream:

$$C_{fx} = \frac{\tau_0}{\frac{1}{2}\rho U^2}$$

Step 5. Average drag coefficient. Compare the total force with the dynamic pressure times the plate area bL:

$$C_D = \frac{F_D}{\frac{1}{2}\rho U^2\, bL}$$

Step 6. Link between the two. Put $\tau_0 = C_{fx}\,\frac{1}{2}\rho U^2$ into Step 3. The constants $\frac{1}{2}\rho U^2$ and b come out of the integral:

$$F_D = \frac{1}{2}\rho U^2\, b \int_0^L C_{fx}\, dx \quad \Rightarrow \quad C_D = \frac{1}{L}\int_0^L C_{fx}\, dx$$

Result.

$$F_D = \int_0^L \tau_0\, b\, dx, \qquad C_{fx} = \frac{\tau_0}{\frac{1}{2}\rho U^2} \ \text{(local)}, \qquad C_D = \frac{F_D}{\frac{1}{2}\rho U^2\, bL} \ \text{(average)}$$

What it tells us. The drag is the sum of the wall shear over the whole wetted area. C_D is simply the average of C_{fx} along the plate. If both sides are wetted, the drag is doubled.

9.4 Von Kármán momentum integral equation (application of the momentum equation) CORE

IN SIMPLE WORDS

This is Newton's second law applied to a thin slice of the boundary layer. Wall friction slows the fluid, so the layer keeps losing momentum along the plate. The equation says that the wall shear stress equals how fast the momentum thickness θ grows. Guess a velocity shape, and the equation tells how δ grows with x.

Derivation from scratch: von Kármán momentum integral equation

Aim. To prove, by applying the momentum equation to a control volume in the boundary layer, that $\tau_0 / \left(\rho U^2 \right) = d\theta/dx$.

Assumptions.

1. Steady, two-dimensional, incompressible flow (ρ constant) over a flat plate; we work per unit width.
2. The free-stream velocity U is constant, so there is no pressure gradient: $dp/dx = 0$.
3. The boundary layer has thickness δ at x and $\delta + d\delta$ at $x + dx$.

The control volume. Take a thin control volume ABCD of length dx and unit width. AB is the upstream vertical face at x, from the wall (A) up to the edge of the layer (B, at $y = \delta$). DC is the downstream vertical face at $x + dx$, from the wall (D) up to C at $y = \delta + d\delta$. BC runs along the edge of the boundary layer, where the velocity is U. AD lies on the plate.

Step 1. Mass through the upstream face AB (entering).

$$\dot{m}_{AB} = \int_0^{\delta} \rho u \, dy$$

Step 2. Mass through the downstream face CD (leaving). It is the mass through AB plus its change over the distance dx:

$$\dot{m}_{CD} = \int_0^{\delta} \rho u \, dy + \frac{d}{dx}\left(\int_0^{\delta} \rho u \, dy \right) dx$$

Step 3. Mass through the wall AD. The plate is solid, so no fluid crosses it: $\dot{m}_{AD} = 0$.

Step 4. Mass through the top face BC, from continuity. In steady flow, mass in = mass out: $\dot{m}_{AB} + \dot{m}_{BC} = \dot{m}_{CD}$. Hence:

$$\dot{m}_{BC} = \dot{m}_{CD} - \dot{m}_{AB} = \frac{d}{dx}\left(\int_0^{\delta} \rho u \, dy \right) dx$$

The layer carries more fluid at $x + dx$ than at x, so this mass enters the control volume through BC. It comes from the free stream outside the boundary layer, which crosses into the growing layer with velocity U in the x-direction.

Step 5. Momentum flux (x-direction) through each face. Each face carries mass flow $\times$ x-velocity.

- Through AB (in): $\int_0^\delta \rho u^2\, dy$
- Through CD (out): $\int_0^\delta \rho u^2\, dy + \frac{d}{dx}\left(\int_0^\delta \rho u^2\, dy \right) dx$
- Through BC (in): $U \times \dot{m}_{BC} = U \dfrac{d}{dx}\left(\int_0^\delta \rho u\, dy \right) dx$
- Through AD: zero.

Step 6. Net rate of momentum outflow. Out minus in:

$$\left[\int_0^\delta \rho u^2\, dy + \frac{d}{dx}\left(\int_0^\delta \rho u^2\, dy \right) dx \right] - \int_0^\delta \rho u^2\, dy - U \frac{d}{dx}\left(\int_0^\delta \rho u\, dy \right) dx$$

The first and second integrals cancel, leaving:

$$\text{net outflow} = \frac{d}{dx}\left(\int_0^\delta \rho u^2\, dy \right) dx - U \frac{d}{dx}\left(\int_0^\delta \rho u\, dy \right) dx$$

Step 7. Forces on the fluid in the x-direction.

- Pressure: since $dp/dx = 0$, the pressure p is the same everywhere. Its push on AB ($p\delta$, forward) is balanced by its push on CD ($p\ (\ \delta + d\delta\ )$, backward) together with the forward push $p\, d\delta$ on the sloping face BC. Net pressure force: $p\delta - p\ (\ \delta + d\delta\ ) + p\, d\delta = 0$.
- Shear on BC: at the edge of the layer $\partial u/\partial y = 0$, so there is no shear there.
- Shear on AD: the wall holds the fluid back with force $\tau_0 \times dx \times 1$, acting in the $-x$ direction.

$$\sum F_x = -\tau_0\, dx$$

Step 8. Momentum equation: net force = net rate of momentum outflow.

$$-\tau_0\, dx = \frac{d}{dx}\left(\int_0^\delta \rho u^2\, dy \right) dx - U \frac{d}{dx}\left(\int_0^\delta \rho u\, dy \right) dx$$

Step 9. Cancel dx and change the sign of both sides.

$$\tau_0 = U \frac{d}{dx} \int_0^\delta \rho u\, dy - \frac{d}{dx} \int_0^\delta \rho u^2\, dy$$

Step 10. Combine into one integral. U is constant, so it can go inside both the derivative and the integral.

$$\tau_0 = \frac{d}{dx} \int_0^\delta \rho\ (\ Uu - u^2\ )\, dy = \rho \frac{d}{dx} \int_0^\delta\ (\ Uu - u^2\ )\, dy$$

Step 11. Take out U^2. Write $Uu - u^2 = U^2 \dfrac{u}{U}\left(1 - \dfrac{u}{U}\right)$. Since U^2 is constant:

$$\tau_0 = \rho U^2 \frac{d}{dx} \int_0^\delta \frac{u}{U}\left(1 - \frac{u}{U}\right) dy$$

Step 12. Recognise the momentum thickness. The integral is exactly θ (Section 9.2). So $\tau_0 = \rho U^2 \, d\theta/dx$; divide by ρU^2.

Result.

$$\tau_0 = \rho \frac{d}{dx} \int_0^\delta \left(Uu - u^2 \right) dy = \rho U^2 \frac{d}{dx} \int_0^\delta \frac{u}{U}\left(1 - \frac{u}{U}\right) dy \quad \Rightarrow \quad \frac{\tau_0}{\rho U^2} = \frac{d\theta}{dx}$$

What it tells us. The wall shear stress equals the rate at which the momentum thickness grows along the plate. Friction keeps taking momentum out of the layer, so θ (and δ) must grow with x. No velocity profile was assumed, so the equation holds for laminar and turbulent layers alike.

This is **von Kármán's momentum integral equation**. It holds for laminar and turbulent layers alike. Its use is to assume a reasonable velocity profile $u/U = F\left(\, y/\delta \,\right)$, from which both θ and τ_0 follow in terms of δ. The equation then becomes a simple differential equation for $\delta\left(\, x \,\right)$.

9.5 Solution for a cubic profile (worked derivation) CORE

IN SIMPLE WORDS

Here the momentum integral is used with one assumed shape, a cubic curve. The steps are always the same. Find θ from the shape and τ_0 from the slope at the wall. Then put both into the equation and integrate. The answer shows that δ grows as $\sqrt{x}$, and the constant 4.64 is within about 6 per cent of the exact 4.91.

Derivation from scratch: laminar boundary layer with a cubic velocity profile

Aim. To find δ, τ_0, C_{fx} and C_D for a laminar flat-plate boundary layer, using the momentum integral equation with the cubic profile.

Assumptions.

1. Laminar, steady, incompressible flow over a flat plate with $dp/dx = 0$; the momentum integral $\tau_0 = \rho U^2 \, d\theta/dx$ applies.
2. The velocity profile is $u/U = F\left(\, \eta \,\right) = \frac{3}{2}\eta - \frac{1}{2}\eta^3$, with $\eta = y/\delta$.
3. The layer starts at the leading edge: $\delta = 0$ at $x = 0$. Write $\nu = \mu/\rho$ and $\mathrm{Re}_x = Ux/\nu$.

Step 1. Check that the profile is sensible. Differentiate: $dF/d\eta = \frac{3}{2} - \frac{3}{2}\eta^2$ and $d^2F/d\eta^2 = -3\eta$.

- At the wall, $\eta = 0$: $F = 0$, so $u = 0$ (no slip); and $d^2F/d\eta^2 = 0$, so $d^2u/dy^2 = 0$.
- At the edge, $\eta = 1$: $F = \frac{3}{2} - \frac{1}{2} = 1$, so $u = U$; and $dF/d\eta = \frac{3}{2} - \frac{3}{2} = 0$, so $du/dy = 0$.

Step 2. Change the variable in θ. Put $y = \delta\eta$, so $dy = \delta\, d\eta$. When y goes from 0 to δ, η goes from 0 to 1.

$$\theta = \int_0^{\delta} \frac{u}{U}\left(1 - \frac{u}{U}\right) dy = \delta \int_0^1 F\,(\,1 - F\,)\; d\eta = \delta \int_0^1 \left(\,F - F^2\,\right) d\eta$$

Step 3. Expand F^2.

$$F^2 = \left(\frac{3}{2}\eta - \frac{1}{2}\eta^3\right)^2 = \frac{9}{4}\eta^2 - 2 \times \frac{3}{2} \times \frac{1}{2}\,\eta^4 + \frac{1}{4}\eta^6 = \frac{9}{4}\eta^2 - \frac{3}{2}\eta^4 + \frac{1}{4}\eta^6$$

Step 4. Write $F - F^2$ term by term.

$$F - F^2 = \frac{3}{2}\eta - \frac{9}{4}\eta^2 - \frac{1}{2}\eta^3 + \frac{3}{2}\eta^4 - \frac{1}{4}\eta^6$$

Step 5. Integrate term by term from 0 to 1. Use $\int_0^1 \eta^n\, d\eta = 1/\,(\,n+1\,)$.

$$\int_0^1 \left(\,F - F^2\,\right) d\eta = \left[\frac{3}{4}\eta^2 - \frac{3}{4}\eta^3 - \frac{1}{8}\eta^4 + \frac{3}{10}\eta^5 - \frac{1}{28}\eta^7\right]_0^1 = \frac{3}{4} - \frac{3}{4} - \frac{1}{8} + \frac{3}{10} - \frac{1}{28}$$

Step 6. Add the fractions. The first two cancel. Use the common denominator 280:

$$-\frac{35}{280} + \frac{84}{280} - \frac{10}{280} = \frac{39}{280} \quad \Rightarrow \quad \theta = \frac{39}{280}\,\delta$$

In the same way, $\delta^* = \delta \int_0^1 \,(\,1 - F\,)\; d\eta = \delta\left[\eta - \frac{3}{4}\eta^2 + \frac{1}{8}\eta^4\right]_0^1 = \delta\left(\,1 - \frac{3}{4} + \frac{1}{8}\right) = \frac{3}{8}\,\delta = 0.375\,\delta$.

Step 7. Wall shear stress. Write $u = U\left(\frac{3y}{2\delta} - \frac{y^3}{2\delta^3}\right)$ and differentiate with respect to y:

$$\frac{du}{dy} = U\left(\frac{3}{2\delta} - \frac{3y^2}{2\delta^3}\right) \quad \Rightarrow \quad \left(\frac{du}{dy}\right)_{\,y\,=\,0} = \frac{3U}{2\delta}$$

$$\tau_0 = \mu\left(\frac{du}{dy}\right)_{\,y\,=\,0} = \frac{3\mu U}{2\delta}$$

Step 8. Put τ_0 and θ into the momentum integral. Since $39/280$ is a constant, $d\theta/dx = \,(\,39/280\,)\; d\delta/dx$.

$$\frac{3\mu U}{2\delta} = \rho U^2 \left(\frac{39}{280}\right)\frac{d\delta}{dx}$$

Step 9. Separate the variables. Multiply both sides by $\delta\, dx$ and by $280/\left(\,39\rho U^2\,\right)$:

$$\delta\, d\delta = \frac{3 \times 280}{2 \times 39}\,\frac{\mu U}{\rho U^2}\, dx = \frac{840}{78}\,\frac{\mu}{\rho U}\, dx = \frac{140}{13}\,\frac{\nu}{U}\, dx$$

Step 10. Integrate both sides.

$$\frac{\delta^2}{2} = \frac{140}{13}\,\frac{\nu x}{U} + C_1$$

Step 11. Boundary condition. At $x = 0$, $\delta = 0$, so $0 = 0 + C_1$ and $C_1 = 0$.

$$\frac{\delta^2}{2} = \frac{140}{13}\frac{\nu x}{U} \quad \Rightarrow \quad \delta^2 = \frac{280}{13}\frac{\nu x}{U} = 21.54\,\frac{\nu x}{U}$$

Step 12. Thickness. Take the square root ($\sqrt{21.54} = 4.64$) and divide by x:

$$\delta = 4.64\sqrt{\frac{\nu x}{U}} \quad \Rightarrow \quad \frac{\delta}{x} = 4.64\sqrt{\frac{\nu}{Ux}} = \frac{4.64}{\sqrt{\mathrm{Re}_x}}$$

Step 13. Wall shear in terms of Re_x. Put $\delta = 4.64x/\sqrt{\mathrm{Re}_x}$ into Step 7:

$$\tau_0 = \frac{3\mu U}{2 \times 4.64\,x}\sqrt{\mathrm{Re}_x} = 0.323\,\frac{\mu U}{x}\sqrt{\mathrm{Re}_x}$$

Now $\dfrac{\mu U}{x} = \rho U^2\,\dfrac{\mu}{\rho U x} = \dfrac{\rho U^2}{\mathrm{Re}_x}$, and $\dfrac{\sqrt{\mathrm{Re}_x}}{\mathrm{Re}_x} = \dfrac{1}{\sqrt{\mathrm{Re}_x}}$. So:

$$\tau_0 = \frac{0.323\,\rho U^2}{\sqrt{\mathrm{Re}_x}}$$

Step 14. Local skin-friction coefficient.

$$C_{fx} = \frac{\tau_0}{\frac{1}{2}\rho U^2} = \frac{2 \times 0.323}{\sqrt{\mathrm{Re}_x}} = \frac{0.646}{\sqrt{\mathrm{Re}_x}}$$

Step 15. Drag on one side and C_D. Write $\tau_0 = 0.323\,\rho U^2\sqrt{\nu/U}\;x^{-1/2}$ and integrate over the plate of length L and width b:

$$F_D = \int_0^L \tau_0\, b\, dx = 0.323\,\rho U^2 b\sqrt{\frac{\nu}{U}}\int_0^L x^{-1/2}\,dx$$

$$= 0.323\,\rho U^2 b\sqrt{\frac{\nu}{U}}\,\left[2x^{1/2}\right]_0^L = 0.646\,\rho U^2 b\sqrt{\frac{\nu L}{U}}$$

$$C_D = \frac{F_D}{\frac{1}{2}\rho U^2\, bL} = 2 \times 0.646\,\sqrt{\frac{\nu}{UL}} = \frac{1.292}{\sqrt{\mathrm{Re}_L}}$$

Result.

$$\frac{\delta}{x} = \frac{4.64}{\sqrt{\mathrm{Re}_x}}, \qquad \tau_0 = \frac{0.323\,\rho U^2}{\sqrt{\mathrm{Re}_x}}, \qquad C_{fx} = \frac{0.646}{\sqrt{\mathrm{Re}_x}}, \qquad C_D = \frac{1.292}{\sqrt{\mathrm{Re}_L}}$$

What it tells us. The layer grows as $\sqrt{x}$ and is thinner in a faster stream ($\delta \propto 1/\sqrt{U}$). The wall shear is largest near the leading edge and falls as $1/\sqrt{x}$. The constant 4.64 is within about 6 per cent of the exact Blasius value 4.91.

The average coefficient is twice the local value at the trailing edge, because τ_0 falls as $1/\sqrt{x}$.

AKTU also asks the same method with a parabolic profile. The steps are identical; only the integrals change.

Derivation from scratch: laminar boundary layer with a parabolic velocity profile

Aim. To find δ/x, C_{fx} and C_D for a laminar flat-plate boundary layer with the profile $u/U = 2\,(\,y/\delta\,) \,-\, (\,y/\delta\,)^{\,2}$.

Assumptions.

1. Laminar, steady, incompressible flow over a flat plate with $dp/dx = 0$; the momentum integral $\tau_0 = \rho U^2\, d\theta/dx$ applies.
2. The profile is $u/U = F(\,\eta\,) = 2\eta - \eta^2$, with $\eta = y/\delta$.
3. $\delta = 0$ at $x = 0$; $\nu = \mu/\rho$ and $\mathrm{Re}_x = Ux/\nu$.

Step 1. Check the profile. At $\eta = 0$: $F = 0$ (no slip). At $\eta = 1$: $F = 2 - 1 = 1$, so $u = U$. Also $dF/d\eta = 2 - 2\eta$, which is 0 at $\eta = 1$, so the profile joins the free stream smoothly.

Step 2. Expand $F - F^2$.

$$F^2 = \left(\,2\eta - \eta^2\,\right)^{\,2} = 4\eta^2 - 4\eta^3 + \eta^4$$

$$F - F^2 = 2\eta - \eta^2 - 4\eta^2 + 4\eta^3 - \eta^4 = 2\eta - 5\eta^2 + 4\eta^3 - \eta^4$$

Step 3. Momentum thickness. With $dy = \delta\, d\eta$ and limits 0 to 1:

$$\theta = \delta \int_0^1 \left(\,2\eta - 5\eta^2 + 4\eta^3 - \eta^4\,\right) d\eta = \delta\left[\eta^2 - \frac{5}{3}\eta^3 + \eta^4 - \frac{1}{5}\eta^5\right]_0^1$$

$$= \delta\left(1 - \frac{5}{3} + 1 - \frac{1}{5}\right)$$

$$1 - \frac{5}{3} + 1 - \frac{1}{5} = \frac{15 - 25 + 15 - 3}{15} = \frac{2}{15} \quad \Rightarrow \quad \theta = \frac{2}{15}\,\delta$$

Step 4. Wall shear stress. $u = U\left(\frac{2y}{\delta} - \frac{y^2}{\delta^2}\right)$, so $\frac{du}{dy} = U\left(\frac{2}{\delta} - \frac{2y}{\delta^2}\right)$. At $y = 0$:

$$\tau_0 = \mu\left(\frac{du}{dy}\right)_{y\,=\,0} = \frac{2\mu U}{\delta}$$

Step 5. Momentum integral.

$$\frac{2\mu U}{\delta} = \rho U^2 \left(\frac{2}{15}\right)\frac{d\delta}{dx}$$

Step 6. Separate the variables. Multiply both sides by $\delta\, dx$ and by $15/\left(\,2\rho U^2\,\right)$:

$$\delta\, d\delta = \frac{2 \times 15}{2}\,\frac{\mu U}{\rho U^2}\, dx = 15\,\frac{\nu}{U}\, dx$$

Step 7. Integrate and apply $\delta = 0$ at $x = 0$.

$$\frac{\delta^2}{2} = 15\,\frac{\nu x}{U} + C_1, \qquad C_1 = 0 \quad \Rightarrow \quad \delta^2 = 30\,\frac{\nu x}{U}$$

Step 8. Thickness. $\sqrt{30} = 5.48$, so:

$$\delta = 5.48\sqrt{\frac{\nu x}{U}} \quad \Rightarrow \quad \frac{\delta}{x} = \frac{5.48}{\sqrt{\mathrm{Re}_x}}$$

Step 9. Wall shear and C_{fx}. Put δ into Step 4 and use $\mu U/x = \rho U^2/\mathrm{Re}_x$:

$$\tau_0 = \frac{2\mu U}{5.48\,x}\sqrt{\mathrm{Re}_x} = \frac{2}{5.48}\frac{\rho U^2}{\sqrt{\mathrm{Re}_x}} = \frac{0.365\,\rho U^2}{\sqrt{\mathrm{Re}_x}}$$

$$C_{fx} = \frac{\tau_0}{\frac{1}{2}\rho U^2} = \frac{2 \times 0.365}{\sqrt{\mathrm{Re}_x}} = \frac{0.730}{\sqrt{\mathrm{Re}_x}}$$

Step 10. Average coefficient. As for the cubic profile, $\tau_0 \propto x^{-1/2}$ and $\int_0^L x^{-1/2}\,dx = 2\sqrt{L}$, so the average is twice the trailing-edge value:

$$C_D = \frac{1}{L}\int_0^L C_{fx}\,dx = \frac{0.730}{L}\sqrt{\frac{\nu}{U}}\,[2x^{1/2}]_0^L = \frac{2 \times 0.730}{\sqrt{\mathrm{Re}_L}} = \frac{1.46}{\sqrt{\mathrm{Re}_L}}$$

Result.

$$\frac{\delta}{x} = \frac{5.48}{\sqrt{\mathrm{Re}_x}}, \qquad C_{fx} = \frac{0.730}{\sqrt{\mathrm{Re}_x}}, \qquad C_D = \frac{1.46}{\sqrt{\mathrm{Re}_L}}$$

What it tells us. The parabolic profile gives the same laws as the cubic one ($\delta \propto \sqrt{x}$, $\tau_0 \propto 1/\sqrt{x}$), with different constants. It overestimates δ by about 12 per cent compared with the exact 4.91, so the cubic profile is the better guess.

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

9.6 Results for other profiles, and Blasius's exact solution ADVANCED

Blasius (1908) solved the boundary layer equations exactly for a flat plate with zero pressure gradient. The momentum integral results with assumed profiles come within a few per cent of it:

Velocity profile u/U ($\eta = y/\delta$)	$(\delta/x)\sqrt{\mathrm{Re}_x}$	δ^*/δ	θ/δ	$C_{fx}\sqrt{\mathrm{Re}_x}$	$C_D\sqrt{\mathrm{Re}_L}$
Linear: η	3.46	0.500	0.167	0.577	1.155
Parabolic: $2\eta - \eta^2$	5.48	0.333	0.133	0.730	1.460
Cubic: $\frac{3}{2}\eta - \frac{1}{2}\eta^3$	4.64	0.375	0.139	0.646	1.292
Sinusoidal: $\sin(\pi\eta/2)$	4.79	0.363	0.137	0.654	1.310
Blasius (exact)	**4.91** (5.0 is often used)	$\delta^* = 1.72x/\sqrt{\mathrm{Re}_x}$	$\theta = 0.664x/\sqrt{\mathrm{Re}_x}$	**0.664**	**1.328**

The laminar results hold only up to $\mathrm{Re}_x \approx 5\times10^5$.

Worked examples

Step-by-step method.

1. Write U, x (or L) and ν (or μ/ρ) in SI units.
2. Find $\mathrm{Re}_x = Ux/\nu$ (or Re_L) and check that it is below 5×10^5 (laminar).
3. Pick the constants for the profile asked: Blasius 4.91, 0.664, 1.328; cubic 4.64, 0.646, 1.292.
4. Find $\delta = (\text{constant}) \times x/\sqrt{\mathrm{Re}_x}$.
5. Find $\tau_0 = C_{fx} \times \frac{1}{2}\rho U^2$.
6. Find the drag $F_D = C_D \times \frac{1}{2}\rho U^2 \times bL$, and double it if both sides are wetted.

Example 9.0 (warm-up) CORE Air ($\nu = 1.5\times10^{-5}\ \mathrm{m^2/s}$) flows at 2 m/s over a flat plate. Using the Blasius result, find the Reynolds number and the boundary layer thickness 0.3 m from the leading edge.

1. $\mathrm{Re}_x = \dfrac{Ux}{\nu} = \dfrac{2\times0.3}{1.5\times10^{-5}} = 40\,000 < 5\times10^5$, so the layer is laminar
2. $\delta = \dfrac{4.91x}{\sqrt{\mathrm{Re}_x}} = \dfrac{4.91\times0.3}{\sqrt{40\,000}} = \dfrac{1.473}{200} = 7.37\ \mathrm{mm}$

Example 9.1 CORE For the parabolic profile $u/U = 2(y/\delta) - (y/\delta)^2$, find δ^*, θ and the shape factor in terms of δ.

1. $\delta^* = \delta\int_0^1 (1 - 2\eta + \eta^2)\,d\eta = \delta\left(1 - 1 + \dfrac{1}{3}\right) = \dfrac{\delta}{3}$
2. $\theta = \delta\int_0^1 (2\eta - \eta^2)(1 - 2\eta + \eta^2)\,d\eta$
   $= \delta\int_0^1 (2\eta - 5\eta^2 + 4\eta^3 - \eta^4)\,d\eta = \delta\left(1 - \dfrac{5}{3} + 1 - \dfrac{1}{5}\right) = \dfrac{2\delta}{15}$

3. $H = \dfrac{\delta^*}{\theta} = \dfrac{1/3}{2/15} = 2.5$

Example 9.2 CORE Air ($\rho = 1.2\,\mathrm{kg/m^3}$, $\nu = 1.5 \times 10^{-5}\,\mathrm{m^2/s}$) flows at 3 m/s over a flat plate 1 m long and 0.5 m wide. Using the Blasius solution, find the boundary layer thickness, the displacement and momentum thicknesses at the trailing edge, the wall shear stress there, and the drag on both sides of the plate.

1. $\mathrm{Re}_L = \dfrac{3 \times 1}{1.5 \times 10^{-5}} = 2 \times 10^5 < 5 \times 10^5$, so the layer is laminar over the whole plate
2. $\delta = \dfrac{4.91 \times 1}{\sqrt{2 \times 10^5}} = 10.98\,\mathrm{mm}$; $\delta^* = \dfrac{1.72}{\sqrt{2 \times 10^5}} = 3.85\,\mathrm{mm}$; $\theta = 1.48\,\mathrm{mm}$
3. $C_{fx} = \dfrac{0.664}{\sqrt{2 \times 10^5}} = 1.485 \times 10^{-3}$, so $\tau_0 = 1.485 \times 10^{-3} \times 0.5 \times 1.2 \times 3^2 = 8.02 \times 10^{-3}\,\mathrm{N/m^2}$
4. $C_D = \dfrac{1.328}{\sqrt{2 \times 10^5}} = 2.97 \times 10^{-3}$; drag on one side
   $= C_D \times \frac{1}{2}\rho U^2 \times bL = 2.97 \times 10^{-3} \times 5.4 \times 0.5 = 8.02 \times 10^{-3}\,\mathrm{N}$
5. Drag on both sides $= 0.016\,\mathrm{N}$

Example 9.3 CORE Oil ($\rho = 900\,\mathrm{kg/m^3}$, $\nu = 10^{-4}\,\mathrm{m^2/s}$) flows at 2 m/s over a plate 1.5 m long and 1 m wide. Assuming the cubic profile, find the boundary layer thickness and wall shear stress 1 m from the leading edge, and the total drag on one side.

1. Re_x at 1 m $= \dfrac{2 \times 1}{10^{-4}} = 2 \times 10^4$ (laminar)
2. $\delta = \dfrac{4.64 \times 1}{\sqrt{2 \times 10^4}} = 32.8\,\mathrm{mm}$; $\delta^* = 0.375\delta = 12.3\,\mathrm{mm}$
3. $\tau_0 = \dfrac{0.323 \times 900 \times 2^2}{\sqrt{2 \times 10^4}} = 8.22\,\mathrm{N/m^2}$
4. $\mathrm{Re}_L = 3 \times 10^4$; $C_D = \dfrac{1.292}{\sqrt{3 \times 10^4}} = 7.46 \times 10^{-3}$
5. $F_D = 7.46 \times 10^{-3} \times 0.5 \times 900 \times 2^2 \times 1.5 \times 1 = 20.1\,\mathrm{N}$

Common mistakes and exam tips.

- Use x (distance from the leading edge) for local values and L (plate length) for C_D and total drag.
- Check that $\mathrm{Re}_L < 5 \times 10^5$ before using laminar formulas.
- Do not mix the constants of different profiles, such as Blasius 4.91 with cubic 0.646.
- The drag on both sides is twice the drag on one side.
- Re_x uses the distance x along the plate, not a pipe diameter.

Check your understanding.

1. CORE What is the fluid velocity at the wall? *(Zero, because of the no-slip condition.)*
2. CORE If x becomes four times larger in a laminar layer, how does δ change? *(It doubles, since $\delta \propto \sqrt{x}$.)*

3. HOTS Why is the average C_D larger than the local C_{fx} at the trailing edge? *(The wall shear is higher near the leading edge, where the layer is thin.)*

10. Turbulent boundary layer and laminar sublayer (L7) CORE

Beyond $\mathrm{Re}_x \approx 5 \times 10^5$, the boundary layer on a flat plate is turbulent. It grows faster (as $x^{0.8}$ instead of $x^{0.5}$), has a fuller velocity profile, and exerts a higher wall shear stress. Next to the wall, a laminar sublayer a fraction of a millimetre thick remains. Most of the velocity change across the layer happens inside it.

> **IN SIMPLE WORDS**
> Further along the plate, the smooth layer breaks up into eddies. Think of incense smoke that rises smoothly and then suddenly turns swirly. The eddies mix fast fluid down towards the wall. So the turbulent layer is thicker, has a fuller velocity shape and rubs harder on the wall. Right at the wall, a very thin smooth film remains: the laminar sublayer. Turbulent formulas use $\mathrm{Re}_x^{1/5}$ (fifth root) where laminar ones use the square root.

10.1 Velocity profile and the momentum integral solution CORE

> **IN SIMPLE WORDS**
> Turbulent flow is too irregular to solve exactly, so two measured rules are used. The one-seventh power law gives the shape of the velocity curve. The Blasius shear relation gives the wall friction. Putting both into the momentum integral gives $\delta = 0.37x/\mathrm{Re}_x^{1/5}$.

The exact equations cannot be solved for turbulent flow, so the momentum integral equation is used with empirical relations. Two are needed:

- the **one-seventh power law** for the velocity profile, $u/U = \left(y/\delta \right)^{1/7}$, which gives $\theta = \frac{7}{72}\delta$ and $\delta^* = \delta/8$;
- the **Blasius wall shear relation**, taken from smooth-pipe data: $\tau_0 = 0.0225\rho U^2 \left(\nu/\left(U\delta \right) \right)^{1/4}$.

> **Derivation from scratch: turbulent boundary layer with the one-seventh power law**
>
> **Aim.** To find δ/x, C_{fx} and C_D for a turbulent boundary layer on a flat plate.
>
> **Assumptions.**
>
> 1. Steady flow over a flat plate with $dp/dx = 0$; the momentum integral $\tau_0 = \rho U^2\, d\theta/dx$ applies (it holds for turbulent layers too).
> 2. The velocity profile is $u/U = F\left(\eta\right) = \eta^{1/7}$, with $\eta = y/\delta$.
> 3. The wall shear is given by the Blasius relation $\tau_0 = 0.0225\rho U^2 \left(\nu/\left(U\delta \right) \right)^{1/4}$. (The 1/7 law cannot be used for τ_0, because its slope at the wall is infinite.)
> 4. The layer is turbulent from the leading edge, so $\delta = 0$ at $x = 0$. $\mathrm{Re}_x = Ux/\nu$.

Step 1. Momentum thickness. With $dy = \delta\, d\eta$, and $F\,(1-F) = \eta^{1/7} - \eta^{2/7}$:

$$\theta = \delta \int_0^1 \left(\eta^{1/7} - \eta^{2/7}\right) d\eta = \delta \left[\frac{7}{8}\eta^{8/7} - \frac{7}{9}\eta^{9/7}\right]_0^1 = \delta\left(\frac{7}{8} - \frac{7}{9}\right) = \delta\,\frac{63-56}{72} = \frac{7}{72}\,\delta$$

Step 2. Displacement thickness (for reference).

$$\delta^* = \delta \int_0^1 \left(1 - \eta^{1/7}\right) d\eta = \delta \left[\eta - \frac{7}{8}\eta^{8/7}\right]_0^1 = \delta\left(1 - \frac{7}{8}\right) = \frac{\delta}{8}$$

Step 3. Put τ_0 and θ into the momentum integral.

$$0.0225\,\rho U^2 \left(\frac{\nu}{U\delta}\right)^{1/4} = \rho U^2\,\frac{7}{72}\,\frac{d\delta}{dx}$$

Cancel ρU^2 on both sides:

$$0.0225 \left(\frac{\nu}{U\delta}\right)^{1/4} = \frac{7}{72}\,\frac{d\delta}{dx}$$

Step 4. Separate the variables. Write $(\nu/(U\delta))^{1/4} = (\nu/U)^{1/4}\,\delta^{-1/4}$. Multiply both sides by $\delta^{1/4}\,dx$ and by $72/7$:

$$\delta^{1/4}\,d\delta = \frac{72}{7} \times 0.0225 \left(\frac{\nu}{U}\right)^{1/4} dx = 0.2314 \left(\frac{\nu}{U}\right)^{1/4} dx$$

Step 5. Integrate both sides. $\int \delta^{1/4}\,d\delta = \delta^{5/4}/(5/4) = \frac{4}{5}\delta^{5/4}$.

$$\frac{4}{5}\,\delta^{5/4} = 0.2314 \left(\frac{\nu}{U}\right)^{1/4} x + C_1$$

Step 6. Boundary condition. At $x = 0$, $\delta = 0$, so $C_1 = 0$. Multiply by $5/4$:

$$\delta^{5/4} = 0.2893 \left(\frac{\nu}{U}\right)^{1/4} x$$

Step 7. Raise both sides to the power $4/5$. $(0.2893)^{4/5} = 0.37$, $\left((\nu/U)^{1/4}\right)^{4/5} = (\nu/U)^{1/5}$ and $x^{4/5}$ stays.

$$\delta = 0.37 \left(\frac{\nu}{U}\right)^{1/5} x^{4/5}$$

Step 8. Divide by x. $x^{4/5}/x = x^{-1/5}$, so:

$$\frac{\delta}{x} = 0.37 \left(\frac{\nu}{Ux}\right)^{1/5} = \frac{0.37}{\mathrm{Re}_x^{1/5}}$$

Step 9. Wall shear stress. Put $\delta = 0.37 x\,\mathrm{Re}_x^{-1/5}$ into the Blasius relation:

$$\frac{\nu}{U\delta} = \frac{\nu}{Ux} \times \frac{1}{0.37\,\mathrm{Re}_x^{-1/5}} = \frac{1}{0.37\,\mathrm{Re}_x^{4/5}} \quad \Rightarrow \quad \left(\frac{\nu}{U\delta}\right)^{1/4} = \frac{0.37^{-1/4}}{\mathrm{Re}_x^{1/5}} = \frac{1.282}{\mathrm{Re}_x^{1/5}}$$

$$\tau_0 = 0.0225 \times 1.282\,\frac{\rho U^2}{\mathrm{Re}_x^{1/5}} = \frac{0.0288\,\rho U^2}{\mathrm{Re}_x^{1/5}}$$

Step 10. Local coefficient.

$$C_{fx} = \frac{\tau_0}{\frac{1}{2}\rho U^2} = \frac{2 \times 0.0288}{\mathrm{Re}_x^{1/5}} = \frac{0.0576}{\mathrm{Re}_x^{1/5}}$$

Step 11. Average coefficient. $C_{fx} = 0.0576\ (\nu/U)^{1/5}\, x^{-1/5}$. Average it over the length L:

$$C_D = \frac{1}{L}\int_0^L C_{fx}\, dx = \frac{0.0576}{L}\left(\frac{\nu}{U}\right)^{1/5} \left[\tfrac{5}{4}x^{4/5}\right]_0^L = \frac{5}{4} \times 0.0576 \left(\frac{\nu}{UL}\right)^{1/5} = \frac{0.072}{\mathrm{Re}_L^{1/5}}$$

Measurements on flat plates fit slightly better with the constant 0.074 in place of 0.072, so 0.074 is the value used in practice.

Result.

$$\frac{\delta}{x} = \frac{0.37}{\mathrm{Re}_x^{1/5}}, \qquad C_{fx} = \frac{0.0576}{\mathrm{Re}_x^{1/5}}, \qquad C_D = \frac{0.074}{\mathrm{Re}_L^{1/5}} \quad \left(5 \times 10^5 < \mathrm{Re}_L < 10^7\right)$$

What it tells us. The turbulent layer grows as $x^{4/5}$, much faster than the laminar $x^{1/2}$. Its wall shear falls only slowly along the plate ($\tau_0 \propto x^{-1/5}$), so turbulent skin friction is much higher. The Blasius shear relation comes from pipe data, which limits the result to Re_L below about 10^7.

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

At higher Reynolds numbers, the **Prandtl–Schlichting** formula $C_D = 0.455/\left(\log_{10}\mathrm{Re}_L\right)^{2.58}$ holds up to $\mathrm{Re}_L \approx 10^9$.

10.2 Mixed boundary layer CORE

IN SIMPLE WORDS

A real plate is laminar at the front and turbulent further back, like a road that starts smooth and becomes bumpy. The turbulent formula alone would wrongly treat the front part as turbulent. The term $1700/\mathrm{Re}_L$ subtracts that extra drag. Use it when the laminar part is a noticeable share of the plate.

On a real plate, the first part (up to x_{cr}, where $\mathrm{Re}_x = 5 \times 10^5$) is laminar and the rest is turbulent. The turbulent formula over-counts the drag of the laminar part, so a correction is subtracted:

$$C_D = \frac{0.074}{\mathrm{Re}_L^{1/5}} - \frac{1700}{\mathrm{Re}_L}$$

Property	Laminar boundary layer	Turbulent boundary layer
Range on a flat plate	$\mathrm{Re}_x < 5 \times 10^5$	$\mathrm{Re}_x > 5 \times 10^5$
Thickness	$\delta = 4.91x/\mathrm{Re}_x^{0.5}$; grows as $x^{0.5}$	$\delta = 0.37x/\mathrm{Re}_x^{0.2}$; grows as $x^{0.8}$
Velocity profile	Near-parabolic	Fuller, about $(y/\delta)^{1/7}$
Shape factor $H = \delta^*/\theta$	About 2.6	About 1.3
Local skin friction	$C_{fx} = 0.664/\mathrm{Re}_x^{0.5}$	$C_{fx} = 0.0576/\mathrm{Re}_x^{0.2}$
Resistance to separation	Low: separates easily	High: stays attached longer
Heat and mass transfer	Lower	Higher, because of mixing

10.3 Laminar sublayer CORE

IN SIMPLE WORDS

Even in turbulent flow, the fluid touching the wall cannot swirl through it, so a thin calm film remains. Inside this sublayer the velocity rises in a straight line from zero. $u_* = \sqrt{\tau_0/\rho}$ is the shear velocity, a speed scale built from the wall stress. The sublayer thickness $\delta' = 11.6\nu/u_*$ is often only a fraction of a millimetre. If the surface bumps are hidden inside it, the wall behaves as smooth.

In a turbulent boundary layer, the velocity fluctuations die out right at the wall because the fluid cannot move through it. A very thin layer remains in which viscous shear dominates and the velocity varies linearly with y. This is the **laminar (viscous) sublayer**. Using the shear velocity $u_* = \sqrt{\tau_0/\rho}$ and the wall coordinate $y^+ = u_* y/\nu$:

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

Region	y^+	Velocity law
Laminar sublayer	0 to about 5	$u/u_* = y^+$ (linear)
Buffer layer	5 to 30	Neither law fits; blends the two
Turbulent (log) layer	30 up to about 0.2δ	$u/u_* = 5.75 \log_{10} y^+ + 5.5$
Outer layer	Rest of the boundary layer	Velocity defect law

For engineering use, the sublayer is taken as extending to the intersection of the linear and log laws, $y^+ \approx 11.6$.

Derivation from scratch: thickness of the laminar sublayer

Aim. To show that the velocity in the sublayer is linear, and that its thickness is $\delta' = 11.6\nu/u_*$.

Assumptions.

1. Inside the sublayer the turbulent fluctuations die out, so the shear stress is purely viscous: $\tau = \mu\, du/dy$.
2. The sublayer is so thin that the shear stress across it equals the wall value: $\tau = \tau_0$.
3. The sublayer ends where the linear law meets the logarithmic law $u/u_* = 5.75 \log_{10} y^+ + 5.5$.

Step 1. Newton's law of viscosity with constant stress.

$$\mu \frac{du}{dy} = \tau_0$$

Step 2. Integrate with respect to y.

$$u = \frac{\tau_0}{\mu}\, y + C_1$$

Step 3. No slip at the wall. At $y = 0$, $u = 0$, so $C_1 = 0$ and $u = \tau_0 y/\mu$: the velocity is linear in y.

Step 4. Write it with the shear velocity. Put $\tau_0 = \rho u_*^2$ and $\mu = \rho\nu$:

$$u = \frac{\rho u_*^2\, y}{\rho\nu} = \frac{u_*^2\, y}{\nu} \quad \Rightarrow \quad \frac{u}{u_*} = \frac{u_*\, y}{\nu} = y^+$$

Step 5. Edge of the sublayer. Set the linear law equal to the log law: $y^+ = 5.75 \log_{10} y^+ + 5.5$. Try $y^+ = 11.6$: the right side is $5.75 \times 1.0645 + 5.5 = 11.62$, which matches. So at $y = \delta'$:

$$\frac{u_*\, \delta'}{\nu} = 11.6$$

Step 6. Solve for δ'. Multiply both sides by ν/u_*.

Result.

$$\delta' = \frac{11.6\, \nu}{u_*}$$

What it tells us. The sublayer is thinner when the wall shear is larger (larger $u_* = \sqrt{\tau_0/\rho}$) and thicker in a more viscous fluid. Faster flow therefore shrinks the sublayer and can expose surface roughness.

The sublayer decides whether a surface acts as smooth or rough. Roughness elements buried inside it do not affect the drag (Section 3.5).

Worked examples

Step-by-step method.

1. Find Re_L and the laminar length $x_{cr} = 5\times10^5\,\nu/U$.
2. For the thickness, use $\delta = 0.37x/\mathrm{Re}_x^{0.2}$.
3. For the drag, use $C_D = 0.074/\mathrm{Re}_L^{0.2}$ (fully turbulent), or subtract $1700/\mathrm{Re}_L$ (mixed layer).
4. Find the drag per side, $F_D = C_D \times \tfrac{1}{2}\rho U^2 \times bL$.
5. For the sublayer, find C_{fx}, then τ_0, then $u_* = \sqrt{\tau_0/\rho}$.
6. Finally, $\delta' = 11.6\nu/u_*$.

Example 10.0 (warm-up) CORE Air ($\nu = 1.5\times10^{-5}\ \mathrm{m^2/s}$) flows at 10 m/s over a long flat plate. Find where the layer turns turbulent, and the turbulent boundary layer thickness 2 m from the leading edge.

1. $x_{cr} = \dfrac{5\times10^5\times1.5\times10^{-5}}{10} = 0.75\ \mathrm{m}$
2. $\mathrm{Re}_x = \dfrac{10\times2}{1.5\times10^{-5}} = 1.33\times10^6$
3. $\delta = \dfrac{0.37\times2}{\left(1.33\times10^6\right)^{0.2}} = \dfrac{0.74}{16.79} = 44.1\ \mathrm{mm}$

Example 10.1 HOTS Water ($\rho = 1000\ \mathrm{kg/m^3}$, $\nu = 10^{-6}\ \mathrm{m^2/s}$) flows at 2 m/s past a flat plate 3 m long and 1 m wide. Find the length of the laminar part, the boundary layer thickness at the trailing edge, and the drag on both sides of the plate.

1. $\mathrm{Re}_L = \dfrac{2\times3}{10^{-6}} = 6\times10^6$; the laminar part ends at $x_{cr} = \dfrac{5\times10^5\times10^{-6}}{2} = 0.25\ \mathrm{m}$
2. At the trailing edge, $\delta = \dfrac{0.37\times3}{\left(6\times10^6\right)^{0.2}} = 48.9\ \mathrm{mm}$
3. Mixed layer: $C_D = \dfrac{0.074}{\left(6\times10^6\right)^{0.2}} - \dfrac{1700}{6\times10^6} = 3.263\times10^{-3} - 0.283\times10^{-3} = 2.980\times10^{-3}$
4. Drag on one side $= 2.980\times10^{-3}\times0.5\times1000\times2^2\times3\times1 = 17.9\ \mathrm{N}$, so the drag on both sides is $35.8\ \mathrm{N}$
5. Treating the whole plate as turbulent would give 19.6 N per side, about 9% more

Example 10.2 CORE For the plate of Example 10.1, find the wall shear stress, the shear velocity and the laminar sublayer thickness 2 m from the leading edge.

1. $\mathrm{Re}_x = \dfrac{2\times2}{10^{-6}} = 4\times10^6$; $C_{fx} = \dfrac{0.0576}{\left(4\times10^6\right)^{0.2}} = 2.754\times10^{-3}$
2. $\tau_0 = 2.754\times10^{-3}\times0.5\times1000\times2^2 = 5.51\ \mathrm{N/m^2}$
3. $u_* = \sqrt{\dfrac{5.51}{1000}} = 0.0742\ \mathrm{m/s}$
4. $\delta' = \dfrac{11.6\times10^{-6}}{0.0742} = 0.156\ \mathrm{mm}$

5. The whole boundary layer there is $\dfrac{0.37 \times 2}{\left(4 \times 10^{6}\right)^{0.2}} = 35.4\,\mathrm{mm}$ thick, about 230 times the sublayer

Common mistakes and exam tips.

- Turbulent formulas use the power 0.2 (fifth root); laminar ones use 0.5 (square root).
- Subtract $1700/\mathrm{Re}_L$ when the question says the layer is mixed, or the laminar front part is to be allowed for.
- x_{cr} uses the plate value 5×10^5, not the pipe value 2000.
- Find u_* with τ_0 in $\mathrm{N/m^2}$ and ρ in $\mathrm{kg/m^3}$; give δ' in mm.
- Use C_{fx} (local) for τ_0 at a point and C_D (average) for the total drag.

Check your understanding.

1. CORE Which grows faster with x, a laminar or a turbulent boundary layer? *(Turbulent: as $x^{0.8}$ against $x^{0.5}$.)*
2. CORE What is the laminar sublayer? *(A very thin layer next to the wall, under a turbulent layer, where viscous shear dominates.)*
3. HOTS If τ_0 increases, what happens to δ'? *(It becomes thinner, since u_* rises and $\delta' = 11.6\nu/u_*$.)*

11. Boundary layer separation and its control (L7) CORE

A boundary layer separates from a surface when it has to flow against a rising pressure (an adverse pressure gradient) for too long. The slow fluid near the wall is brought to rest and then pushed backwards. The main flow leaves the surface and a wide, eddying wake forms behind it. Separation is the main cause of drag on bluff bodies, of stall on aircraft wings, and of losses in diffusers and bends.

IN SIMPLE WORDS

Imagine cycling up a steep hill at the end of a long ride. Tired legs give up before the top. The slow fluid near the wall is the tired cyclist, and a rising pressure is the hill. When it can no longer move forward, it stops and turns back, and the main flow leaves the surface. This leaves a wide, swirling wake that causes large drag and loss of lift. Engineers control it to save fuel and to stop wings from stalling.

11.1 Mechanism CORE

IN SIMPLE WORDS

On the front of a body the pressure falls in the flow direction and pushes the fluid along, like rolling downhill. Behind the widest point the pressure rises, and the fluid must roll uphill. The fluid near the wall has already lost speed to friction, so it stops first. The place where it stops is the separation point; behind it lies the wake.

- Over the front of a curved body, the flow speeds up and the pressure falls ($dp/dx < 0$, a **favourable gradient**). The boundary layer is helped along and stays thin.
- Past the widest point, the flow slows and the pressure rises ($dp/dx > 0$, an **adverse gradient**). The fluid near the wall, which has already lost momentum to friction, now also has to climb a pressure hill.
- The near-wall fluid slows to rest at the **point of separation**. Downstream of it, the flow next to the wall reverses. The boundary layer lifts off the surface, and the region behind it fills with large eddies: the **wake**.

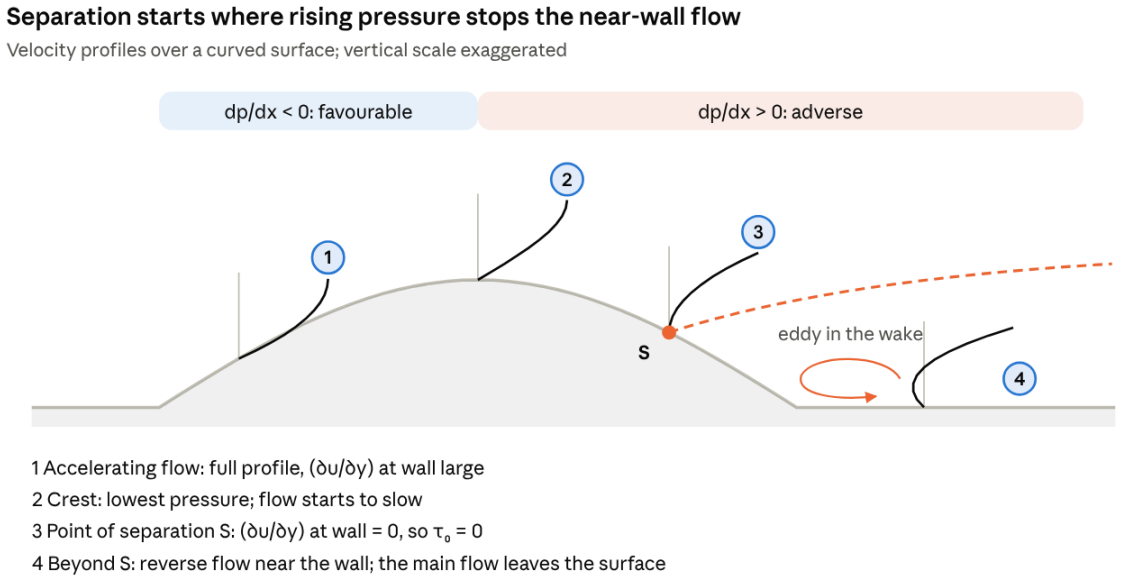


Figure 5. Velocity profiles approaching and beyond the point of separation

The wall velocity gradient falls along the adverse-gradient side until it reaches zero at S. Beyond S the near-wall flow reverses.

11.2 Criterion for separation CORE

> **IN SIMPLE WORDS**
>
> Look at the slope of the velocity curve right at the wall, $\left(\partial u/\partial y\right)_{y=0}$. A positive slope means the near-wall fluid still moves forward: attached. A zero slope means it has just stopped: the separation point. A negative slope means it flows backwards: separated. Since $\tau_0 = \mu\left(\partial u/\partial y\right)$ at the wall, the wall shear is zero at separation.

The state of the layer is read from the velocity gradient at the wall.

Derivation from scratch: criterion for separation from the wall shear stress

Aim. To show why an adverse pressure gradient causes separation, and that separation occurs where $\left(\partial u/\partial y\right)_{y=0} = 0$, that is, where $\tau_0 = 0$.

Assumptions.

1. Steady, two-dimensional flow along a wall, with the pressure $p\left(x\right)$ imposed by the outer flow.
2. At the wall the fluid is at rest (no slip), so a fluid element touching the wall has no acceleration.

Step 1. Wall shear stress. By Newton's law of viscosity at the wall:

$$\tau_0 = \mu \left(\frac{\partial u}{\partial y}\right)_{y=0}$$

If the slope is positive, the fluid just above the wall moves forward. If it is negative, that fluid moves backward.

Step 2. Force balance on a tiny element at the wall. Take an element of length dx, height dy and unit width resting on the wall. The forces in the x-direction are:

- pressure on the left face, $p\,dy$ (forward); on the right face, $\left(p + \dfrac{\partial p}{\partial x}dx\right)dy$ (backward);
- shear on the bottom face, $\tau\,dx$ (backward); on the top face, $\left(\tau + \dfrac{\partial \tau}{\partial y}dy\right)dx$ (forward).

The element is at rest, so the sum is zero:

$$p\,dy - \left(p + \frac{\partial p}{\partial x}dx\right)dy - \tau\,dx + \left(\tau + \frac{\partial \tau}{\partial y}dy\right)dx = 0$$

Step 3. Simplify. The terms $p\,dy$ and $\tau\,dx$ cancel. Divide by $dx\,dy$ and use $\tau = \mu\,\partial u/\partial y$:

$$\frac{\partial \tau}{\partial y} = \frac{\partial p}{\partial x} \quad \Rightarrow \quad \mu\left(\frac{\partial^2 u}{\partial y^2}\right)_{y=0} = \frac{dp}{dx}$$

Step 4. Meaning of the sign. With a favourable gradient ($dp/dx < 0$) the profile bends forward and the wall slope stays healthy. With an adverse gradient ($dp/dx > 0$) the curvature at the wall is positive: the profile is pushed back near the wall. As the fluid moves further against the rising pressure, the wall slope keeps falling, reaches zero, and then turns negative.

Step 5. The three states.

$$\left(\frac{\partial u}{\partial y}\right)_{y=0} > 0 \ \text{ attached}$$

$$\left(\frac{\partial u}{\partial y}\right)_{y=0} = 0 \ \text{ on the verge of separation}$$

$$\left(\frac{\partial u}{\partial y}\right)_{y=0} < 0 \ \text{ separated (reverse flow)}$$

Result.

$$\text{Separation: } \left(\frac{\partial u}{\partial y}\right)_{y=0} = 0, \quad \text{that is, } \tau_0 = 0$$

What it tells us. Separation is possible only under an adverse pressure gradient. On a flat plate with $dp/dx = 0$ the wall curvature is zero, and the layer never separates. Since $\tau_0 = \mu\,(\,\partial u/\partial y\,)$ at the wall, the wall shear stress is zero at the separation point.

A turbulent layer has much more momentum near the wall than a laminar one, so it can push further against an adverse gradient before separating.

The circular cylinder. In ideal flow, the pressure coefficient on a cylinder is $C_p = 1 - 4\sin^2\theta$, measured from the front stagnation point. The pressure falls to its minimum at $\theta = 90^\circ$ and rises again over the rear half. A laminar boundary layer separates at about $\theta = 82^\circ$, just past the shoulder, leaving a wide wake. A turbulent layer holds on to about $\theta = 120^\circ$, so the wake is narrower and the drag lower. This is the cause of the **drag crisis** (Section 12.4).

11.3 Effects of separation CORE

IN SIMPLE WORDS

Separation leaves a low-pressure region behind the body, which pulls the body backwards (pressure drag). On a wing it destroys lift (stall). In diffusers and bends it wastes energy. The eddies shed from the body can also shake chimneys and cables.

Effect	Where it matters
Large pressure (form) drag from the low-pressure wake	Bluff bodies: trucks, buildings, cylinders, spheres
Loss of lift (stall) when the angle of attack is too high	Aircraft wings, fan and compressor blades
Energy loss and poor pressure recovery	Diffusers, sudden enlargements, pipe bends, draft tubes
Periodic vortex shedding (Kármán vortex street) and vibration	Chimneys, cables, heat-exchanger tubes, bridges
Noise and buffeting	Vehicles, wind-tunnel models

11.4 Methods of controlling separation CORE

IN SIMPLE WORDS

To prevent separation, give the tired near-wall fluid more energy, remove it, or make the pressure hill gentler. Streamlining makes the hill gentle. Suction removes slow fluid, and blowing adds fast fluid. The dimples on a golf ball make the layer turbulent, so it clings to the surface longer.

All the methods either add momentum to the slow fluid near the wall, remove that fluid, or reduce the adverse pressure gradient:

Method	How it works
Streamlining	A long, gently tapered tail keeps the adverse gradient mild, so the layer stays attached almost to the end
Suction	Slow fluid is drawn off through slots or a porous surface, leaving a thinner, faster layer
Blowing (injection)	A high-speed jet along the surface re-energises the boundary layer
Slotted wing and slats	Air from the high-pressure lower side flows through a slot and energises the upper layer; delays stall
Vortex generators	Small vanes on a wing or car roof create vortices that mix fast outer air into the layer
Rotating cylinder	A moving wall gives the near-wall fluid extra momentum (also the Magnus effect, Section 12)
Tripping the layer	Roughness, a trip wire or dimples make the layer turbulent early, as on a golf ball
Gentle divergence and guide vanes	Diffuser cone angles kept below about 7° to 8°; turning vanes in bends and elbows

Worked examples

Step-by-step method.

1. Write u as a function of y from the given profile ($u = U \times$ the profile).
2. Differentiate u with respect to y.
3. Put $y = 0$ to get the velocity gradient at the wall.
4. Positive: attached; zero: on the verge of separation; negative: separated.
5. If the wall shear is asked, multiply by μ: $\tau_0 = \mu \left(\partial u / \partial y \right)_{y=0}$.

Example 11.0 (warm-up) CORE Water ($\mu = 10^{-3}\ \mathrm{N{\cdot}s/m^2}$) in a boundary layer has the linear profile $u/U = y/\delta$, with $U = 2\ \mathrm{m/s}$ and $\delta = 4\ \mathrm{mm}$. Is the layer attached? Find the wall shear stress.

1. $\left(\dfrac{\partial u}{\partial y}\right)_{y=0} = \dfrac{U}{\delta} = \dfrac{2}{0.004} = 500\ \mathrm{s^{-1}} > 0$, so the layer is **attached**
2. $\tau_0 = \mu\left(\dfrac{\partial u}{\partial y}\right)_{y=0} = 10^{-3} \times 500 = 0.5\ \mathrm{N/m^2}$

Example 11.1 HOTS Check whether each of these boundary layer profiles is attached, on the verge of separation, or separated: (a) $u/U = 2\,(y/\delta) - (y/\delta)^2$; (b) $u/U = (y/\delta)^2$; (c) $u/U = 2\,(y/\delta)^2 - (y/\delta)$.

1. (a) $\left(\dfrac{\partial u}{\partial y}\right)_{y=0} = \dfrac{2U}{\delta} > 0$: **attached**
2. (b) $\dfrac{\partial u}{\partial y} = \dfrac{2Uy}{\delta^2}$, which is 0 at $y = 0$: **on the verge of separation**
3. (c) $\left(\dfrac{\partial u}{\partial y}\right)_{y=0} = -\dfrac{U}{\delta} < 0$: **separated**, with reverse flow near the wall

Example 11.2 HOTS Using the ideal-flow pressure distribution $C_p = 1 - 4\sin^2\theta$ on a circular cylinder, find where the pressure is lowest and over what range the pressure gradient is adverse.

1. C_p is lowest where $\sin^2\theta = 1$, at $\theta = 90^\circ$, where $C_p = -3$
2. From 0° to 90° the pressure falls (favourable). From 90° to 180° it rises back to $C_p = 1$ at the rear (adverse)
3. A real boundary layer cannot climb the full rise from $C_p = -3$ to 1. It separates soon after 90°, so the rear pressure stays low, and the front-to-rear difference produces the form drag

Common mistakes and exam tips.

- Evaluate the velocity gradient at $y = 0$ (the wall), not at $y = \delta$.
- Separation needs an adverse gradient ($dp/dx > 0$); a flat plate with $dp/dx = 0$ does not separate.
- A turbulent layer separates later than a laminar one, not earlier.
- For control methods, name each method and say how it works: adds momentum, removes slow fluid, or eases the gradient.
- Sketch the velocity profiles before, at and after separation; the sketch earns marks.

Check your understanding.

1. CORE What is an adverse pressure gradient? *(Pressure rising in the flow direction, $dp/dx > 0$.)*
2. CORE What is the wall shear stress at the point of separation? *(Zero.)*

3. HOTS Why do golf balls have dimples? *(They make the layer turbulent, which delays separation, narrows the wake and lowers the drag.)*

12. Drag and lift (L8) CORE

A body in a moving fluid feels a resultant force from the pressure and the shear stress over its surface. The component of this force along the flow is **drag**, and the component across it is **lift**. Both are written as a coefficient times the dynamic pressure $\frac{1}{2}\rho U^2$ times an area. The coefficients depend on the body's shape, its Reynolds number and, for lift, its angle to the flow.

IN SIMPLE WORDS

Hold your hand out of a moving car window. You feel a push backwards (drag) and, if you tilt your hand, a push upwards (lift). Drag acts along the flow and lift acts across it. Both are written as $F = C \times \frac{1}{2}\rho U^2 \times A$. Here $\frac{1}{2}\rho U^2$ is the dynamic pressure, A is a reference area and C is a coefficient that depends on the shape. Engineers use them to size car engines, aircraft wings and the supports of signboards.

12.1 Definitions CORE

IN SIMPLE WORDS

On each tiny piece of surface, pressure pushes straight in and shear rubs along it. Add up their parts along the flow to get drag, and the parts across the flow to get lift. In practice we do not integrate; we use measured coefficients C_D and C_L. Note the area used: frontal area for bluff bodies, wing area for aerofoils.

Derivation from scratch: drag and lift as integrals of pressure and shear

Aim. To express the drag and lift on a body as the sum of the pressure and shear forces on its surface, and to define C_D and C_L.

Assumptions.

1. A body is held in a uniform stream of velocity U in the x-direction. Drag is the force along x; lift is the force across it, in the y-direction (upward).
2. On every small surface element dA the fluid exerts a pressure p and a wall shear stress τ_0.

Step 1. Forces on one element. The pressure pushes on the surface, along the inward normal, with force $p\,dA$. The shear drags the surface along its tangent, with force $\tau_0\,dA$. These two forces are at right angles to each other.

Step 2. Fix the angle. Take an element on the upper front part of the body. Let θ be the angle that the pressure force $p\,dA$ makes with the flow direction. The pressure force then points downstream and downward; the shear force, at right angles to it, points downstream and upward.

Step 3. Resolve the pressure force.

$$\text{along the flow: } p\,dA\cos\theta, \qquad \text{across the flow: } -p\,dA\sin\theta$$

The minus sign shows that pressure on this element pushes the body downward.

Step 4. Resolve the shear force. It is turned 90° from the pressure force, so the sine and cosine swap places:

$$\text{along the flow: } \tau_0\, dA \sin\theta, \qquad \text{across the flow: } \tau_0\, dA \cos\theta$$

Step 5. Add the parts for one element.

$$dF_D = p\cos\theta\, dA + \tau_0 \sin\theta\, dA, \qquad dF_L = -p\sin\theta\, dA + \tau_0 \cos\theta\, dA$$

Step 6. Add over the whole surface A. On other parts of the body θ takes other values, and the same expressions give each element's share with the correct sign.

$$F_D = \int_A p\cos\theta\, dA + \int_A \tau_0 \sin\theta\, dA$$

$$F_L = -\int_A p\sin\theta\, dA + \int_A \tau_0 \cos\theta\, dA$$

Step 7. Coefficients. The integrals are hard to evaluate for a real body. So each force is compared with the dynamic pressure $\frac{1}{2}\rho U^2$ times a reference area A, which also has the units of force. The ratios are the drag and lift coefficients:

$$C_D = \frac{F_D}{\frac{1}{2}\rho U^2 A}, \qquad C_L = \frac{F_L}{\frac{1}{2}\rho U^2 A}$$

Result.

$$F_D = \int_A p\cos\theta\, dA + \int_A \tau_0 \sin\theta\, dA$$

$$F_L = -\int_A p\sin\theta\, dA + \int_A \tau_0 \cos\theta\, dA$$

$$F_D = C_D \frac{1}{2}\rho U^2 A, \qquad F_L = C_L \frac{1}{2}\rho U^2 A$$

What it tells us. The first integral in F_D is the pressure (form) drag and the second is the skin friction drag. Once C_D and C_L are measured, both forces grow as the square of the speed and in proportion to the area.

A is the projected frontal area for bluff bodies, and the plan (wing) area for aerofoils and flat plates.

12.2 Pressure drag and friction drag CORE

IN SIMPLE WORDS

Drag has two parts. Pressure (form) drag comes from high pressure in front and low pressure in the wake, like pushing an open umbrella. Friction drag comes from fluid rubbing on the surface, like a thin plate moving edge-on. Bluff bodies suffer mainly pressure drag; streamlined bodies mainly friction drag.

The first integral above is the **pressure (form) drag**, from the difference between the high pressure at the front and the low pressure in the wake. The second is the **skin friction drag**. Their shares depend on shape:

Body	Pressure drag	Friction drag	Remark
Flat plate parallel to the flow	None	All	Section 9 and 10 results
Streamlined body (aerofoil, fish, airship)	Small	Most	Boundary layer stays attached; thin wake
Bluff body (cylinder, sphere, truck)	Most	Small	Early separation; wide wake
Flat plate normal to the flow	All	None	Separation fixed at the edges; C_D independent of Re

Streamlining reduces drag by turning a wide wake into a thin one, at the cost of more wetted area. A streamlined strut can have one-tenth the drag of a round tube of the same thickness.

12.3 Typical drag coefficients CORE

> **IN SIMPLE WORDS**
>
> C_D is a number that tells how much drag a shape produces. A flat plate facing the flow has about 2, a sphere about 0.47 and a streamlined body less than 0.1. Multiply C_D by the dynamic pressure and the frontal area to get the force. These values are measured in wind tunnels.

Body (frontal area, Re about 10^4 to 10^5)	C_D
Flat plate normal to the flow, long (2D)	2.0
Circular disc normal to the flow	1.1 to 1.2
Circular cylinder, axis across the flow, long	1.2
Sphere	0.47
Hemisphere, open side facing the flow	1.4
Modern car	0.25 to 0.35
Truck or bus	0.6 to 0.9
Streamlined body	0.04 to 0.1

12.4 Drag on a sphere; Stokes' law CORE

IN SIMPLE WORDS

A tiny particle moving very slowly, like dust settling in still air, is in creeping flow ($\mathrm{Re} < 1$). Its drag is then simply $3\pi\mu DU$, where μ is the viscosity, D the diameter and U the speed. When drag plus buoyancy balances the weight, the particle falls at a steady terminal velocity. Measuring that velocity lets us find the viscosity.

For very slow, "creeping" flow ($\mathrm{Re} = \rho UD/\mu < 1$), inertia is negligible and Stokes solved the flow exactly. The total drag is:

$$F_D = 3\pi\mu DU \quad (\ \pi\mu DU \text{ from pressure} + 2\pi\mu DU \text{ from skin friction}\ ), \qquad C_D = \frac{24}{\mathrm{Re}}$$

Terminal velocity. A small sphere falling through a still fluid accelerates until the drag plus the buoyancy equals its weight.

Derivation from scratch: Stokes' drag coefficient and terminal velocity of a sphere

Aim. To show that $C_D = 24/\mathrm{Re}$ in Stokes flow, and to find the terminal velocity of a small sphere from a force balance.

Assumptions.

1. A sphere of diameter D and density ρ_s falls through a still fluid of density ρ and viscosity μ.
2. The flow is creeping ($\mathrm{Re} = \rho UD/\mu < 1$), so Stokes' drag $F_D = 3\pi\mu DU$ applies.
3. At terminal velocity the sphere moves steadily: its acceleration is zero.

Step 1. Drag coefficient. Divide the drag by the dynamic pressure times the frontal area $\pi D^2/4$:

$$C_D = \frac{3\pi\mu DU}{\frac{1}{2}\rho U^2 \times \frac{\pi}{4}D^2} = \frac{3\pi\mu DU \times 8}{\pi\rho U^2 D^2} = \frac{24\,\mu}{\rho UD} = \frac{24}{\mathrm{Re}}$$

Step 2. List the forces on the sphere. The volume of the sphere is $\frac{\pi}{6}D^3$.

- Weight, downward: $W = \frac{\pi}{6}D^3\,\rho_s g$.
- Buoyancy (weight of displaced fluid, by Archimedes), upward: $F_B = \frac{\pi}{6}D^3\,\rho g$.
- Drag, upward (it opposes the downward motion): $F_D = 3\pi\mu DU$.

Step 3. Force balance at terminal velocity. Upward forces equal downward forces.

$$3\pi\mu DU + \frac{\pi}{6}D^3\rho g = \frac{\pi}{6}D^3\rho_s g$$

Step 4. Collect the density terms. Move the buoyancy to the right side.

$$3\pi\mu DU = \frac{\pi}{6}D^3\left(\rho_s - \rho\right)g$$

Step 5. Solve for U. Divide both sides by $3\pi\mu D$. Note that $\frac{\pi/6}{3\pi} = \frac{1}{18}$ and $\frac{D^3}{D} = D^2$.

$$U = \frac{D^2\,\left(\rho_s - \rho\right)\,g}{18\,\mu}$$

Step 6. Solve for μ instead (falling-sphere viscometer). Multiply both sides by μ/U.

$$\mu = \frac{D^2\,\left(\rho_s - \rho\right)\,g}{18\,U}$$

Result.

$$C_D = \frac{24}{\mathrm{Re}}, \qquad U = \frac{D^2\,\left(\rho_s - \rho\right)\,g}{18\,\mu}, \qquad \mu = \frac{D^2\,\left(\rho_s - \rho\right)\,g}{18\,U}$$

What it tells us. The terminal velocity grows as the square of the diameter: a particle twice as large settles four times as fast. It is proportional to the density difference and inversely proportional to the viscosity. Always check afterwards that $\mathrm{Re} < 1$, or Stokes' law does not apply.

This is the basis of the falling-sphere viscometer, and of the settling of dust, sediment and droplets.

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

Re **range**	C_D **of a sphere**	**Flow behaviour**
< 1	$24/\mathrm{Re}$	Stokes flow; no separation
1 to 10^3	Falls gradually from $24/\mathrm{Re}$ to about 0.4; Oseen: $\left(24/\mathrm{Re}\right)\left(1 + 3\mathrm{Re}/16\right)$ up to $\mathrm{Re} \approx 5$	Separation begins near $\mathrm{Re} = 20$; a wake forms
10^3 to 3×10^5	About 0.4 to 0.5	Laminar separation near 82°; wide wake
About 3×10^5	Falls suddenly to about 0.1 to 0.2	**Drag crisis**: the boundary layer turns turbulent, separation moves back to about 120°, and the wake narrows

The dimples on a golf ball trip the layer to turbulence at a lower Re, bringing the drag crisis into the speed range of a drive.

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

12.5 Lift: circulation and the Magnus effect ADVANCED

Lift needs an asymmetric flow, with faster flow and lower pressure on one side. It is described by the **circulation** Γ, the line integral of velocity around the body. The **Kutta-Joukowski theorem** gives the lift per unit span:

$$F_L' = \rho U \Gamma$$

Rotating cylinder (Magnus effect). A cylinder of radius R spinning at ω in a stream U drags fluid round with it. The flow speeds up on the side where the surface moves with the stream and slows on the other, so the cylinder is pushed sideways. Taking the circulation as that of the surface, $\Gamma = 2\pi R\,(\,\omega R\,)$:

$$C_L = \frac{F_L'}{\frac{1}{2}\rho U^2\,(\,2R\,)} = \frac{\Gamma}{RU}, \qquad \sin\theta_s = \frac{\Gamma}{4\pi RU}$$

Here θ_s locates the two stagnation points, which move towards each other as the spin increases. The same effect makes a spinning cricket or tennis ball swerve, and has been used for Flettner rotor ships.

Aerofoils. An aerofoil produces circulation through its shape and its angle of attack α. The sharp trailing edge forces the flow to leave smoothly there (the Kutta condition). For a thin aerofoil, $C_L \approx 2\pi\alpha$ (α in radians) up to the **stall**, at about 12° to 16°. At the stall, the boundary layer separates from the upper surface, the lift falls sharply and the drag rises. A good aerofoil reaches a lift-to-drag ratio of 50 to 100.

Worked examples

Step-by-step method.

1. Convert the speed to m/s (divide km/h by 3.6) and all lengths to metres.
2. Find the dynamic pressure $\frac{1}{2}\rho U^2$.
3. Choose the area: frontal area for bluff bodies, wing (plan) area for aerofoils.
4. Find the force $F = C \times \frac{1}{2}\rho U^2 \times A$, and the power as $F_D \times U$.
5. For Stokes' law, use $U = D^2\left(\rho_s - \rho\right) g/\,(\,18\mu\,)$ and check that $\mathrm{Re} = \rho UD/\mu < 1$.

Example 12.0 (warm-up) CORE A ball 0.2 m in diameter ($C_D = 0.47$) is held in a 10 m/s air stream ($\rho = 1.2\ \mathrm{kg/m^3}$). Find the drag on it.

1. $A = \dfrac{\pi}{4} \times 0.2^2 = 0.0314\ \mathrm{m^2}$
2. $\dfrac{1}{2}\rho U^2 = \dfrac{1}{2} \times 1.2 \times 10^2 = 60\ \mathrm{N/m^2}$
3. $F_D = C_D \times \dfrac{1}{2}\rho U^2 \times A = 0.47 \times 60 \times 0.0314 = \ 0.886\ \mathrm{N}$

Example 12.1 HOTS A car has $C_D = 0.3$ and a frontal area of 2.2 m². Find the aerodynamic drag and the power needed to overcome it at 100 km/h and 120 km/h in still air ($\rho = 1.2\ \mathrm{kg/m^3}$).

Speed	U (m/s)	Drag $F = C_D \tfrac{1}{2}\rho U^2 A$ (N)	Power FU (kW)
100 km/h	27.8	306	8.5
120 km/h	33.3	440	14.7

A 20% rise in speed raises the drag by 44% and the power by 73%, since drag rises as U^2 and power as U^3.

Example 12.2 CORE In a falling-sphere viscometer, a steel ball ($\rho_s = 7800\ \mathrm{kg/m^3}$) of 2 mm diameter falls through oil ($\rho = 900\ \mathrm{kg/m^3}$) at a steady 20 mm/s. Find the viscosity of the oil and check that Stokes' law applies.

1. $\mu = \dfrac{D^2\left(\rho_s - \rho\right) g}{18U} = \dfrac{(\,0.002\,)^{\,2} \times 6900 \times 9.81}{18 \times 0.02} = \ 0.752\ \mathrm{N{\cdot}s/m^2}$
2. $\mathrm{Re} = \dfrac{900 \times 0.02 \times 0.002}{0.752} = 0.048 < 1$, so Stokes' law holds
3. Drag $\ = 3\pi\mu DU = 2.84 \times 10^{-4}\ \mathrm{N}$, equal to the net weight $\tfrac{\pi}{6}D^3\left(\rho_s - \rho\right) g$, as it should be

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

Example 12.3 ADVANCED A cylinder 1.2 m in diameter and 10 m long rotates at 200 rpm with its axis across a 20 m/s air stream ($\rho = 1.2\ \mathrm{kg/m^3}$). Find the circulation, the lift force, the lift coefficient and the position of the stagnation points.

1. $\omega = \dfrac{2\pi \times 200}{60} = 20.94\ \mathrm{rad/s}$; surface speed $\omega R = 12.57\ \mathrm{m/s}$
2. $\Gamma = 2\pi R\,(\,\omega R\,) \ = 2\pi \times 0.6 \times 12.57 = \ 47.4\ \mathrm{m^2/s}$
3. Lift $\ = \rho U \Gamma \times \mathrm{length} = 1.2 \times 20 \times 47.4 \times 10 = \ 11.37\ \mathrm{kN}$
4. $C_L = \dfrac{\Gamma}{RU} = \dfrac{47.4}{0.6 \times 20} = \ 3.95$
5. $\sin\theta_s = \dfrac{47.4}{4\pi \times 0.6 \times 20} = 0.314$, so $\theta_s = 18.3^\circ$, measured from the horizontal axis toward the side where the surface moves against the stream

Example 12.4 HOTS A light aircraft of weight 12 kN has a wing area of 16 m². Find the lift coefficient needed for level flight at 200 km/h ($\rho = 1.2\ \mathrm{kg/m^3}$). If the drag coefficient based on wing area is 0.035, find the drag, the power needed and the lift-to-drag ratio.

1. $U = 55.6\ \mathrm{m/s}$; $\tfrac{1}{2}\rho U^2 = 1852\ \mathrm{N/m^2}$
2. $C_L = \dfrac{W}{\frac{1}{2}\rho U^2 A} = \dfrac{12\,000}{1852 \times 16} = \ 0.405$. By thin-aerofoil theory this needs $\alpha \approx 0.405/\,(\,2\pi\,)\ \mathrm{rad} \approx 3.7^\circ$, well below the stall
3. Drag $\ = 0.035 \times 1852 \times 16 = \ 1037\ \mathrm{N}$; power $\ = 1037 \times 55.6 = \ 57.7\ \mathrm{kW}$

4. $L/D = \dfrac{12\,000}{1037} = 11.6$

Common mistakes and exam tips.

- Convert km/h to m/s (divide by 3.6) before squaring the speed.
- Drag rises as U^2 but power as U^3: doubling the speed needs eight times the power.
- Use the frontal area for cars and spheres, and the wing area for aircraft.
- In Stokes' law, use the net density $\left(\rho_s - \rho\right)$ and the diameter D, not the radius; then check $\mathrm{Re} < 1$.
- In level flight, lift equals weight; do not equate drag to weight.

Check your understanding.

1. CORE What is the difference between drag and lift? *(Drag acts along the flow; lift acts across it.)*
2. HOTS If a car's speed doubles, how does its drag change? *(It becomes four times as large, since $F_D \propto U^2$.)*
3. CORE When does Stokes' law apply? *(In creeping flow, when $\mathrm{Re} < 1$.)*

13. Formula sheet and practice questions CORE

13.1 Formula sheet CORE

Topic	Result
Reynolds number	$\mathrm{Re} = \rho VD/\mu$; pipe laminar < 2000, turbulent > 4000 ; plate transition $\mathrm{Re}_x \approx 5 \times 10^5$
Laminar pipe: shear	$\tau = \left(-\partial p/\partial x \right) r/2$; $\tau_0 = \left(p_1 - p_2 \right) D/\left(4L \right)$
Laminar pipe: velocity	$u = u_{\max}\left(1 - r^2/R^2 \right)$; $u_{\max} = 2V$; $u = V$ at $r = 0.707R$
Hagen–Poiseuille	$p_1 - p_2 = 32\mu VL/D^2 = 128\mu QL/\left(\pi D^4 \right)$; $f = 64/\mathrm{Re}$
Correction factors, laminar	$\alpha = 2$, $\beta = 4/3$
Fixed parallel plates	$u = \frac{1}{2\mu}\left(-\frac{\partial p}{\partial x} \right)\left(ty - y^2 \right)$; $V = \frac{2}{3}u_{\max}$; $p_1 - p_2 = 12\mu VL/t^2$
Couette flow	$u = Uy/t$ (no pressure gradient); $\tau = \mu U/t$
Darcy–Weisbach	$h_f = fLV^2/\left(2gD \right) = 8fLQ^2/\left(\pi^2 gD^5 \right)$; $f = 4f_1$
Wall shear, shear velocity	$\tau_0 = f\rho V^2/8$; $u_* = V\sqrt{f/8}$
Mixing length	$\tau = \rho l^2 \left(du/dy \right)^2$, $l = 0.4y$
Turbulent pipe profile	$\left(u_{\max} - u \right)/u_* = 5.75 \log\left(R/y \right)$; $u_{\max} = V + 3.75u_*$
Laminar sublayer	$\delta' = 11.6\nu/u_*$; smooth if $k/\delta' < 0.25$, rough if > 6
Blasius (smooth pipe)	$f = 0.316/\mathrm{Re}^{0.25}$, $4000 < \mathrm{Re} < 10^5$
Colebrook	$1/\sqrt{f} = -2 \log\left[\frac{k/D}{3.7} + \frac{2.51}{\mathrm{Re}\sqrt{f}} \right]$
Sudden enlargement	$\left(V_1 - V_2 \right)^2/\left(2g \right)$
Sudden contraction	$\left(1/C_c - 1 \right)^2 V_2^2/\left(2g \right) \approx 0.5V_2^2/\left(2g \right)$
Entrance, exit	$0.5V^2/\left(2g \right)$, $V^2/\left(2g \right)$
Equivalent length	$L_e = KD/f$
Pipes in series	Same Q; $H = \sum$ losses; $L/D^5 = \sum L_i/D_i^5$ (Dupuit)
Pipes in parallel	Same h_f; $Q = \sum Q_i$
Power transmission	$P = \rho gQ\left(H - h_f \right)$; maximum at $h_f = H/3$, $\eta = 66.7\%$
Nozzle for maximum power	$d = \left(D^5/\left(2fL \right) \right)^{1/4}$
Siphon summit	$p_s/\left(\rho g \right) = p_a/\left(\rho g \right) - h_s - \left(1.5 + fL_1/D \right) V^2/\left(2g \right)$
Water hammer, gradual	$p = \rho LV/T$, when $T > 2L/C$
Water hammer, sudden	$p = \rho VC$, $C = \sqrt{K/\rho}$; elastic pipe $p = V\sqrt{\rho/\left(1/K + D/\left(Et \right) \right)}$
BL thicknesses	$\delta^* = \int \left(1 - u/U \right) dy$; $\theta = \int \left(u/U \right)\left(1 - u/U \right) dy$

Topic	Result
Momentum integral	$\tau_0 / \left(\rho U^2 \right) = d\theta/dx$
Blasius (laminar plate)	$\delta = 4.91x/\sqrt{\mathrm{Re}_x}$; $C_{fx} = 0.664/\sqrt{\mathrm{Re}_x}$; $C_D = 1.328/\sqrt{\mathrm{Re}_L}$
Turbulent plate	$\delta = 0.37x/\mathrm{Re}_x^{0.2}$; $C_{fx} = 0.0576/\mathrm{Re}_x^{0.2}$; $C_D = 0.074/\mathrm{Re}_L^{0.2}$ ($-1700/\mathrm{Re}_L$ if mixed)
Separation	$\left(\partial u/\partial y \right)$ at $y = 0$ equal to zero
Drag and lift	$F_D = C_D \frac{1}{2}\rho U^2 A$; $F_L = C_L \frac{1}{2}\rho U^2 A$
Stokes' law	$F_D = 3\pi\mu DU$; $C_D = 24/\mathrm{Re}$; terminal $U = D^2 \left(\rho_s - \rho \right) g / \left(18\mu \right)$
Kutta–Joukowski	Lift per unit span $= \rho U \Gamma$; rotating cylinder $C_L = \Gamma / \left(RU \right)$

13.2 Short-answer questions (2 marks) CORE

1. CORE Define the Reynolds number and state its critical values for pipe flow.
2. CORE What is meant by fully developed flow? Give the entrance length for laminar flow.
3. CORE Show that the mean velocity of laminar flow in a pipe is half the maximum velocity.
4. CORE At what radius does the local velocity equal the mean velocity in laminar pipe flow?
5. CORE State the laws of fluid friction for laminar and turbulent flow.
6. CORE Distinguish between hydraulically smooth and hydraulically rough pipes.
7. CORE What is the Moody chart? Name its four zones.
8. CORE Define the hydraulic gradient line and the total energy line. How far apart are they?
9. CORE What is an equivalent pipe? State Dupuit's equation.
10. CORE State the condition for maximum power transmission through a pipe, and the efficiency at that condition.
11. HOTS Why is the height of a siphon summit limited?
12. CORE What is water hammer? Define the critical time of closure.
13. CORE Define displacement thickness and momentum thickness.
14. CORE State the condition for boundary layer separation in terms of the velocity gradient.
15. CORE List four methods of controlling boundary layer separation.
16. HOTS Distinguish between pressure drag and friction drag, with an example of each.

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

17. ADVANCED What is the drag crisis on a sphere?
18. ADVANCED Explain the Magnus effect.

13.3 Long-answer questions (10 marks) CORE

1. CORE Derive the Hagen–Poiseuille equation for laminar flow through a circular pipe. Show that $f = 64/\mathrm{Re}$.
2. CORE Derive the velocity distribution and the pressure drop for laminar flow between two fixed parallel plates.
3. CORE Derive the Darcy–Weisbach equation for the loss of head due to friction in a pipe.
4. CORE Explain Prandtl's mixing length theory and derive the logarithmic velocity distribution near a wall.
5. CORE Derive an expression for the loss of head due to sudden enlargement of a pipe.
6. HOTS Show that the power transmitted through a pipe is greatest when the friction loss is one-third of the supply head. Derive the nozzle diameter for maximum power.
7. HOTS Derive the pressure rise due to sudden closure of a valve in an elastic pipe.
8. CORE Derive von Kármán's momentum integral equation. Use it with a cubic velocity profile to find δ/x and C_D for a laminar flat plate.
9. CORE Explain boundary layer separation with a sketch, its effects, and the methods of controlling it.
10. CORE Explain drag and lift on immersed bodies. Discuss the variation of the drag coefficient of a sphere with Reynolds number.

13.4 Numerical problems (with answers) CORE

1. CORE Oil ($\rho = 850\ \mathrm{kg/m^3}$, $\mu = 0.2\ \mathrm{N{\cdot}s/m^2}$) flows through a 50 mm pipe at 1.5 L/s. For 100 m of pipe, find the pressure drop, the head loss and the power required. *(Answer: $\mathrm{Re} = 162$; 195.6 kPa; 23.5 m; 293 W.)*
2. CORE Find the head lost to friction when $0.08\ \mathrm{m^3/s}$ of water flows through 800 m of 250 mm pipe with $f = 0.022$. *(Answer: 9.53 m.)*
3. CORE A pipe carrying 30 L/s of water enlarges suddenly from 100 mm to 200 mm diameter. Find the loss of head. *(Answer: 0.418 m.)*
4. CORE Find the diameter of a single pipe equivalent to 400 m of 250 mm pipe followed by 300 m of 150 mm pipe. *(Answer: 174 mm.)*
5. HOTS Two pipes, each 1000 m long and of diameters 300 mm and 200 mm, are laid in parallel and carry a total of $0.2\ \mathrm{m^3/s}$. Taking $f = 0.02$, find the flow in each and the head loss. *(Answer: 0.147 and $0.053\ \mathrm{m^3/s}$; 14.6 m.)*
6. HOTS A pipe 1 km long and 200 mm in diameter ($f = 0.024$) is supplied under a head of 100 m. Find the maximum power it can transmit and the nozzle diameter for maximum power at its outlet. *(Answer: 48.0 kW at 73.3 L/s; 50.8 mm.)*

7. **HOTS** Water flows at 1.5 m/s through a 1500 m pipe. The valve at its end is closed in 2 s. Taking $K = 2.1\ \text{GPa}$ and a rigid pipe, find the pressure rise. *(Answer: $2L/C = 2.07\ \text{s}$, so the closure is sudden; 2.17 MPa.)*
8. **CORE** A siphon of 150 mm diameter and 300 m length ($f = 0.02$) joins two reservoirs whose levels differ by 3 m. Its summit is 4 m above the upper level, 60 m from the inlet. Find the discharge and the absolute summit pressure (atmosphere = 10.3 m of water). *(Answer: 21.0 L/s; 5.61 m of water.)*
9. **CORE** Air ($\rho = 1.2\ \text{kg/m}^3$, $\nu = 1.5 \times 10^{-5}\ \text{m}^2/\text{s}$) flows at 5 m/s past a plate 0.6 m long. Find the boundary layer thickness at the trailing edge and the drag per metre width on both sides. *(Answer: 6.59 mm; 0.053 N.)*
10. **HOTS** Water ($\nu = 10^{-6}\ \text{m}^2/\text{s}$) flows at 4 m/s past a plate 5 m long and 2 m wide. Treating the layer as mixed, find the drag on one side. *(Answer: $\text{Re}_L = 2 \times 10^7$, $C_D = 2.48 \times 10^{-3}$, 198 N; the 0.074 formula is used slightly beyond its range.)*
11. **CORE** Find the terminal velocity of a dust particle of 50 µm diameter and density 2500 kg/m³ falling in air ($\mu = 1.8 \times 10^{-5}\ \text{N·s/m}^2$, $\rho = 1.2\ \text{kg/m}^3$). *(Answer: 0.189 m/s; $\text{Re} = 0.63$, so Stokes' law applies.)*
12. **CORE** A hoarding 4 m × 2 m faces a 90 km/h wind. Taking $C_D = 1.2$ and $\rho = 1.2\ \text{kg/m}^3$, find the wind force. *(Answer: 3.6 kN.)*

13.5 Key terms in one line each CORE

Term	Meaning in plain words
Siphon	A bent pipe that carries liquid over a ridge from a higher to a lower level, with no pump
Priming	Filling a siphon (or pump) with liquid before it can start working
Summit pressure	The lowest pressure in a siphon, at its highest point; it must stay above about 2.5 m of water absolute
Water hammer	The pressure surge when a flowing column of water is stopped quickly
Critical time of closure	The wave round-trip time $2L/C$; closing faster than this counts as sudden
Bulk modulus K	How hard a fluid is to compress; for water about $2.1\ \mathrm{GPa}$
Surge tank	An open standpipe that lets the water level rise instead of the pressure
Boundary layer	The thin layer next to a surface where the speed rises from zero to the stream speed
No-slip condition	Fluid touching a solid wall moves with the wall
Displacement thickness δ^*	How far the boundary layer pushes the outer flow away from the wall
Momentum thickness θ	A thickness that measures the momentum lost to wall friction
Shape factor H	The ratio δ^*/θ; a rising value warns of separation
Momentum integral equation	Wall shear stress equals ρU^2 times the growth rate of θ
Transition	The change from laminar to turbulent flow, near $\mathrm{Re}_x = 5 \times 10^5$ on a plate
Laminar sublayer	A very thin smooth film next to the wall under a turbulent layer
Shear velocity u_*	A velocity scale from the wall stress, $\sqrt{\tau_0/\rho}$
Adverse pressure gradient	Pressure rising in the flow direction, which slows the near-wall fluid
Separation	The boundary layer leaving the surface after the near-wall flow stops and reverses
Wake	The low-pressure, eddying region behind a body
Drag	The fluid force on a body along the flow direction

Term	Meaning in plain words
Lift	The fluid force on a body across the flow direction
Drag coefficient C_D	Drag divided by dynamic pressure times reference area
Stokes' law	Drag on a slow, small sphere: $3\pi\mu DU$, valid for $\mathrm{Re} < 1$
Terminal velocity	The steady speed at which drag plus buoyancy balance the weight
Stall	Sudden loss of lift when the flow separates from the upper surface of a wing
Magnus effect	The sideways force on a spinning ball or cylinder in a stream